EXPLAROTARY DATA ANALYSIS

BCSE331L

Course Project

23BDS0081

Submitted To:

Prof. Balaji GN

Submitted By:

Vyom Khurana

23BDS0081

Video Link:
https://drive.google.com/drive/folders/1-E9nqPz4qoCq0UACKuIwQ9jodzxtQR_N

GitHub Link:
https://github.com/Vyomkhurana/EDA

23BDS0081

Exploratory Data Analysis

BCSE331P

Assessment – I

23BDS0081

Submitted To:

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23BDS0081

DATAFRAME OPERATIONS

1) Append the records of dataframe1 and dataframe3.

CODE:

```
#23BDS0081
#VYOMKHURANA

dataframe1 <- data.frame(
ProductID=c(101,102,103,104,105,106,107,108,109,110),
NameofProd=c("Mouse","Keyboard","Monitor","Laptop","Printer","Speaker","
Router","SSD","Webcam","Headset"),
Price=c(500,1200,8000,55000,6500,1800,2500,4200,1600,2200)
)

dataframe3 <- data.frame(
ProductID=c(111,112,113,114,115,116,117,118,119,120),
NameofProd=c("Tablet","Charger","Cable","Projector","Scanner","Pendrive","
PowerBank","Microphone","Smartwatch","Camera"),
Price=c(18000,900,300,25000,7000,700,1500,2800,12000,35000)
)

append_df <- rbind(dataframe1,dataframe3)

append_df
```

SCREENSHOT OF THE OUPUT:

```
18       118 Microphone  2800
19       119 Smartwatch 12000
20       120     Camera 35000
> #23BDS0081
> #VYOMKHURANA
>
> dataframe1 <- data.frame(
+     ProductID=c(101,102,103,104,105,106,107,108,109,110),
+     NameofProd=c("Mouse","Keyboard","Monitor","Laptop","Printer","Speaker","Router","SSD","Webca
m","Headset"),
+     Price=c(500,1200,8000,55000,6500,1800,2500,4200,1600,2200)
+ )
>
> dataframe2 <- data.frame(
+     ProductID=c(101,102,103,104,105,106,107,108,109,110),
+     Company_Location=c("Delhi","Mumbai","Chennai","Bangalore","Hyderabad","Pune","Noida","Kolkat
a","Jaipur","Ahmedabad")
+ )
>
> concat_df <- cbind(dataframe1,dataframe2["Company_Location"])
>
> concat_df
   ProductID NameofProd Price Company_Location
1        101      Mouse   500            Delhi
2        102   Keyboard  1200           Mumbai
3        103    Monitor  8000          Chennai
4        104     Laptop 55000        Bangalore
5        105    Printer  6500        Hyderabad
6        106    Speaker  1800             Pune
7        107     Router  2500            Noida
8        108        SSD  4200          Kolkata
9        109     Webcam  1600           Jaipur
10       110    Headset  2200        Ahmedabad
>
```

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2) **Concatenate the data frames wherever applicable.**

CODE:

```
#23BDS0081
#VYOMKHURANA

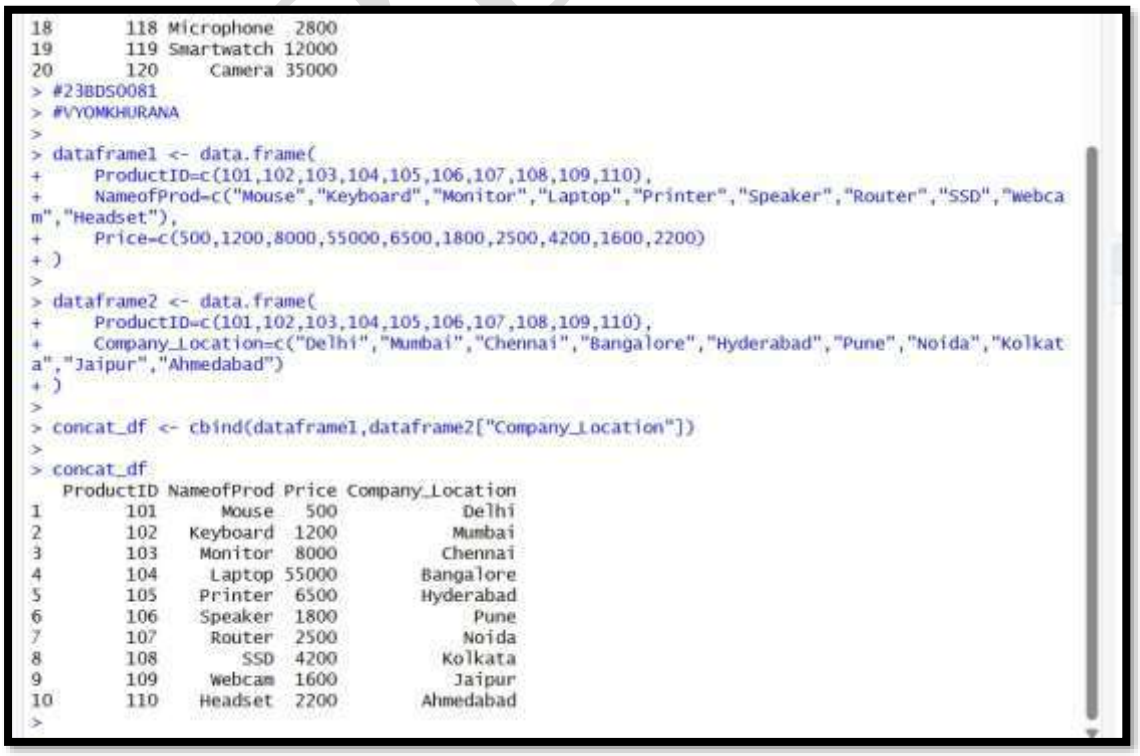
dataframe1 <- data.frame(
ProductID=c(101,102,103,104,105,106,107,108,109,110),
NameofProd=c("Mouse","Keyboard","Monitor","Laptop","Printer","Speaker","
Router","SSD","Webcam","Headset"),
Price=c(500,1200,8000,55000,6500,1800,2500,4200,1600,2200)
)

dataframe2 <- data.frame(
ProductID=c(101,102,103,104,105,106,107,108,109,110),
Company_Location=c("Delhi","Mumbai","Chennai","Bangalore","Hyderabad",
"Pune","Noida","Kolkata","Jaipur","Ahmedabad")
)

concat_df <- cbind(dataframe1,dataframe2["Company_Location"])

concat_df
```

SCREENSHOT OF THE OUTPUT:



```
18       118 Microphone  2800
19       119 Smartwatch 12000
20       120     Camera 35000
> #23BDS0081
> #VYOMKHURANA
>
> dataframe1 <- data.frame(
+     ProductID=c(101,102,103,104,105,106,107,108,109,110),
+     NameofProd=c("Mouse","Keyboard","Monitor","Laptop","Printer","Speaker","Router","SSD","Webca
m","Headset"),
+     Price=c(500,1200,8000,55000,6500,1800,2500,4200,1600,2200)
+ )
>
> dataframe2 <- data.frame(
+     ProductID=c(101,102,103,104,105,106,107,108,109,110),
+     Company_Location=c("Delhi","Mumbai","Chennai","Bangalore","Hyderabad","Pune","Noida","Kolkat
a","Jaipur","Ahmedabad")
+ )
>
> concat_df <- cbind(dataframe1,dataframe2["Company_Location"])
>
> concat_df
   ProductID NameofProd Price Company_Location
1        101      Mouse   500            Delhi
2        102   Keyboard  1200           Mumbai
3        103    Monitor  8000          Chennai
4        104     Laptop 55000        Bangalore
5        105    Printer  6500        Hyderabad
6        106    Speaker  1800             Pune
7        107     Router  2500            Noida
8        108        SSD  4200          Kolkata
9        109     Webcam  1600           Jaipur
10       110    Headset  2200        Ahmedabad
>
```

3) **Merge dataframe1 and dataframe2 using the ProductID attribute.**

CODE:

```
#23BDS0081
#VYOMKHURANA

dataframe1 <- data.frame(
ProductID=c(101,102,103,104,105,106,107,108,109,110),
NameofProd=c("Mouse","Keyboard","Monitor","Laptop","Printer","Speaker","
Router","SSD","Webcam","Headset"),
Price=c(500,1200,8000,55000,6500,1800,2500,4200,1600,2200)
)

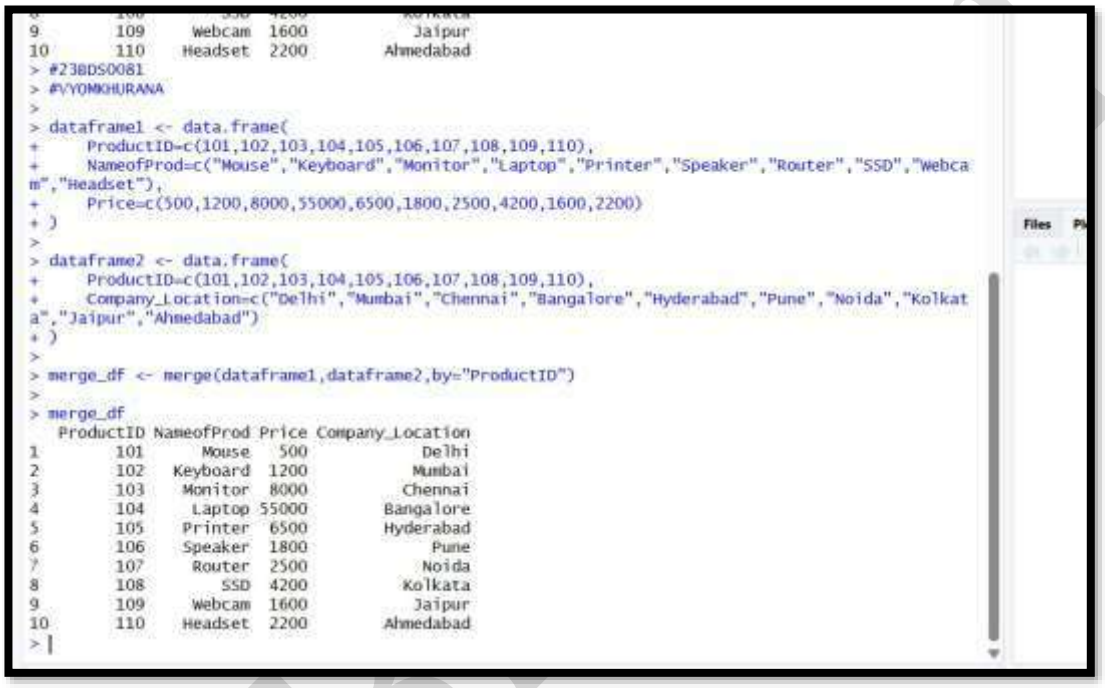
dataframe2 <- data.frame(
ProductID=c(101,102,103,104,105,106,107,108,109,110),
Company_Location=c("Delhi","Mumbai","Chennai","Bangalore","Hyderabad",
"Pune","Noida","Kolkata","Jaipur","Ahmedabad")
)

merge_df <- merge(dataframe1,dataframe2,by="ProductID")

merge_df
```

SCREENSHOT OF THE OUTPUT:

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```
9         109     Webcam  1600          Jaipur
10        110    Headset  2200       Ahmedabad
> #23BDS0081
> #VYOMKHURANA
>
> dataframe1 <- data.frame(
+     ProductID=c(101,102,103,104,105,106,107,108,109,110),
+     NameofProd=c("Mouse","Keyboard","Monitor","Laptop","Printer","Speaker","Router","SSD","Webca
m","Headset"),
+     Price=c(500,1200,8000,55000,6500,1800,2500,4200,1600,2200)
+ )
>
> dataframe2 <- data.frame(
+     ProductID=c(101,102,103,104,105,106,107,108,109,110),
+     Company_Location=c("Delhi","Mumbai","Chennai","Bangalore","Hyderabad","Pune","Noida","Kolkat
a","Jaipur","Ahmedabad")
+ )
>
> merge_df <- merge(dataframe1,dataframe2,by="ProductID")
>
> merge_df
   ProductID NameofProd Price Company_Location
1        101      Mouse   500            Delhi
2        102   Keyboard  1200           Mumbai
3        103    Monitor  8000          Chennai
4        104     Laptop 55000        Bangalore
5        105    Printer  6500        Hyderabad
6        106    Speaker  1800             Pune
7        107     Router  2500            Noida
8        108        SSD  4200          Kolkata
9        109     Webcam  1600           Jaipur
10       110    Headset  2200        Ahmedabad
>
```

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4) **Perform the following join operations between dataframe1 and dataframe2:**

- **Inner Join**
- **Left Join**
- **Right Join**
- **Full Outer Join**

CODE:

```
#23BDS0081
#VYOMKHURANA

dataframe1 <- data.frame(
ProductID=c(101,102,103,104,105,106,107,108,109,110),
NameofProd=c("Mouse","Keyboard","Monitor","Laptop","Printer","Speaker","
Router","SSD","Webcam","Headset"),
Price=c(500,1200,8000,55000,6500,1800,2500,4200,1600,2200)
)

dataframe2 <- data.frame(
ProductID=c(101,102,103,104,105,106,107,108,109,110),
Company_Location=c("Delhi","Mumbai","Chennai","Bangalore","Hyderabad",
"Pune","Noida","Kolkata","Jaipur","Ahmedabad")
)

inner_join <- merge(dataframe1,dataframe2,by="ProductID")

left_join <- merge(dataframe1,dataframe2,by="ProductID",all.x=TRUE)

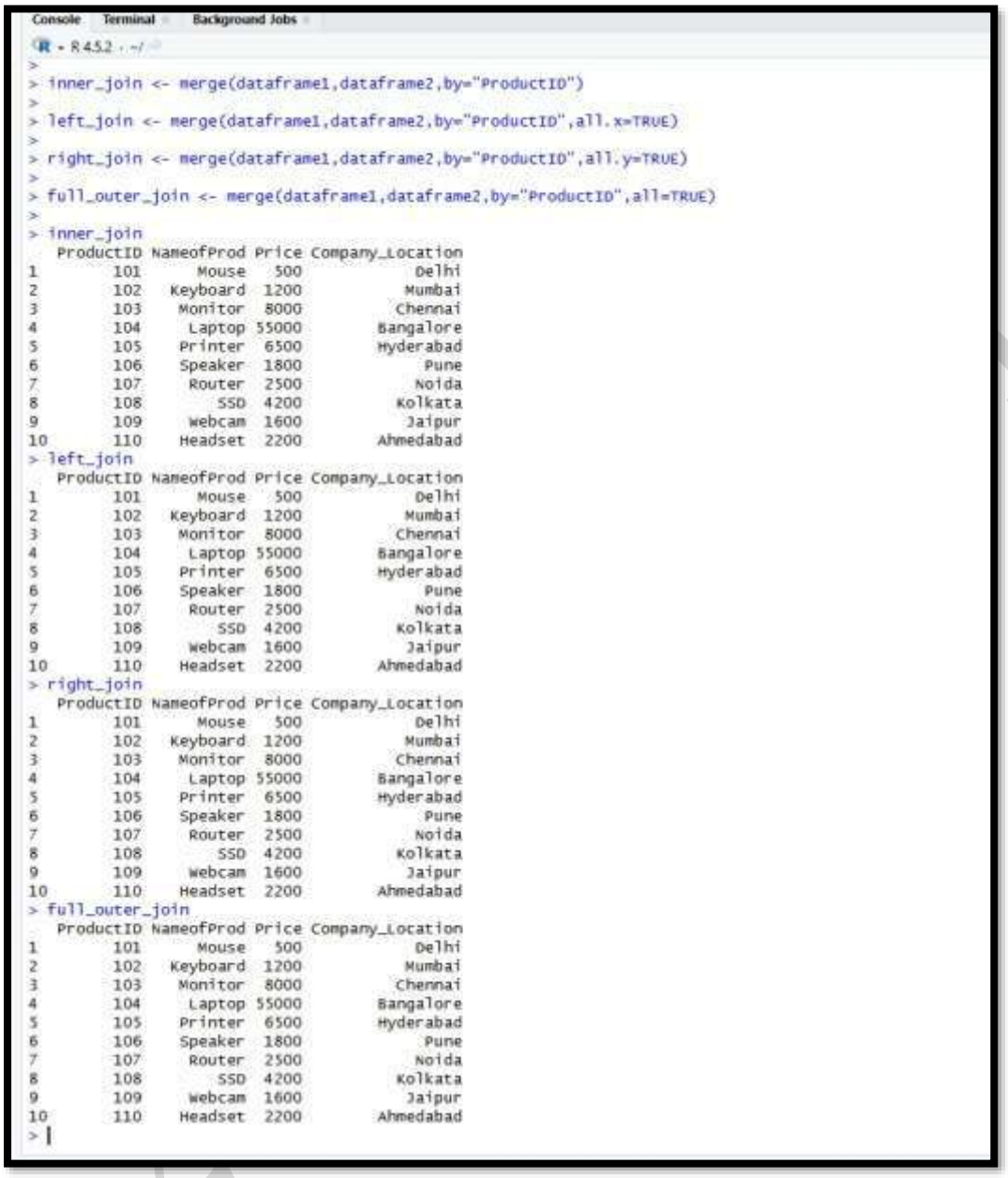
right_join <- merge(dataframe1,dataframe2,by="ProductID",all.y=TRUE)

full_outer_join <- merge(dataframe1,dataframe2,by="ProductID",all=TRUE)

inner_join
left_join
right_join
full_outer_join
```

23BDS0081

SCREENSHOT OF THE OUTPUT:



```
Console  Terminal  Background Jobs
R · R 4.5.2 · ~/
>
> inner_join <- merge(dataframe1,dataframe2,by="ProductID")
>
> left_join <- merge(dataframe1,dataframe2,by="ProductID",all.x=TRUE)
>
> right_join <- merge(dataframe1,dataframe2,by="ProductID",all.y=TRUE)
>
> full_outer_join <- merge(dataframe1,dataframe2,by="ProductID",all=TRUE)
>
> inner_join
   ProductID NameofProd Price Company_Location
1        101      Mouse   500            Delhi
2        102   Keyboard  1200           Mumbai
3        103    Monitor  8000          Chennai
4        104     Laptop 55000        Bangalore
5        105    Printer  6500        Hyderabad
6        106    Speaker  1800             Pune
7        107     Router  2500            Noida
8        108        SSD  4200          Kolkata
9        109     Webcam  1600           Jaipur
10       110    Headset  2200        Ahmedabad
> left_join
   ProductID NameofProd Price Company_Location
1        101      Mouse   500            Delhi
2        102   Keyboard  1200           Mumbai
3        103    Monitor  8000          Chennai
4        104     Laptop 55000        Bangalore
5        105    Printer  6500        Hyderabad
6        106    Speaker  1800             Pune
7        107     Router  2500            Noida
8        108        SSD  4200          Kolkata
9        109     Webcam  1600           Jaipur
10       110    Headset  2200        Ahmedabad
> right_join
   ProductID NameofProd Price Company_Location
1        101      Mouse   500            Delhi
2        102   Keyboard  1200           Mumbai
3        103    Monitor  8000          Chennai
4        104     Laptop 55000        Bangalore
5        105    Printer  6500        Hyderabad
6        106    Speaker  1800             Pune
7        107     Router  2500            Noida
8        108        SSD  4200          Kolkata
9        109     Webcam  1600           Jaipur
10       110    Headset  2200        Ahmedabad
> full_outer_join
   ProductID NameofProd Price Company_Location
1        101      Mouse   500            Delhi
2        102   Keyboard  1200           Mumbai
3        103    Monitor  8000          Chennai
4        104     Laptop 55000        Bangalore
5        105    Printer  6500        Hyderabad
6        106    Speaker  1800             Pune
7        107     Router  2500            Noida
8        108        SSD  4200          Kolkata
9        109     Webcam  1600           Jaipur
10       110    Headset  2200        Ahmedabad
>
```

Vyom khurana
23BDS0081
LAB ASSIGNMENT – 2

CODE -

```
# Registration Number: #23BDS0081

if (!require(zoo)) install.packages("zoo", quiet = TRUE)
library(zoo)

# 1
url <- "https://raw.githubusercontent.com/salemprakash/EDA/main/Data/SuicideChina.csv?
plain=1"
df <- read.csv(url, stringsAsFactors = FALSE)

# 2
head(df, 10)

# 3
tail(df, 10)

# 4
nrow(df)
ncol(df)

# 5
colnames(df)

# 6
str(df)

# 7
summary(df)

# 8
colSums(is.na(df))

# 9
sum(is.na(df))

# 10
colSums(is.na(df))

# 11
df[!complete.cases(df), ]

# 12
df_clean <- na.omit(df)
head(df_clean, 10)
```

```r
# 13
df_imputed <- df
df_imputed$Age[is.na(df_imputed$Age)] <- mean(df_imputed$Age, na.rm = TRUE)

# 14
df_imputed$Edu[is.na(df_imputed$Edu)] <- median(df_imputed$Edu, na.rm = TRUE)

# 15
get_mode <- function(v) {
  uniqv <- unique(na.omit(v))
  uniqv[which.max(tabulate(match(v, uniqv)))]
}
df_imputed$Sex[is.na(df_imputed$Sex)] <- get_mode(df_imputed$Sex)

# 16
df_ffill <- na.locf(df, na.rm = FALSE)
head(df_ffill, 10)

# 17
df_bfill <- na.locf(df, fromLast = TRUE, na.rm = FALSE)
head(df_bfill, 10)

# 18
df_constant <- df
for(col in names(df_constant)) {
  if(is.numeric(df_constant[[col]])) {
    df_constant[[col]][is.na(df_constant[[col]])] <- 0
  } else {
    df_constant[[col]][is.na(df_constant[[col]])] <- "Unknown"
  }
}
head(df_constant, 10)

# 19
df_unique <- unique(df)

# 20
df$Sex <- as.factor(df$Sex)

# 21
names(df)[names(df) == "Sex"] <- "Gender"

# 22
df$Age_Group <- ifelse(df$Age < 35, "Young", "Senior")
head(df, 10)

# 23
summary(df)

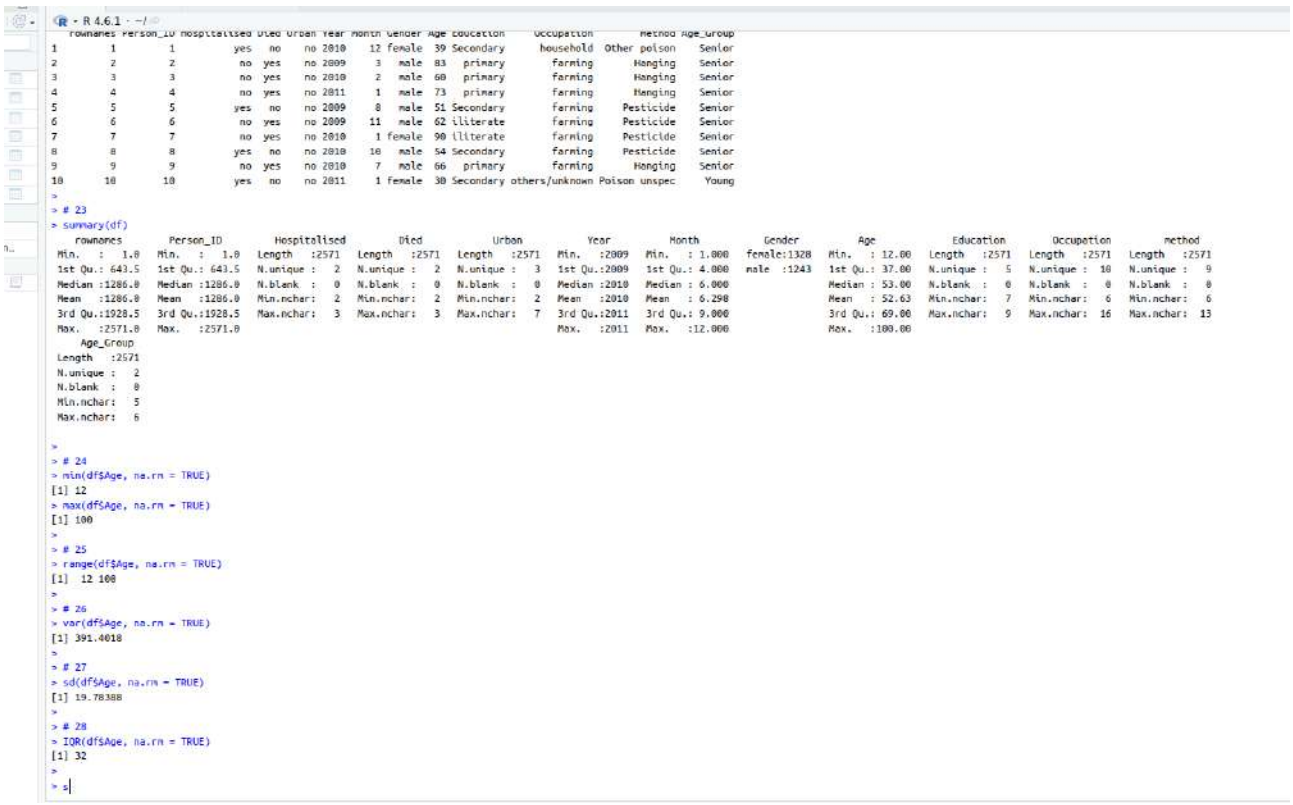
# 24
min(df$Age, na.rm = TRUE)
max(df$Age, na.rm = TRUE)
```

```
# 25
range(df$Age, na.rm = TRUE)

# 26
var(df$Age, na.rm = TRUE)

# 27
sd(df$Age, na.rm = TRUE)

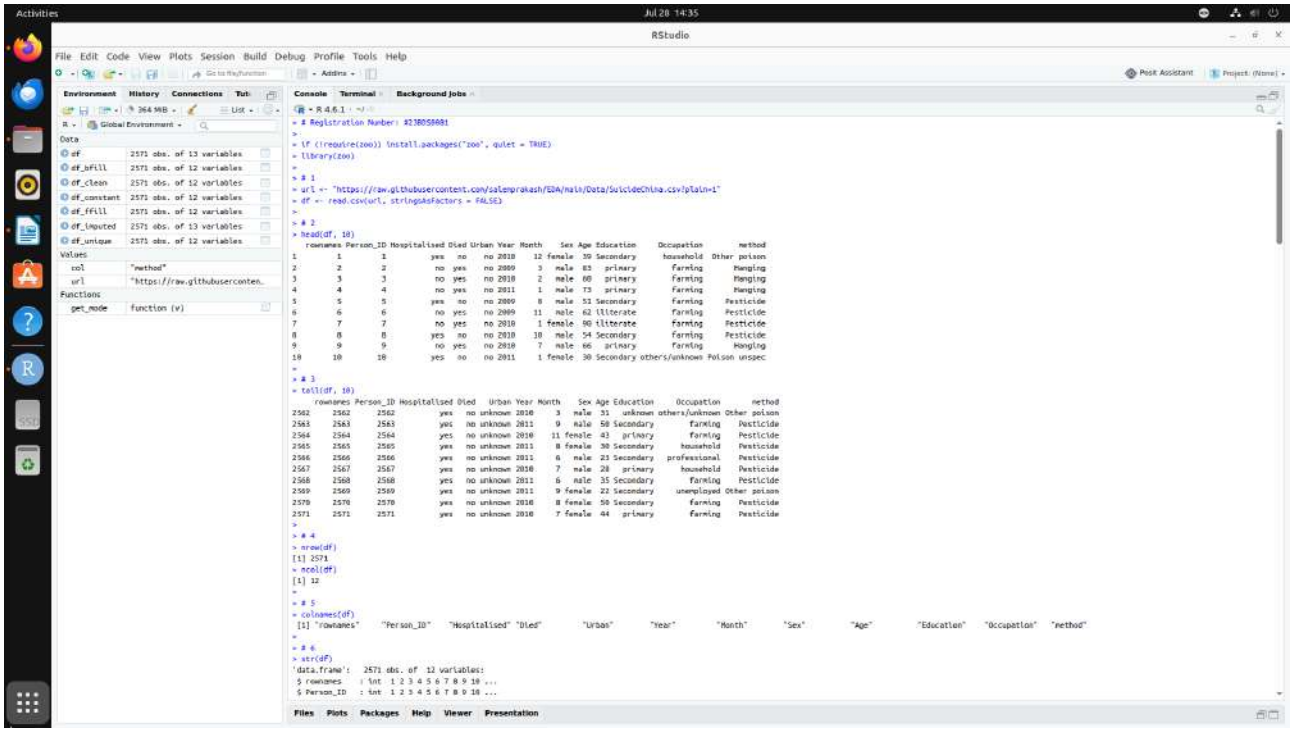
# 28
IQR(df$Age, na.rm = TRUE)
```



```
- R 4.6.1 · ~/

  rownames Person_ID Hospitalised Died Urban Year Month Gender Age Education     Occupation       Method Age_Group
1        1         1          yes   no    no 2010    12 female  39 Secondary      household  Other poison    Senior
2        2         2           no  yes    no 2009     3   male  83   primary        farming      Hanging    Senior
3        3         3           no  yes    no 2010     2   male  60   primary        farming      Hanging    Senior
4        4         4           no  yes    no 2011     1   male  73   primary        farming      Hanging    Senior
5        5         5          yes   no    no 2009     8   male  51 Secondary        farming    Pesticide    Senior
6        6         6           no  yes    no 2009    11   male  62 illiterate       farming    Pesticide    Senior
7        7         7           no  yes    no 2010     1 female  90 illiterate       farming    Pesticide    Senior
8        8         8          yes   no    no 2010    10   male  54 Secondary        farming    Pesticide    Senior
9        9         9           no  yes    no 2010     7   male  66   primary        farming      Hanging    Senior
10      10        10          yes   no    no 2011     1 female  30 Secondary others/unknown Poison unspec      Young
>
> # 23
> summary(df)
    rownames       Person_ID       Hospitalised          Died              Urban               Year          Month          Gender          Age          Education          Occupation          method
 Min.   :   1.0   Min.   :   1.0   Length  :2571   Length  :2571   Length  :2571   Min.   :2009   Min.   : 1.000   female:1328   Min.   : 12.00   Length  :2571   Length  :2571   Length  :2571
 1st Qu.: 643.5   1st Qu.: 643.5   N.unique :   2   N.unique :   2   N.unique :   3   1st Qu.:2009   1st Qu.: 4.000   male  :1243   1st Qu.: 37.00   N.unique :   5   N.unique :  10   N.unique :   9
 Median :1286.0   Median :1286.0   N.blank  :   0   N.blank  :   0   N.blank  :   0   Median :2010   Median : 6.000                  Median : 53.00   N.blank  :   0   N.blank  :   0   N.blank  :   0
 Mean   :1286.0   Mean   :1286.0   Min.nchar:   2   Min.nchar:   2   Min.nchar:   2   Mean   :2010   Mean   : 6.298                  Mean   : 52.63   Min.nchar:   7   Min.nchar:   6   Min.nchar:   6
 3rd Qu.:1928.5   3rd Qu.:1928.5   Max.nchar:   3   Max.nchar:   3   Max.nchar:   7   3rd Qu.:2011   3rd Qu.: 9.000                  3rd Qu.: 69.00   Max.nchar:   9   Max.nchar:  16   Max.nchar:  13
 Max.   :2571.0   Max.   :2571.0                                                     Max.   :2011   Max.   :12.000                  Max.   :100.00
    Age_Group
 Length  :2571
 N.unique :   2
 N.blank  :   0
 Min.nchar:   5
 Max.nchar:   6

>
> # 24
> min(df$Age, na.rm = TRUE)
[1] 12
> max(df$Age, na.rm = TRUE)
[1] 100
>
> # 25
> range(df$Age, na.rm = TRUE)
[1]  12 100
>
> # 26
> var(df$Age, na.rm = TRUE)
[1] 391.4018
>
> # 27
> sd(df$Age, na.rm = TRUE)
[1] 19.78388
>
> # 28
> IQR(df$Age, na.rm = TRUE)
[1] 32
>
> s
```



```
Activities                                  Jul 28 14:35
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help

Environment  History  Connections  Tut
Data
  df            2571 obs. of 13 variables
  df_bfill      2571 obs. of 12 variables
  df_clean      2571 obs. of 12 variables
  df_constant   2571 obs. of 12 variables
  df_ffill      2571 obs. of 12 variables
  df_imputed    2571 obs. of 13 variables
  df_unique     2571 obs. of 12 variables
Values
  col           "method"
  url           "https://raw.githubuserconten.
Functions
  get_mode      function (v)

Console  Terminal  Background Jobs
> # Registration Number: 23BDS0081
>
> if (!require(zoo)) install.packages("zoo", quiet = TRUE)
> library(zoo)
>
> # 1
> url <- "https://raw.githubusercontent.com/salemprakash/EDA/main/Data/SuicideChina.csv?plain=1"
> df <- read.csv(url, stringsAsFactors = FALSE)
>
> # 2
> head(df, 10)
   rownames Person_ID Hospitalised Died Urban Year Month    Sex Age Education     Occupation       method
1         1         1          yes   no    no 2010    12 female  39 Secondary      household  Other poison
2         2         2           no  yes    no 2009     3   male  83   primary        farming      Hanging
3         3         3           no  yes    no 2010     2   male  60   primary        farming      Hanging
4         4         4           no  yes    no 2011     1   male  73   primary        farming      Hanging
5         5         5          yes   no    no 2009     8   male  51 Secondary        farming    Pesticide
6         6         6           no  yes    no 2009    11   male  62 illiterate       farming    Pesticide
7         7         7           no  yes    no 2010     1 female  90 illiterate       farming    Pesticide
8         8         8          yes   no    no 2010    10   male  54 Secondary        farming    Pesticide
9         9         9           no  yes    no 2010     7   male  66   primary        farming      Hanging
10       10        10          yes   no    no 2011     1 female  30 Secondary others/unknown Poison unspec
>
> # 3
> tail(df, 10)
     rownames Person_ID Hospitalised Died    Urban Year Month    Sex Age Education     Occupation       method
2562     2562      2562          yes   no unknown 2010     3   male  51   unknown others/unknown Other poison
2563     2563      2563          yes   no unknown 2011     9   male  58 Secondary        farming    Pesticide
2564     2564      2564          yes   no unknown 2010    11 female  43   primary        farming    Pesticide
2565     2565      2565          yes   no unknown 2011     8 female  30 Secondary      household    Pesticide
2566     2566      2566          yes   no unknown 2011     6   male  23 Secondary   professional    Pesticide
2567     2567      2567          yes   no unknown 2010     7   male  28   primary      household    Pesticide
2568     2568      2568          yes   no unknown 2011     6   male  35 Secondary        farming    Pesticide
2569     2569      2569          yes   no unknown 2011     9 female  22 Secondary     unemployed Other poison
2570     2570      2570          yes   no unknown 2010     8 female  50 Secondary        farming    Pesticide
2571     2571      2571          yes   no unknown 2010     7 female  44   primary        farming    Pesticide
>
> # 4
> nrow(df)
[1] 2571
> ncol(df)
[1] 12
>
> # 5
> colnames(df)
 [1] "rownames"   "Person_ID"   "Hospitalised" "Died"        "Urban"       "Year"        "Month"       "Sex"         "Age"         "Education"   "Occupation"  "method"
>
> # 6
> str(df)
'data.frame':   2571 obs. of  12 variables:
 $ rownames   : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Person_ID  : int  1 2 3 4 5 6 7 8 9 10 ...

Files  Plots  Packages  Help  Viewer  Presentation
```

```
 $ Age          : int  39 83 60 73 51 62 90 54 66 30 ...
 $ Education    : chr  "Secondary" "primary" "primary" "primary" ...
 $ Occupation   : chr  "household" "farming" "farming" "farming" ...
 $ method       : chr  "Other poison" "Hanging" "Hanging" "Hanging" ...
>
> # 7
> summary(df)
    rownames        Person_ID        Hospitalised          Died              Urban              Year          Month             Sex                Age          Education          Occupation
 Min.   :   1.0   Min.   :   1.0   Length  :2571   Length  :2571   Length  :2571   Min.   :2009   Min.   : 1.000   Length  :2571   Min.   : 12.00   Length  :2571   Length  :2571   Le
 1st Qu.: 643.5   1st Qu.: 643.5   N.unique :   2   N.unique :   2   N.unique :   3   1st Qu.:2009   1st Qu.: 4.000   N.unique :   2   1st Qu.: 37.00   N.unique :   5   N.unique :  10   N.
 Median :1286.0   Median :1286.0   N.blank  :   0   N.blank  :   0   N.blank  :   0   Median :2010   Median : 6.000   N.blank  :   0   Median : 53.00   N.blank  :   0   N.blank  :   0   N.
 Mean   :1286.0   Mean   :1286.0   Min.nchar:   2   Min.nchar:   2   Min.nchar:   2   Mean   :2010   Mean   : 6.298   Min.nchar:   4   Mean   : 52.63   Min.nchar:   7   Min.nchar:   6   Mi
 3rd Qu.:1928.5   3rd Qu.:1928.5   Max.nchar:   3   Max.nchar:   3   Max.nchar:   7   3rd Qu.:2011   3rd Qu.: 9.000   Max.nchar:   6   3rd Qu.: 69.00   Max.nchar:   9   Max.nchar:  16   Ma
 Max.   :2571.0   Max.   :2571.8                                                     Max.   :2011   Max.   :12.000                    Max.   :100.00
>
> # 8
> colSums(is.na(df))
    rownames    Person_ID Hospitalised         Died        Urban         Year        Month          Sex          Age    Education   Occupation       method
           0            0            0            0            0            0            0            0            0            0            0            0
>
> # 9
> sum(is.na(df))
[1] 0
>
> # 10
> colSums(is.na(df))
    rownames    Person_ID Hospitalised         Died        Urban         Year        Month          Sex          Age    Education   Occupation       method
           0            0            0            0            0            0            0            0            0            0            0            0
>
> # 11
> df[!complete.cases(df), ]
 [1] rownames     Person_ID    Hospitalised Died         Urban        Year         Month        Sex          Age          Education    Occupation   method
<0 rows> (or 0-length row.names)
>
> # 12
> df_clean <- na.omit(df)
> head(df_clean, 10)
   rownames Person_ID Hospitalised Died Urban Year Month    Sex Age Education     Occupation        method
1         1         1          yes   no    no 2010    12 female  39 Secondary      household  Other poison
2         2         2           no  yes    no 2009     3   male  83   primary        farming       Hanging
3         3         3           no  yes    no 2010     2   male  60   primary        farming       Hanging
4         4         4           no  yes    no 2011     1   male  73   primary        farming       Hanging
5         5         5          yes   no    no 2009     8   male  51 Secondary        farming     Pesticide
6         6         6           no  yes    no 2009    11   male  62 iliterate        farming     Pesticide
7         7         7           no  yes    no 2010     1 female  90 iliterate        farming     Pesticide
8         8         8          yes   no    no 2010    10   male  54 Secondary        farming     Pesticide
9         9         9           no  yes    no 2010     7   male  66   primary        farming       Hanging
10       10        10          yes   no    no 2011     1 female  30 Secondary others/unknown Poison unspec
>
> # 13
> df_imputed <- df
> df_imputed$Age[is.na(df_imputed$Age)] <- mean(df_imputed$Age, na.rm = TRUE)
```

EXPLORATORY DATA ANALYSIS LAB

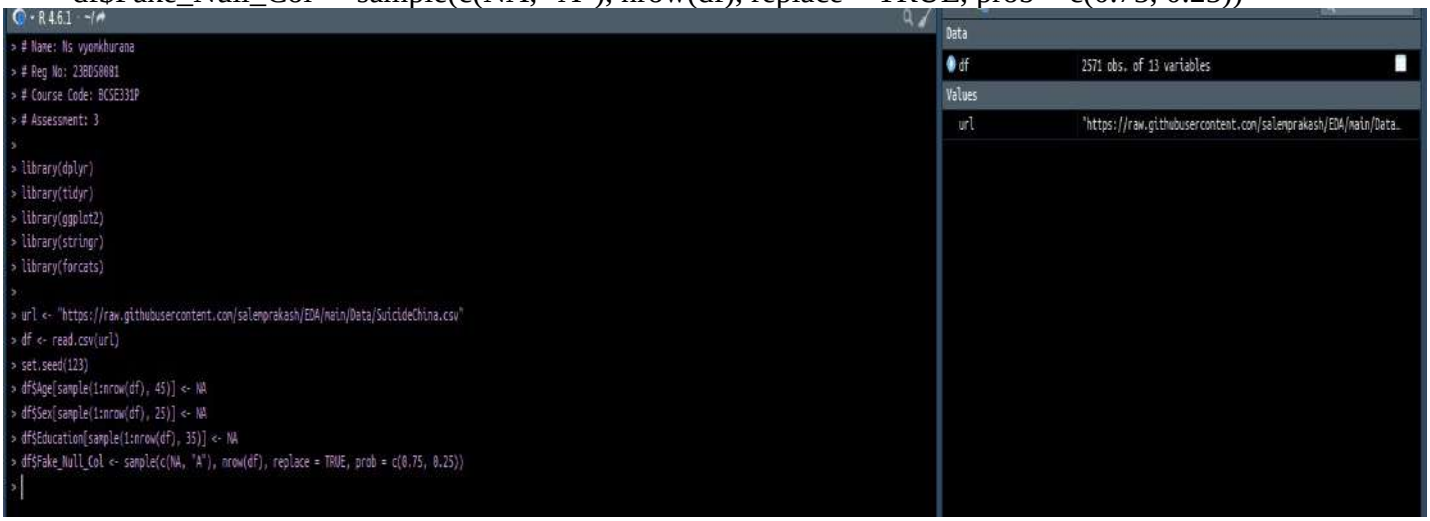
DA – 3

23BDS0081
VYOM KHURANA

```
# Name: vyomkhurana
# Reg No: 23BDS0081
# Course Code: BCSE331P
# Assessment: 3

library(dplyr)
library(tidyr)
library(ggplot2)
library(stringr)
library(forcats)


# 1
url <- "https://raw.githubusercontent.com/salemprakash/EDA/main/Data/SuicideChina.csv"
df <- read.csv(url)
set.seed(123)
df$Age[sample(1:nrow(df), 45)] <- NA
df$Sex[sample(1:nrow(df), 25)] <- NA
df$Education[sample(1:nrow(df), 35)] <- NA
df$Fake_Null_Col <- sample(c(NA, "A"), nrow(df), replace = TRUE, prob = c(0.75, 0.25))
```




```
# 2
head(df, 6)
tail(df, 6)
```

```
> head(df, 6)
  rownames Person_ID Hospitalised Died Urban Year Month    Sex Age Education Occupation       method Fake_Null_Col
1        1         1          yes   no    no 2010    12 female  39 Secondary  household Other poison          <NA>
2        2         2           no  yes    no 2009     3   male  83   primary    farming      Hanging          <NA>
3        3         3           no  yes    no 2010     2   male  60   primary    farming      Hanging          <NA>
4        4         4           no  yes    no 2011     1   male  73   primary    farming      Hanging          <NA>
5        5         5          yes   no    no 2009     8   male  51 Secondary    farming    Pesticide             A
6        6         6           no  yes    no 2009    11   male  62 iliterate    farming    Pesticide          <NA>
> tail(df, 6)
     rownames Person_ID Hospitalised Died   Urban Year Month    Sex Age Education   Occupation       method Fake_Null_Col
2566     2566      2566          yes   no unknown 2011     6   male  23 Secondary professional    Pesticide          <NA>
2567     2567      2567          yes   no unknown 2010     7   male  28   primary    household    Pesticide             A
2568     2568      2568          yes   no unknown 2011     6   male  35 Secondary      farming    Pesticide          <NA>
2569     2569      2569          yes   no unknown 2011     9 female  22 Secondary   unemployed Other poison          <NA>
2570     2570      2570          yes   no unknown 2010     8 female  50 Secondary      farming    Pesticide             A
2571     2571      2571          yes   no unknown 2010     7 female  44   primary      farming    Pesticide          <NA>
>
```

3
sum(is.na(df))

```
>
> sum(is.na(df))
[1] 2042
>
```

4
na_rows <- df[!complete.cases(df),]
head(na_rows)

```
> head(na_rows)
  rownames Person_ID Hospitalised Died Urban Year Month    Sex Age Education Occupation       method Fake_Null_Col
1        1         1          yes   no    no 2010    12 female  39 Secondary  household Other poison          <NA>
2        2         2           no  yes    no 2009     3   male  83   primary    farming      Hanging          <NA>
3        3         3           no  yes    no 2010     2   male  60   primary    farming      Hanging          <NA>
4        4         4           no  yes    no 2011     1   male  73   primary    farming      Hanging          <NA>
6        6         6           no  yes    no 2009    11   male  62 iliterate    farming    Pesticide          <NA>
7        7         7           no  yes    no 2010     1 female  90 iliterate    farming    Pesticide          <NA>
>
```

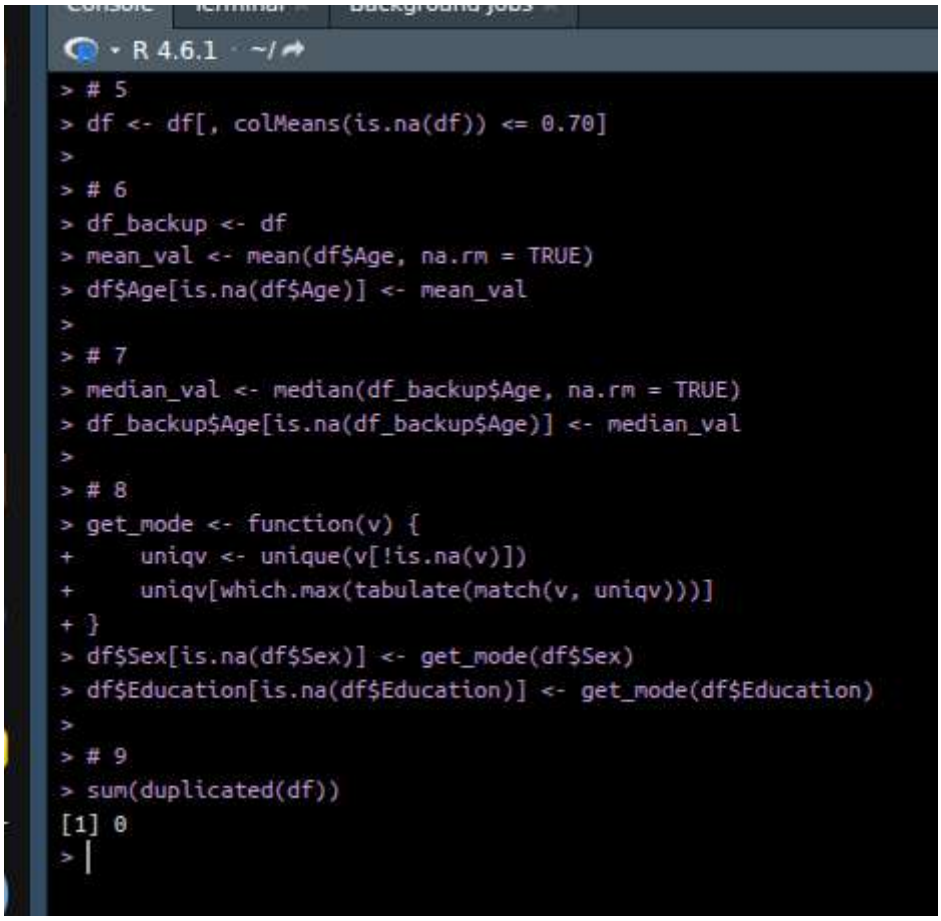
5
df <- df[, colMeans(is.na(df)) <= 0.70]

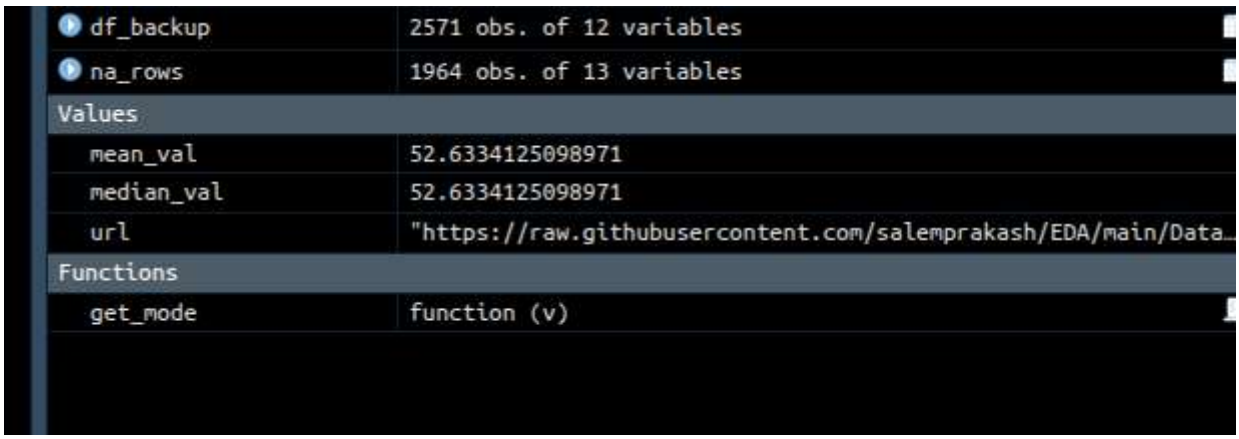
6
df_backup <- df
mean_val <- mean(df$Age, na.rm = TRUE)
df$Age[is.na(df$Age)] <- mean_val

7
median_val <- median(df_backup$Age, na.rm = TRUE)
df_backup$Age[is.na(df_backup$Age)] <- median_val

8
```
get_mode <- function(v) {
  uniqv <- unique(v[!is.na(v)])
  uniqv[which.max(tabulate(match(v, uniqv)))]
}
df$Sex[is.na(df$Sex)] <- get_mode(df$Sex)
df$Education[is.na(df$Education)] <- get_mode(df$Education)
```

9
sum(duplicated(df))


R 4.6.1 ~/
> # 5
> df <- df[, colMeans(is.na(df)) <= 0.70]
>
> # 6
> df_backup <- df
> mean_val <- mean(df$Age, na.rm = TRUE)
> df$Age[is.na(df$Age)] <- mean_val
>
> # 7
> median_val <- median(df_backup$Age, na.rm = TRUE)
> df_backup$Age[is.na(df_backup$Age)] <- median_val
>
> # 8
> get_mode <- function(v) {
+ uniqv <- unique(v[!is.na(v)])
+ uniqv[which.max(tabulate(match(v, uniqv)))]
+ }
> df$Sex[is.na(df$Sex)] <- get_mode(df$Sex)
> df$Education[is.na(df$Education)] <- get_mode(df$Education)
>
> # 9
> sum(duplicated(df))
[1] 0
>


df_backup
2571 obs. of 12 variables
na_rows
1964 obs. of 13 variables
Values
mean_val
52.6334125098971
median_val
52.6334125098971
url
"https://raw.githubusercontent.com/salemprakash/EDA/main/Data
Functions
get_mode
function (v)

```r
# 10
df <- unique(df)

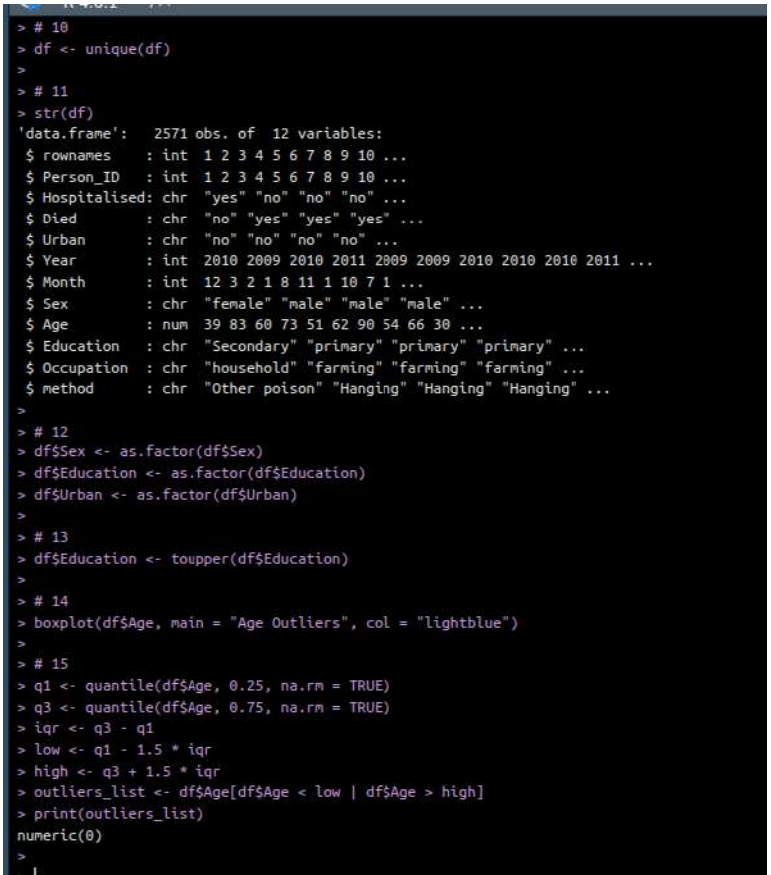
# 11
str(df)

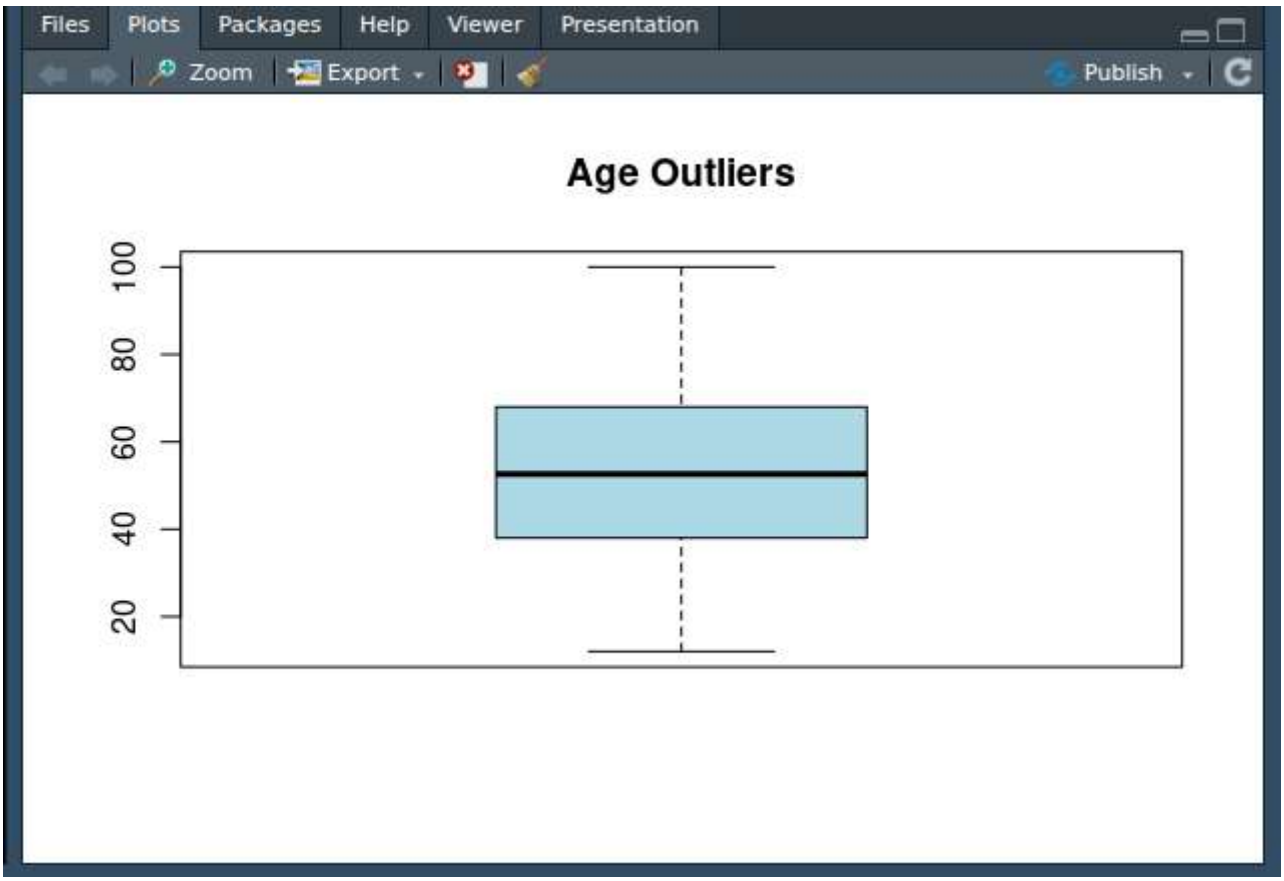
# 12
df$Sex <- as.factor(df$Sex)
df$Education <- as.factor(df$Education)
df$Urban <- as.factor(df$Urban)

# 13
df$Education <- toupper(df$Education)

# 14
boxplot(df$Age, main = "Age Outliers", col = "lightblue")

# 15
q1 <- quantile(df$Age, 0.25, na.rm = TRUE)
q3 <- quantile(df$Age, 0.75, na.rm = TRUE)
iqr <- q3 - q1
low <- q1 - 1.5 * iqr
high <- q3 + 1.5 * iqr
outliers_list <- df$Age[df$Age < low | df$Age > high]
print(outliers_list)
```






```
# 16
low_ext <- q1 - 3 * iqr
high_ext <- q3 + 3 * iqr
df <- df[df$Age >= low_ext & df$Age <= high_ext, ]

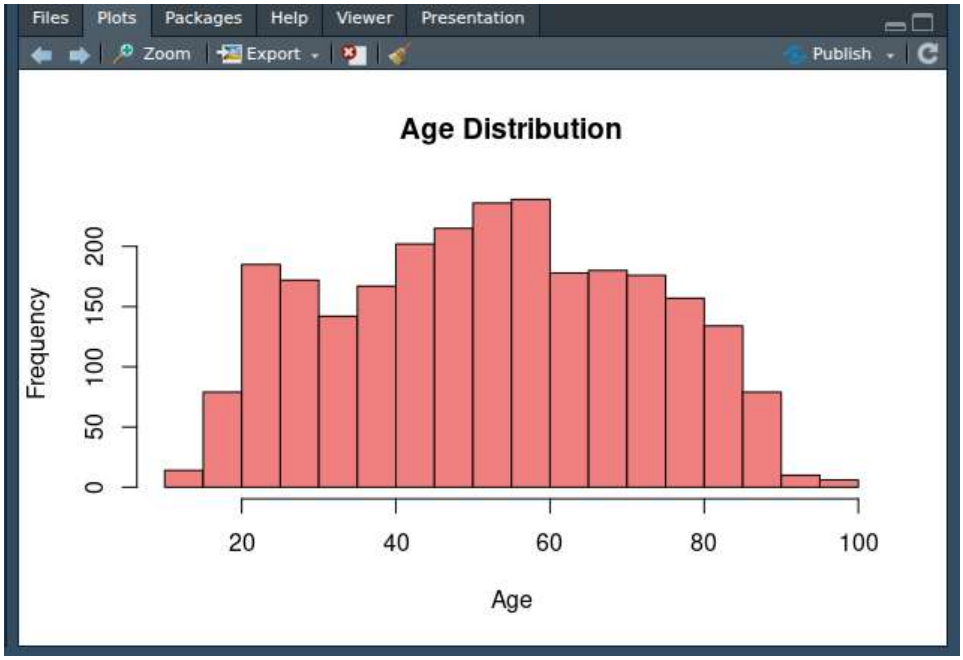
# 17
hist(df$Age, main = "Age Distribution", xlab = "Age", col = "lightcoral")
# Interpretation: The histogram indicates the distribution of cases across ages. Peaks show most vulnerable age groups.

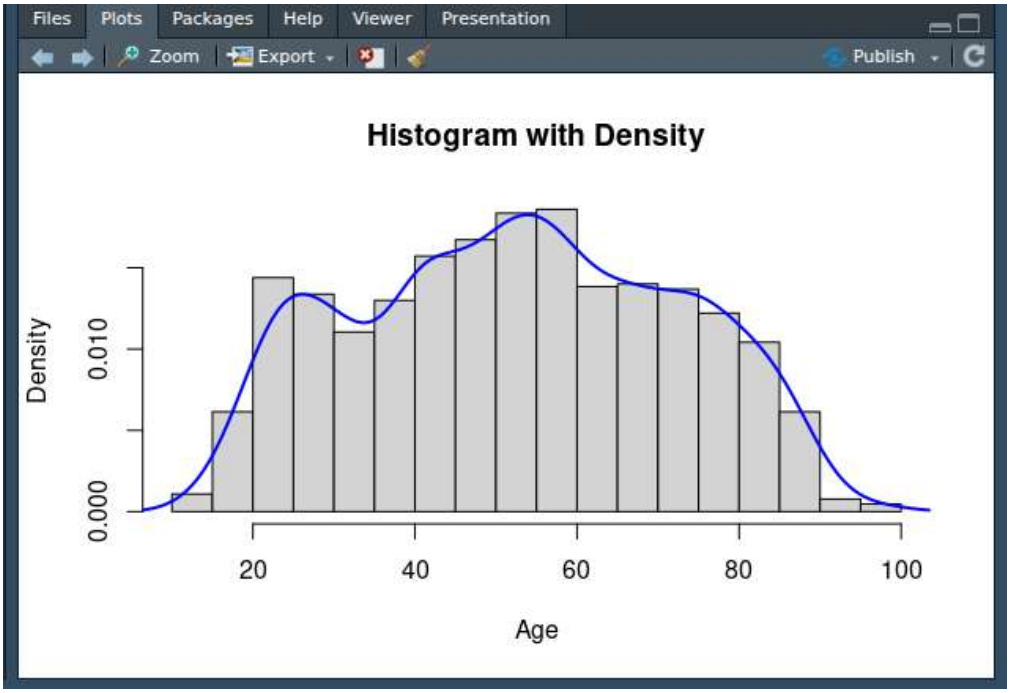
# 18
hist(df$Age, prob = TRUE, main = "Histogram with Density", xlab = "Age", col = "lightgray")
lines(density(df$Age), col = "blue", lwd = 2)

# 19
ggplot(df, aes(x = Sex, y = Age, fill = Sex)) +
  geom_violin() +
  labs(title = "Age Distribution by Sex")

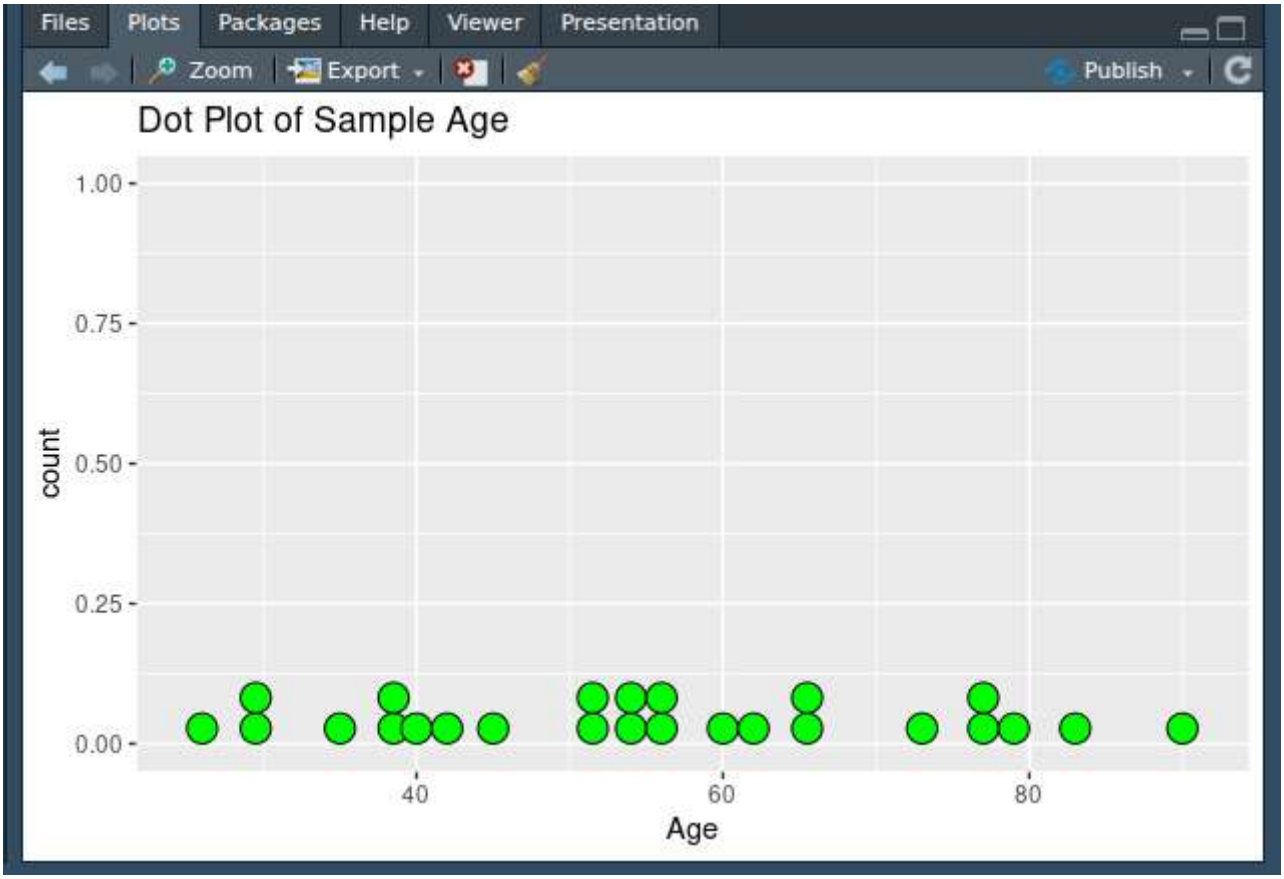
# 20
sub_df <- head(df, 25)
ggplot(sub_df, aes(x = Age)) +
  geom_dotplot(binwidth = 2, fill = "green") +
```

labs(title = "Dot Plot of Sample Age")






Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Age Distribution by Sex
100
75
50
25
Age
female
male
Sex
Sex
female
male

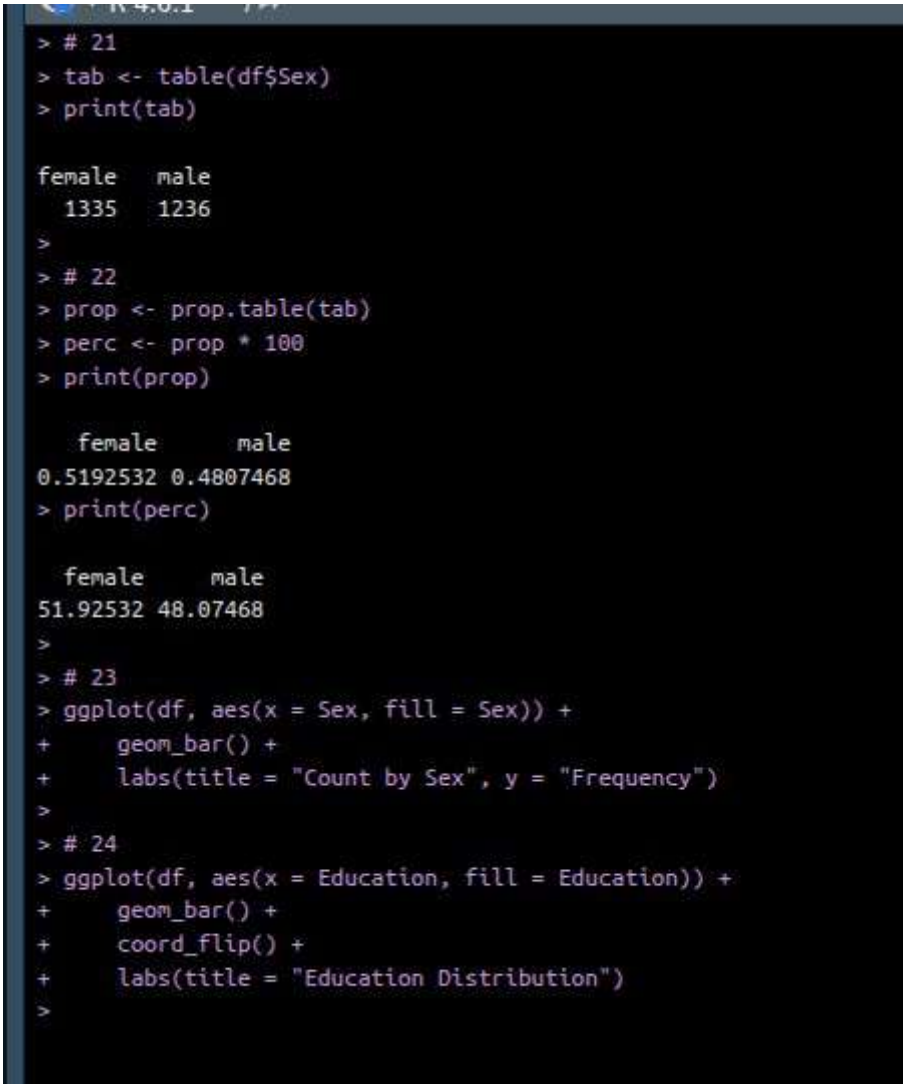

Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Dot Plot of Sample Age
1.00
0.75
0.50
0.25
0.00
count
40
60
80
Age

```r
# 21
tab <- table(df$Sex)
print(tab)

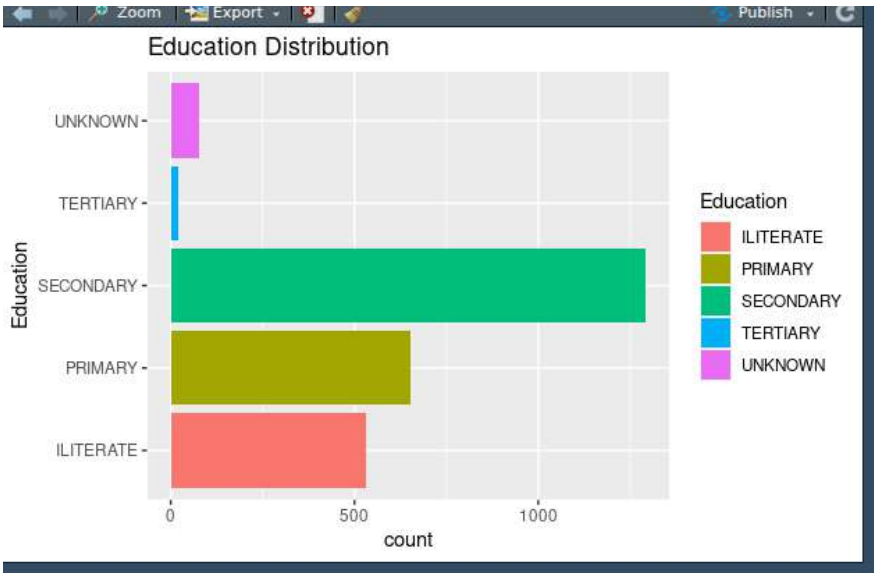
# 22
prop <- prop.table(tab)
perc <- prop * 100
print(prop)
print(perc)

# 23
ggplot(df, aes(x = Sex, fill = Sex)) +
  geom_bar() +
  labs(title = "Count by Sex", y = "Frequency")

# 24
ggplot(df, aes(x = Education, fill = Education)) +
  geom_bar() +
  coord_flip() +
  labs(title = "Education Distribution")
```





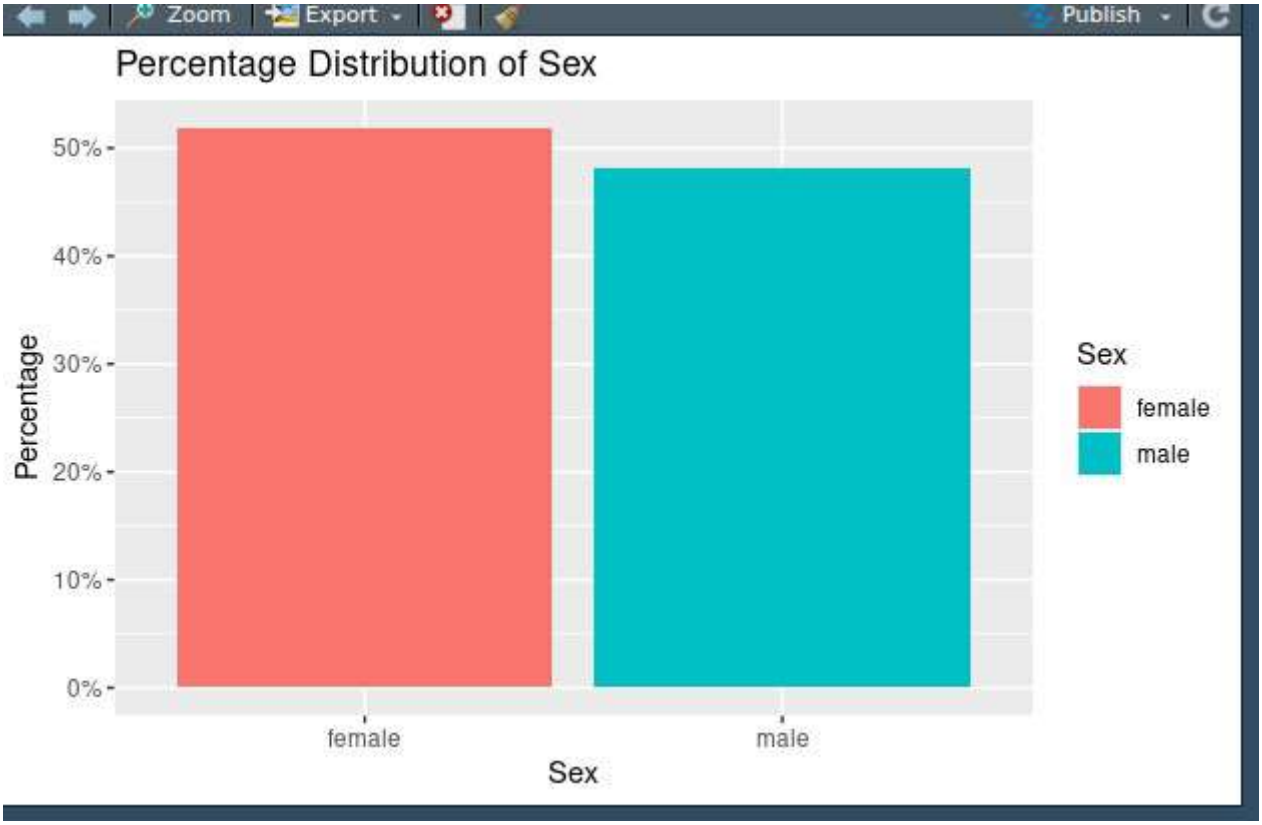


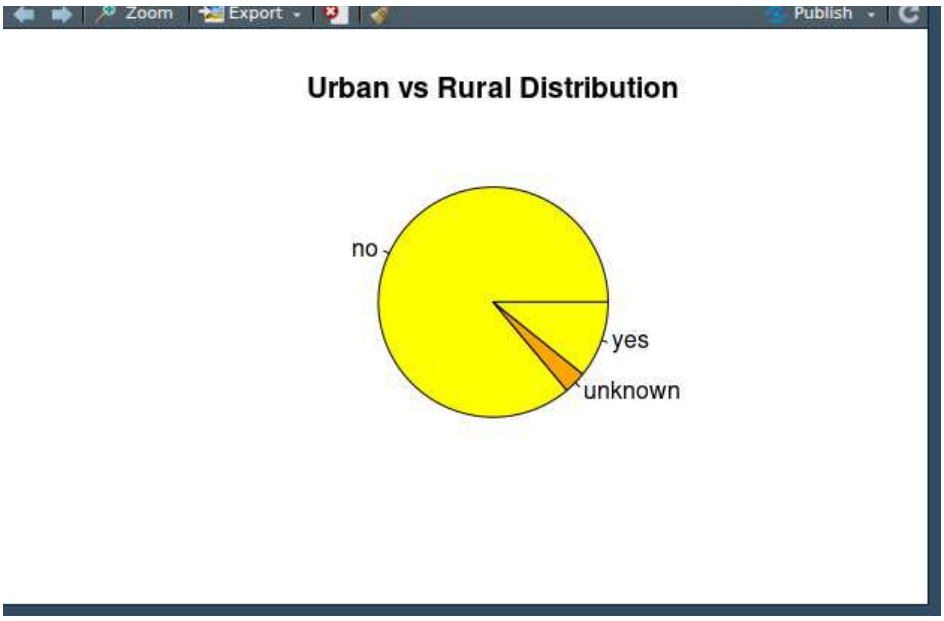

```
# 25
ggplot(df, aes(x = Sex, y = after_stat(count/sum(count)), fill = Sex)) +
  geom_bar() +
  scale_y_continuous(labels = scales::percent) +
  labs(title = "Percentage Distribution of Sex", y = "Percentage")

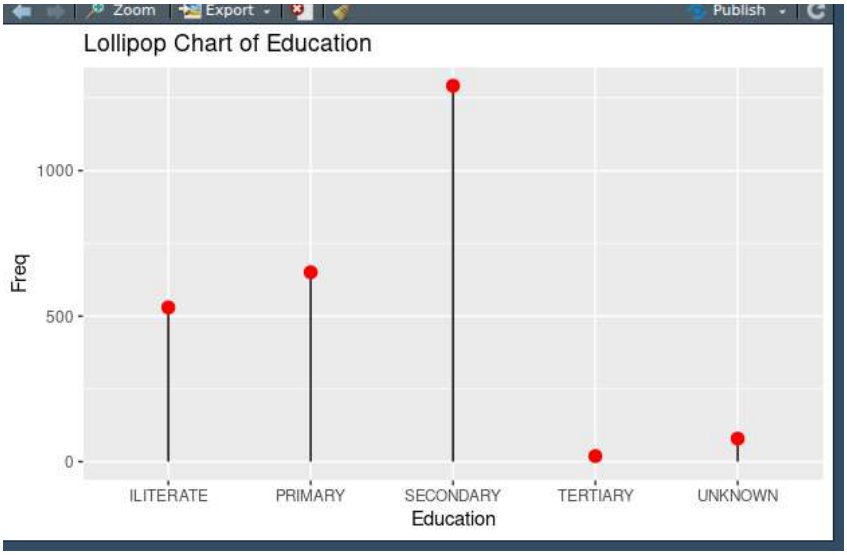
# 26
pie(table(df$Urban), main = "Urban vs Rural Distribution", col = c("yellow", "orange"))

# 27
freq_df <- as.data.frame(table(df$Education))
colnames(freq_df) <- c("Education", "Freq")
```

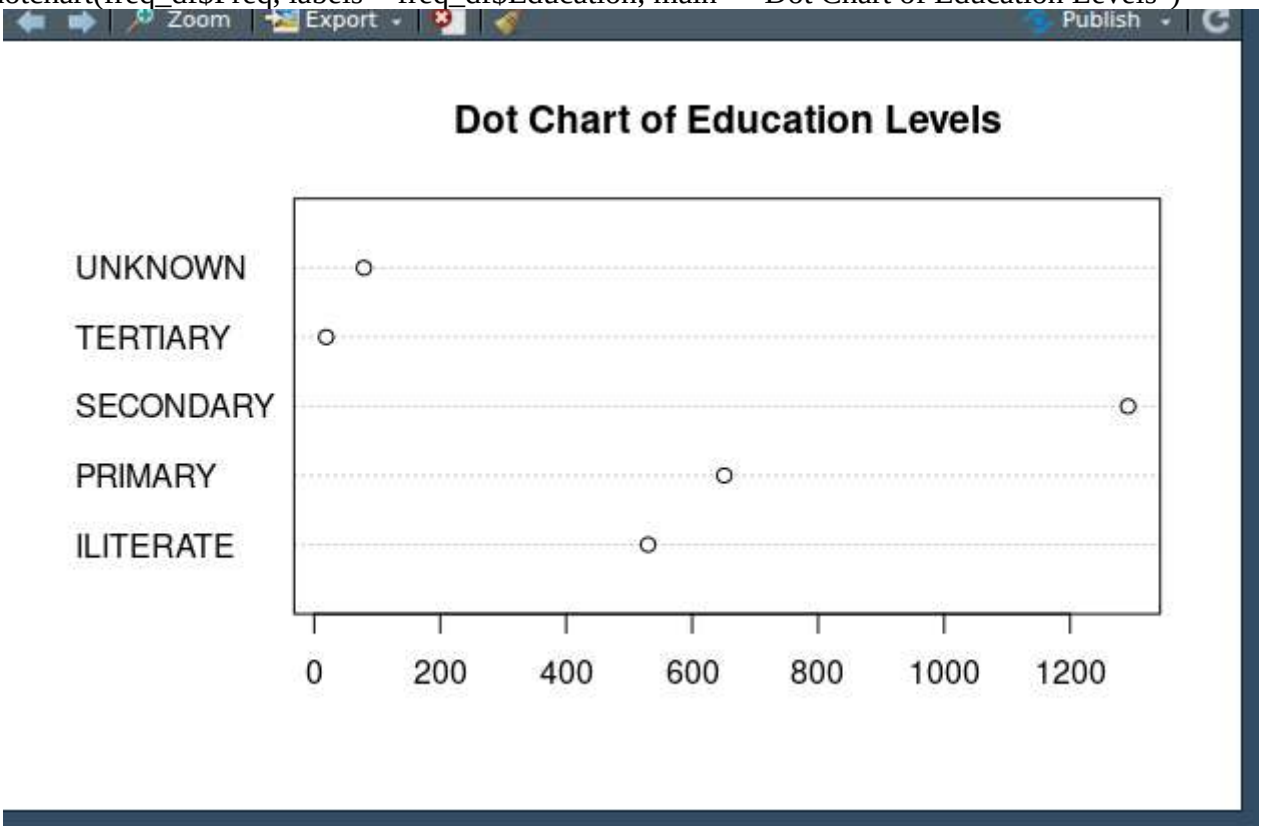
```
ggplot(freq_df, aes(x = Education, y = Freq)) +
  geom_segment(aes(x = Education, xend = Education, y = 0, yend = Freq), col = "black") +
  geom_point(size = 3, col = "red") +
  labs(title = "Lollipop Chart of Education")
```



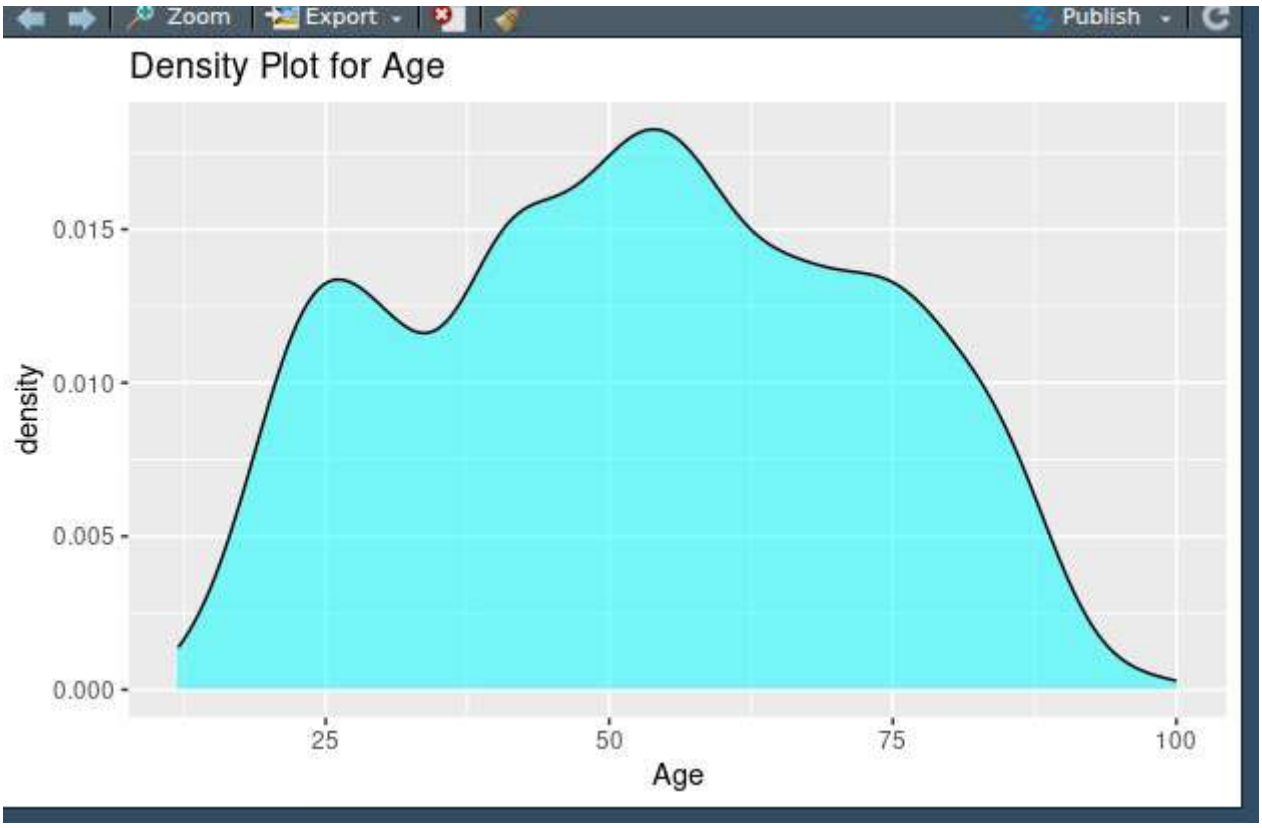




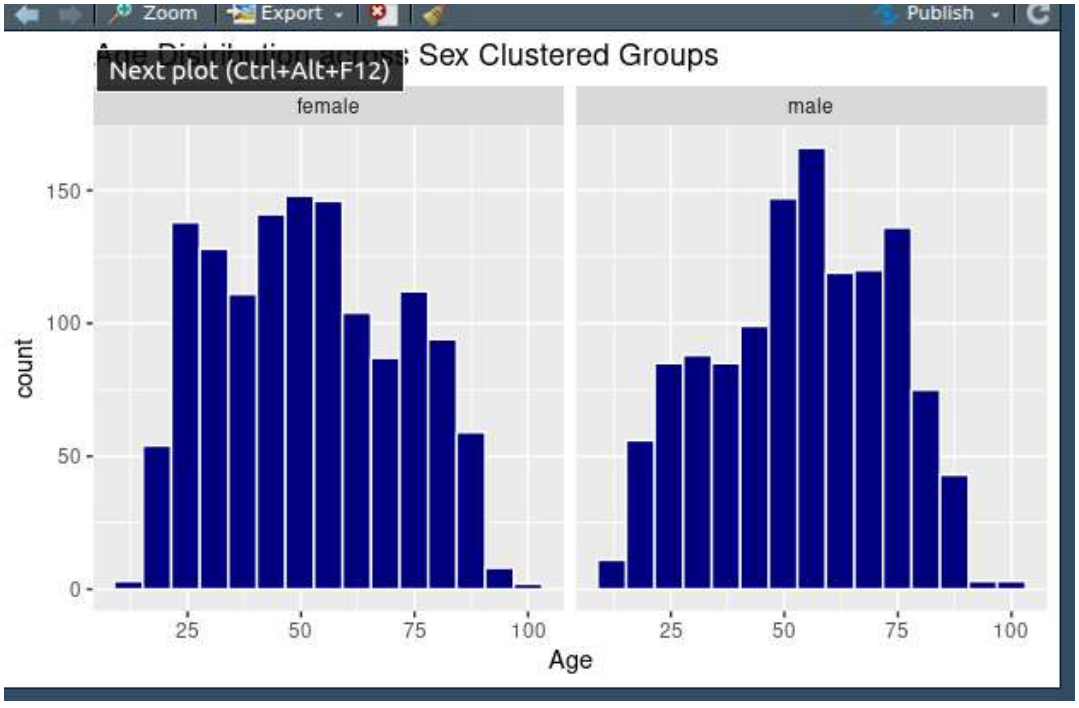

```
# 28
dotchart(freq_df$Freq, labels = freq_df$Education, main = "Dot Chart of Education Levels")
```




```
# 29
ggplot(df, aes(x = Age)) +
  geom_density(fill = "cyan", alpha = 0.5) +
  labs(title = "Density Plot for Age")
```




```
# 30
ggplot(df, aes(x = Age)) +
  geom_histogram(bins = 15, fill = "navy", col = "white") +
  facet_wrap(~ Sex) +
  labs(title = "Age Distribution across Sex Clustered Groups")
# Interpretation: Faceting splits the visual comparison between groups. We can see if the count patterns differ between specific gender subsets.
```



Exploratory Data Analysis

BCSE331P

Assessment – 4

23BDS0081

Submitted To:

Prof. Balaji GN

Submitted By:

Vyom Khurana

23BDS0081

Data Transformations

1. Normalize the data to the following ranges: o [-3, +3] o [-1, +1] o [0, 1] (Min-Max Normalization)

CODE:

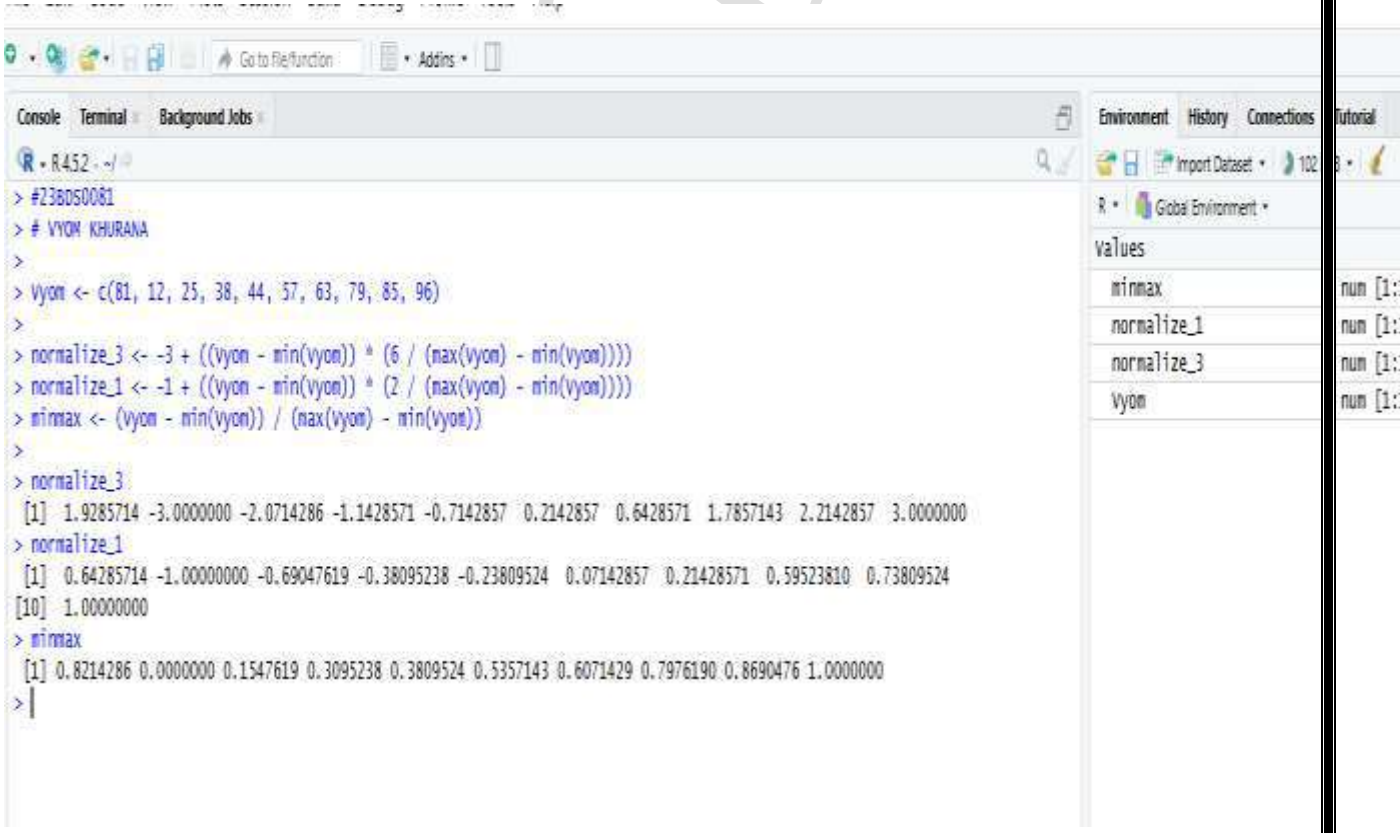
```
#23BDS0081
# VYOM KHURANA

Vyom <- c(81, 12, 25, 38, 44, 57, 63, 79, 85, 96)

normalize_3 <- -3 + ((Vyom - min(Vyom)) * (6 / (max(Vyom) - min(Vyom))))
normalize_1 <- -1 + ((Vyom - min(Vyom)) * (2 / (max(Vyom) - min(Vyom))))
minmax <- (Vyom - min(Vyom)) / (max(Vyom) - min(Vyom))

normalize_3
normalize_1
minmax
```

SCREENSHOT OF THE OUPUT:



Transform the data into Z-score format.
CODE:

#23BDS0081
VYOM KHURANA

Vyom <- c(81, 12, 25, 38, 44, 57, 63, 79, 85, 96)

z_score <- (Vyom - mean(Vyom)) / sd(Vyom)

z_score
SCREENSHOT OF THE OUTPUT:

```
 [1]  0.64285714 -1.00000000 -0.69047619 -0.38095238 -0.23809524  0.07142857  0.21428571  0.59523810  0.73809524
[10]  1.00000000
> minmax
 [1] 0.8214286 0.0000000 0.1547619 0.3095238 0.3809524 0.5357143 0.6071429 0.7976190 0.8690476 1.0000000
> #23BDS0081
> # VYOM KHURANA
>
> vyom <- c(81, 12, 25, 38, 44, 57, 63, 79, 85, 96)
>
> z_score <- (vyom - mean(vyom)) / sd(vyom)
>
> z_score
 [1]  0.82648067 -1.65296134 -1.18582009 -0.71867885 -0.50307519 -0.03593394  0.17966971  0.75461279  0.97021644
[10]  1.36548981
>
```

Transform the data using Decimal Scaling Normalization.

CODE:

#23BDS0081

VYOM KHURANA

Vyom <- c(81, 12, 25, 38, 44, 57, 63, 79, 85, 96)

j <- ceiling(log10(max(abs(Vyom)) + 1))

decimal_scaling <- Vyom / (10^j)

decimal_scaling

23BDS0081

SCREENSHOT OF THE OUTPUT:

```
 [1]  0.82648067 -1.65296134 -1.18582009 -0.71867885 -0.50307519 -0.035
[10]  1.36548981
> #23BDS0081
> # VYOM KHURANA
>
> vyom <- c(81, 12, 25, 38, 44, 57, 63, 79, 85, 96)
>
> j <- ceiling(log10(max(abs(vyom)) + 1))
>
> decimal_scaling <- vyom / (10^j)
>
> decimal_scaling
 [1] 0.81 0.12 0.25 0.38 0.44 0.57 0.63 0.79 0.85 0.96
>
```

23BDS0081

EDA LAB 5
VYOM KHURANA
23BDS0081

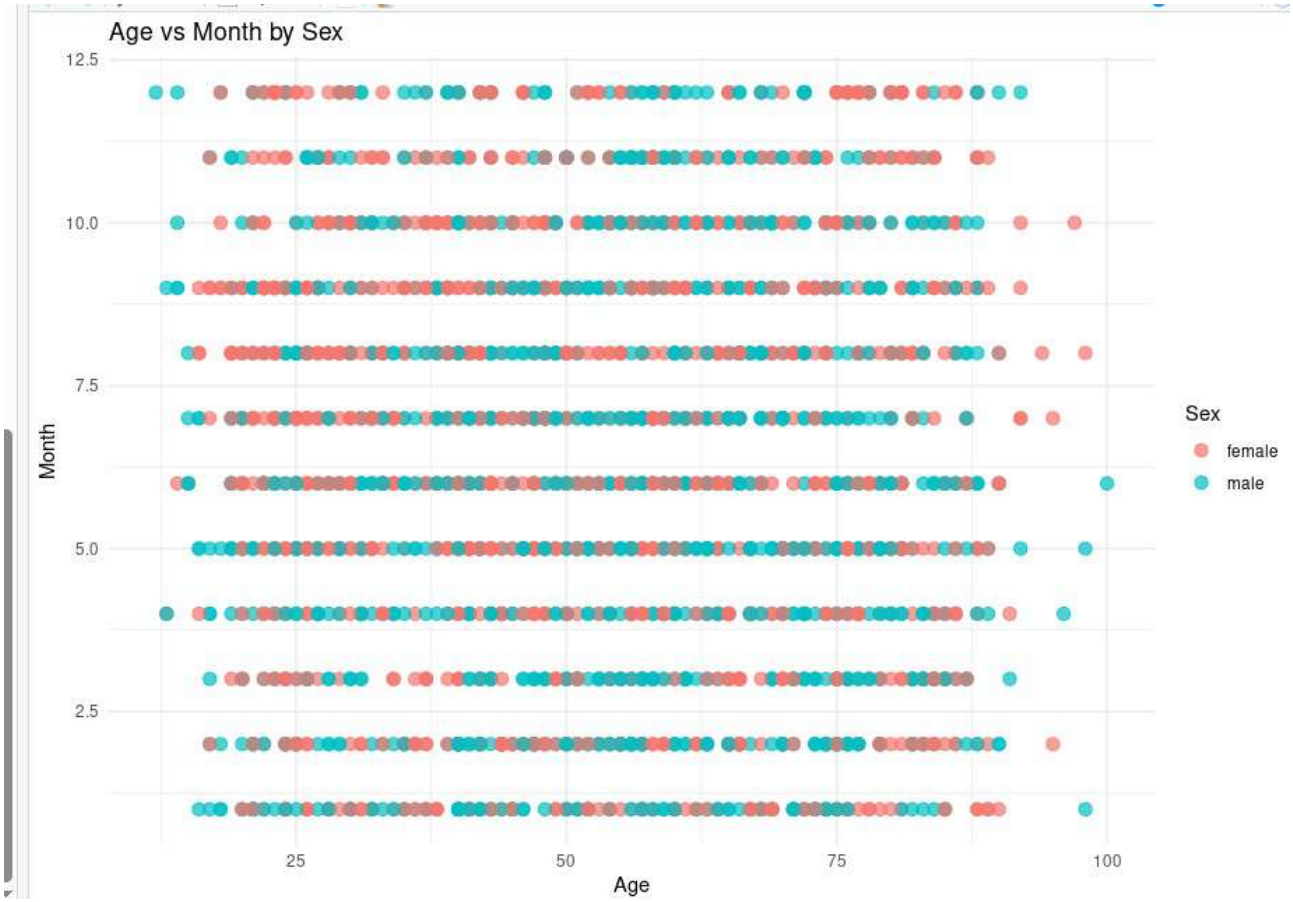
1. Load the dataset shared through e-mail into the R environment. Examine the dataset and identify the numerical and categorical variables.

```
> #23BDS0081
> #VYOM KHURANA
> suicide_data <- read.csv(
+     "https://raw.githubusercontent.com/salemprakash/EDA/main/Data/SuicideChina.csv",
+     stringsAsFactors = FALSE
+ )
>
> head(suicide_data)
  rownames Person_ID Hospitalised Died Urban Year Month    Sex Age Education Occupation
method
1        1         1          yes   no    no 2010    12 female  39 Secondary  household Other
poison
2        2         2           no  yes    no 2009     3   male  83   primary    farming
Hanging
3        3         3           no  yes    no 2010     2   male  60   primary    farming
Hanging
4        4         4           no  yes    no 2011     1   male  73   primary    farming
Hanging
5        5         5          yes   no    no 2009     8   male  51 Secondary    farming
Pesticide
6        6         6           no  yes    no 2009    11   male  62 iliterate    farming
Pesticide
> str(suicide_data)
'data.frame':   2571 obs. of  12 variables:
 $ rownames    : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Person_ID   : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Hospitalised: chr  "yes" "no" "no" "no" ...
 $ Died        : chr  "no" "yes" "yes" "yes" ...
 $ Urban       : chr  "no" "no" "no" "no" ...
 $ Year        : int  2010 2009 2010 2011 2009 2009 2010 2010 2010 2011 ...
 $ Month       : int  12 3 2 1 8 11 1 10 7 1 ...
 $ Sex         : chr  "female" "male" "male" "male" ...
 $ Age         : int  39 83 60 73 51 62 90 54 66 30 ...
 $ Education   : chr  "Secondary" "primary" "primary" "primary" ...
 $ Occupation  : chr  "household" "farming" "farming" "farming" ...
 $ method      : chr  "Other poison" "Hanging" "Hanging" "Hanging" ...
> dim(suicide_data)
[1] 2571   12
> names(suicide_data)
 [1] "rownames"     "Person_ID"    "Hospitalised" "Died"         "Urban"        "Year"
"Month"        "Sex"          "Age"          "Education"    "Occupation"   "method"
> summary(suicide_data)
     rownames        Person_ID         Hospitalised         Died              Urban              Year
Month               Sex                Age              Education         Occupation          method
 Min.   :   1.0   Min.   :   1.0   Length   :2571   Length   :2571   Length   :2571   Min.   :2009
Min.   : 1.000   Length   :2571   Min.   : 12.00   Length   :2571   Length   :2571
Length   :2571
```

```
 1st Qu.: 643.5   1st Qu.: 643.5   N.unique :   2   N.unique :   2   N.unique :   3   1st Qu.:2009
1st Qu.: 4.000   N.unique :   2   1st Qu.: 37.00   N.unique :   5   N.unique :  10   N.unique :
9
 Median :1286.0   Median :1286.0   N.blank  :   0   N.blank  :   0   N.blank  :   0   Median :2010
Median : 6.000   N.blank  :   0   Median : 53.00   N.blank  :   0   N.blank  :   0   N.blank  :
0
 Mean   :1286.0   Mean   :1286.0   Min.nchar:   2   Min.nchar:   2   Min.nchar:   2   Mean   :2010
Mean   : 6.298   Min.nchar:   4   Mean   : 52.63   Min.nchar:   7   Min.nchar:   6   Min.nchar:
6
 3rd Qu.:1928.5   3rd Qu.:1928.5   Max.nchar:   3   Max.nchar:   3   Max.nchar:   7   3rd Qu.:2011
3rd Qu.: 9.000   Max.nchar:   6   3rd Qu.: 69.00   Max.nchar:   9   Max.nchar:  16   Max.nchar:
13
 Max.   :2571.0   Max.   :2571.0                                                     Max.   :2011
Max.   :12.000                    Max.   :100.00
> 
> numerical_variables <- names(suicide_data[sapply(suicide_data, is.numeric)])
> categorical_variables <- names(suicide_data[sapply(suicide_data, function(x) is.character(x) |
is.factor(x))])
> 
> numerical_variables
[1] "rownames"  "Person_ID" "Year"      "Month"     "Age"
> categorical_variables
[1] "Hospitalised" "Died"         "Urban"        "Sex"          "Education"    "Occupation"
"method"
> 
```

2. Create a scatter plot involving three variables, where two numerical variables are represented on the X- and Y-axes, and the third variable is represented using color.

```
> library(ggplot2)
> ggplot(suicide_data, aes(x = Age, y = Month, color = Sex)) +
+     geom_point(size = 3, alpha = 0.7) +
+     labs(
+         title = "Age vs Month by Sex",
+         x = "Age",
+         y = "Month",
+         color = "Sex"
+     ) +
+     theme_minimal()
> 
> 
```



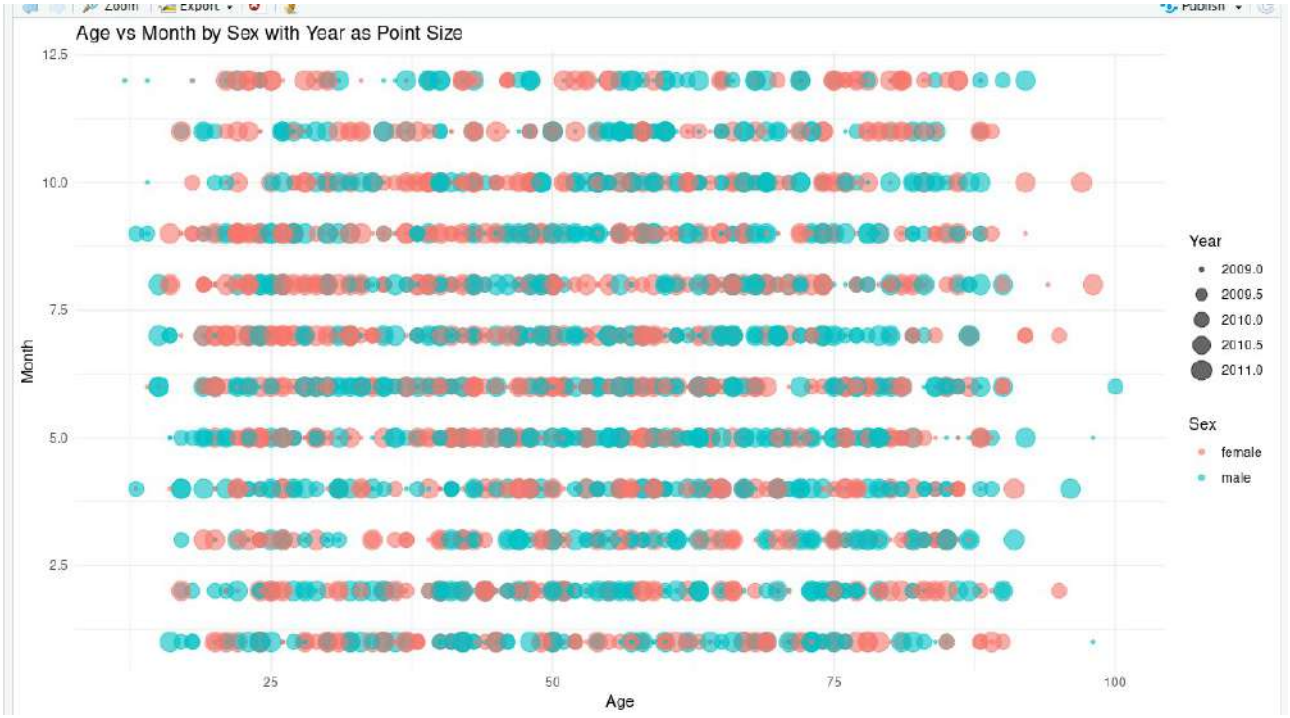
3. Create a scatter plot involving four variables, where two numerical variables are represented on the X- and Y-axes, the third variable is represented using color, and the fourth variable is represented using shape.

```
> p <- ggplot(
+     suicide_data,
+     aes(x = Age, y = Month, color = Sex, shape = Hospitalised)
+ ) +
+     geom_point(size = 3, alpha = 0.7) +
+     labs(
+         title = "Age vs Month by Sex and Hospitalisation",
+         x = "Age",
+         y = "Month",
+         color = "Sex",
+         shape = "Hospitalised"
+     ) +
+     theme_minimal()
>
> print(p)
```



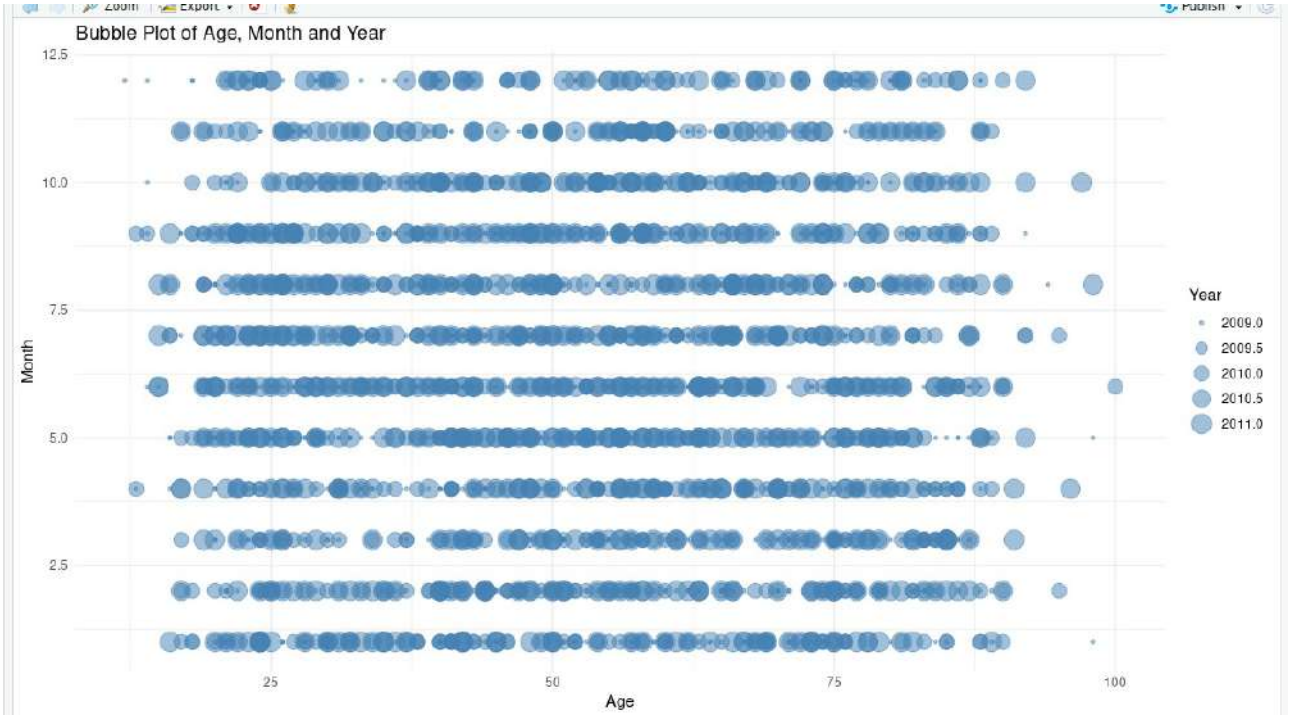
4. Create a scatter plot involving four variables, where two numerical variables are represented on the X- and Y-axes, the third variable is represented using color, and the fourth numerical variable is represented using point size.

```
> p4 <- ggplot(
+     suicide_data,
+     aes(x = Age, y = Month, color = Sex, size = Year)
+ ) +
+     geom_point(alpha = 0.6) +
+     labs(
+         title = "Age vs Month by Sex with Year as Point Size",
+         x = "Age",
+         y = "Month",
+         color = "Sex",
+         size = "Year"
+     ) +
+     theme_minimal()
>
> print(p4)
>
```



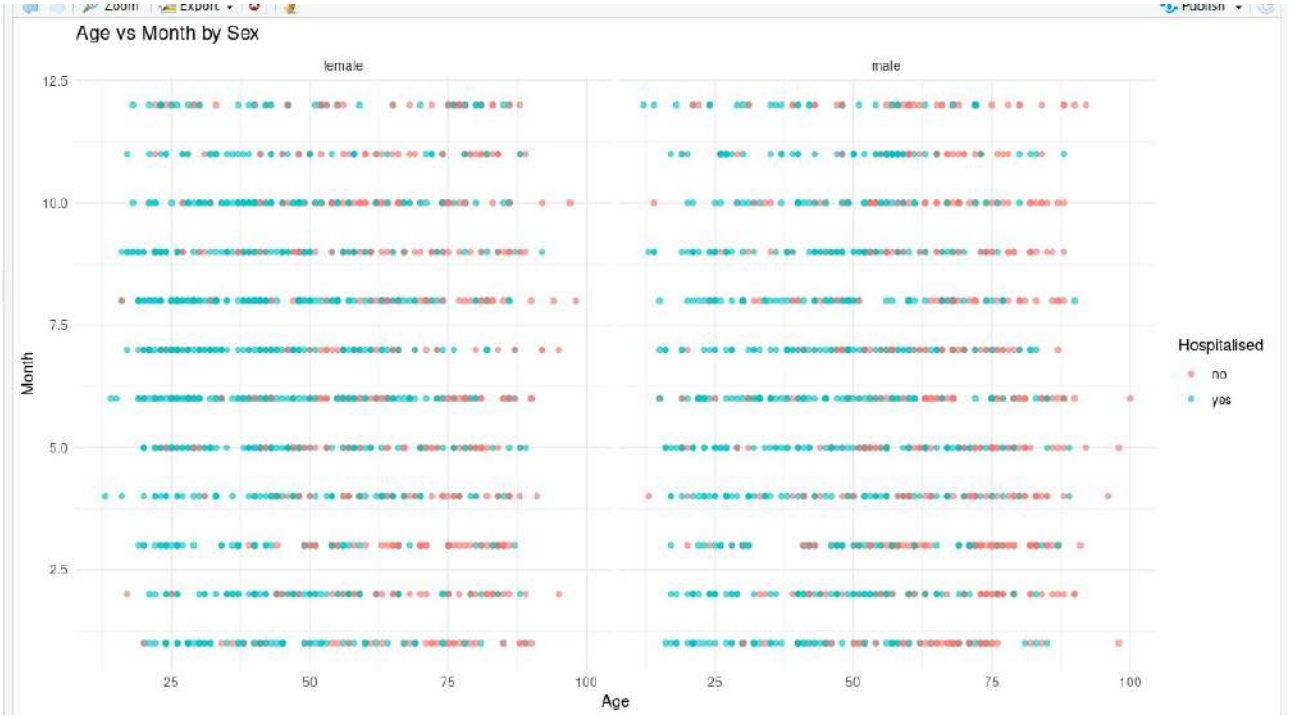
5. Construct a bubble plot using three numerical variables, where two variables are represented on the X- and Y-axes, and the third variable determines the size of the bubbles.

```
> p5 <- ggplot(
+     suicide_data,
+     aes(x = Age, y = Month, size = Year)
+ ) +
+     geom_point(alpha = 0.5, color = "steelblue") +
+     labs(
+         title = "Bubble Plot of Age, Month and Year",
+         x = "Age",
+         y = "Month",
+         size = "Year"
+     ) +
+     theme_minimal()
>
> print(p5)
>
```



6. Generate subplots using facet_wrap() by selecting an appropriate categorical variable from the dataset.

```
> p6 <- ggplot(
+     suicide_data,
+     aes(x = Age, y = Month, color = Hospitalised)
+ ) +
+     geom_point(alpha = 0.6) +
+     facet_wrap(~Sex) +
+     labs(
+         title = "Age vs Month by Sex",
+         x = "Age",
+         y = "Month",
+         color = "Hospitalised"
+     ) +
+     theme_minimal()
>
> print(p6)
>
```



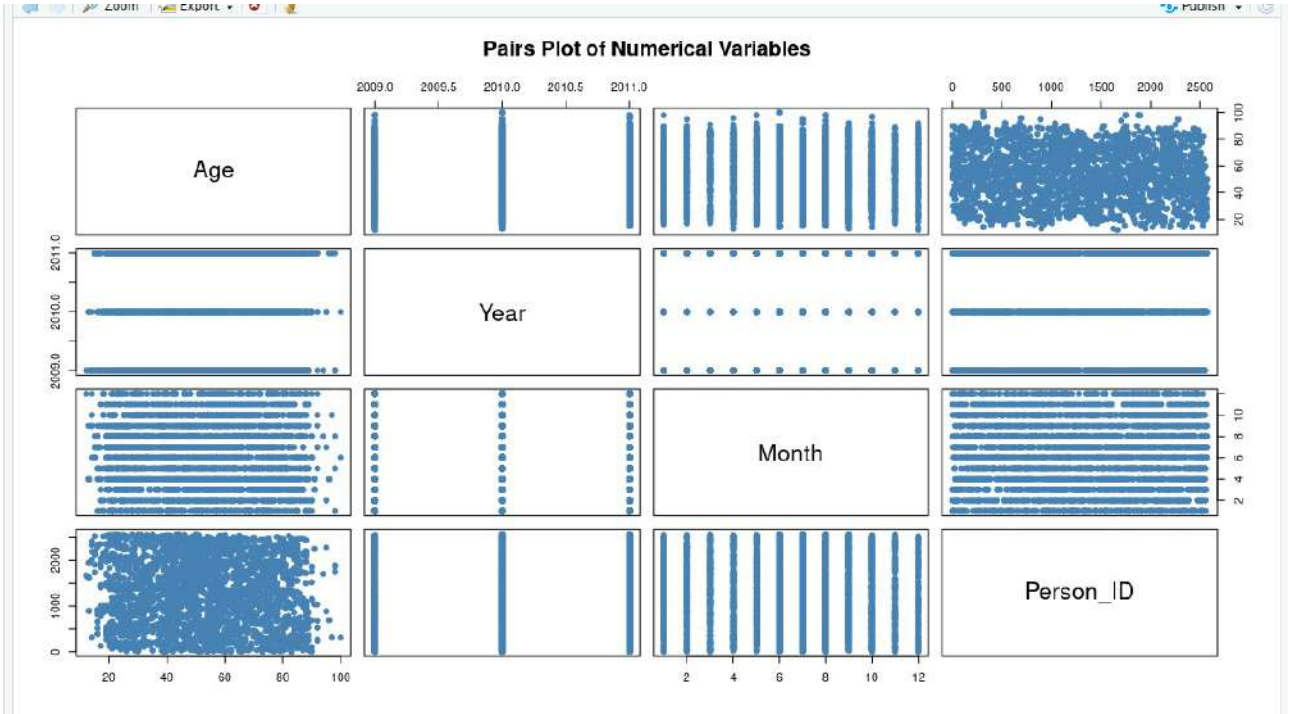
7. Generate subplots using facet_grid() by selecting two appropriate categorical variables from the dataset. Arrange the plots in rows and columns and compare the patterns observed across the different groups.

```
> p7 <- ggplot(
+     suicide_data,
+     aes(x = Age, y = Month, color = Died)
+ ) +
+     geom_point(alpha = 0.6) +
+     facet_grid(Education ~ Sex) +
+     labs(
+         title = "Age vs Month by Education and Sex",
+         x = "Age",
+         y = "Month",
+         color = "Died"
+     ) +
+     theme_minimal()
>
> print(p7)
>
>
```



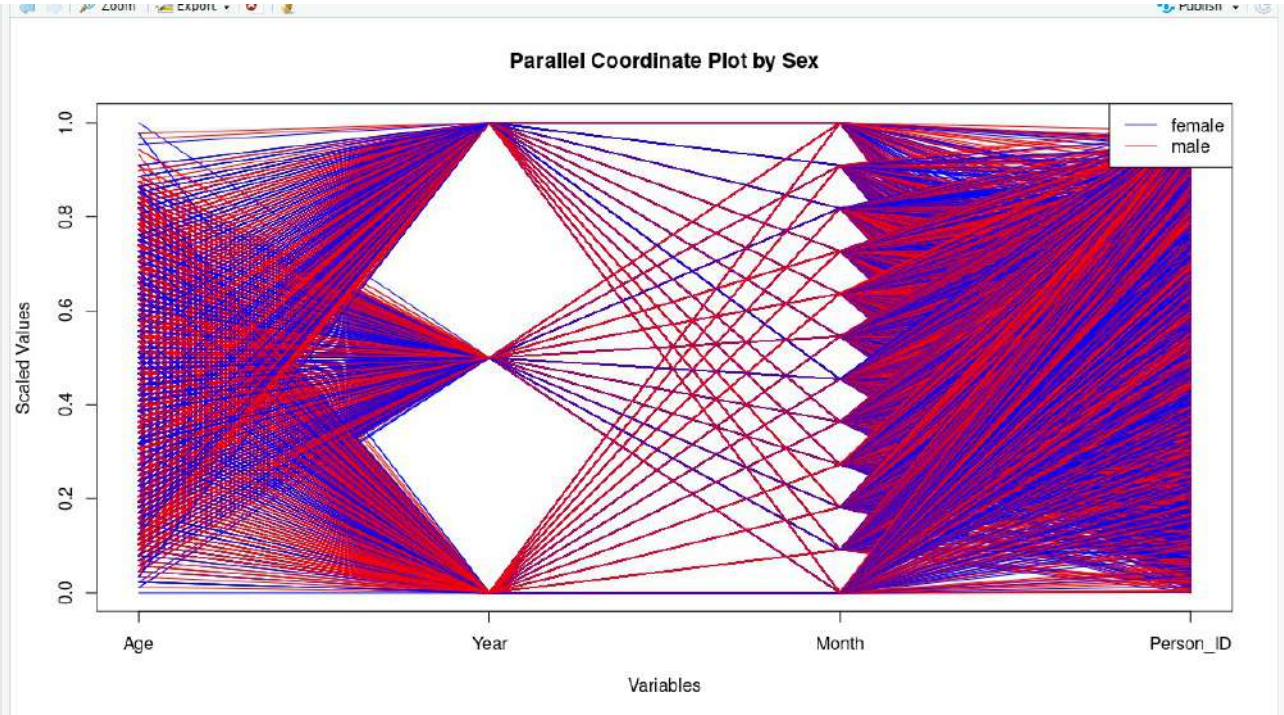
8. Construct a pairs plot using at least four numerical variables from the dataset. Use the plot to identify possible relationships and correlations among the selected variables.

```
> p8 <- pairs(
+     suicide_data[, c("Age", "Year", "Month", "Person_ID")],
+     main = "Pairs Plot of Numerical Variables",
+     pch = 19,
+     col = "steelblue"
+ )
>
```



9. Create a parallel coordinate plot using at least four numerical variables from the dataset. Differentiate the observations using an appropriate categorical variable.

```
> parallel_data <- suicide_data[
+     ,
+     c("Age", "Year", "Month", "Person_ID")
+ ]
>
> parallel_data <- as.data.frame(lapply(
+     parallel_data,
+     function(x) (x - min(x)) / (max(x) - min(x))
+ ))
>
> colors <- ifelse(
+     suicide_data$Sex == "male",
+     "blue",
+     "red"
+ )
>
> matplot(
+     t(parallel_data),
+     type = "l",
+     lty = 1,
+     col = colors,
+     xaxt = "n",
+     xlab = "Variables",
+     ylab = "Scaled Values",
+     main = "Parallel Coordinate Plot by Sex"
+ )
>
> axis(
+     1,
+     at = 1:4,
+     labels = c("Age", "Year", "Month", "Person_ID")
+ )
>
> legend(
+     "topright",
+     legend = unique(suicide_data$Sex),
+     col = c("blue", "red"),
+     lty = 1
+ )
>
>
```


Parallel Coordinate Plot by Sex
female
male
Scaled Values
0.0
0.2
0.4
0.6
0.8
1.0
Age
Year
Month
Person_ID
Variables

EDA LAB
ASSESSMENT - 6

VYOM KHURANA
23BDS0081

TIME SERIES ANALYSIS

1 .Load the built-in AirPassengers dataset in R.

```
#23BDS0081
> #VYOM KHURANA
>
> data("AirPassengers")
> data_air <- AirPassengers
>
```

2. Explore the first five and last five observations of the dataset

```
> head(data_air,5)
      Jan Feb Mar Apr May
1949 112 118 132 129 121
> tail(data_air,5)
      Aug Sep Oct Nov Dec
1960 606 508 461 390 432
>
```

3. Identify the starting and ending time periods of the dataset. (start, end)

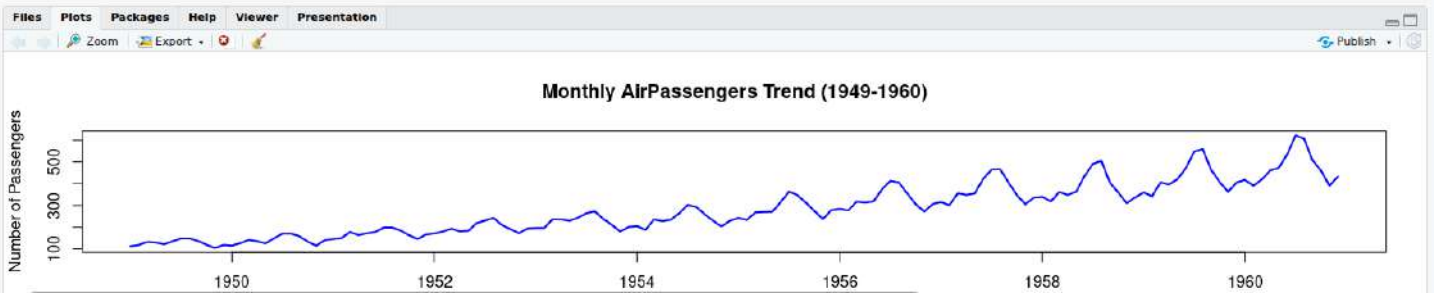
```
> start(data_air)
[1] 1949    1
> end(data_air)
[1] 1960   12
>
```

4. Determine the frequency and time-series characteristics of the dataset. (frequency, class)

```
> frequency(data_air)
[1] 12
> class(data_air)
[1] "ts"
```

5. Visualize the monthly passenger trend over the entire period. (plot)

```
> plot(data_air,
+      main = "Monthly AirPassengers Trend (1949-1960)",
+      xlab = "Year",
+      ylab = "Number of Passengers",
+      col = "blue",
+      lwd = 2)
>
>
```



6. Decompose the time series into trend, seasonal, and random components. (decompose)

```
> decomp <- decompose(AirPassengers)
> print(decomp)
$x
      Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
1949 112 118 132 129 121 135 148 148 136 119 104 118
1950 115 126 141 135 125 149 170 170 158 133 114 140
1951 145 150 178 163 172 178 199 199 184 162 146 166
1952 171 180 193 181 183 218 230 242 209 191 172 194
1953 196 196 236 235 229 243 264 272 237 211 180 201
1954 204 188 235 227 234 264 302 293 259 229 203 229
1955 242 233 267 269 270 315 364 347 312 274 237 278
1956 284 277 317 313 318 374 413 405 355 306 271 306
1957 315 301 356 348 355 422 465 467 404 347 305 336
1958 340 318 362 348 363 435 491 505 404 359 310 337
1959 360 342 406 396 420 472 548 559 463 407 362 405
1960 417 391 419 461 472 535 622 606 508 461 390 432

$seasonal
              Jan         Feb         Mar         Apr         May         Jun         Jul         Aug
Sep         Oct         Nov         Dec
1949 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232
16.520202 -20.642677 -53.593434 -28.619949
1950 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232
16.520202 -20.642677 -53.593434 -28.619949
1951 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232
16.520202 -20.642677 -53.593434 -28.619949
1952 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232
16.520202 -20.642677 -53.593434 -28.619949
1953 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232
16.520202 -20.642677 -53.593434 -28.619949
1954 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232
16.520202 -20.642677 -53.593434 -28.619949
1955 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232
16.520202 -20.642677 -53.593434 -28.619949
```

```
1956 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232
16.520202 -20.642677 -53.593434 -28.619949
1957 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232
16.520202 -20.642677 -53.593434 -28.619949
1958 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232
16.520202 -20.642677 -53.593434 -28.619949
1959 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232
16.520202 -20.642677 -53.593434 -28.619949
1960 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232
16.520202 -20.642677 -53.593434 -28.619949

$trend
             Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct
Nov      Dec
1949        NA       NA       NA       NA       NA       NA 126.7917 127.2500 127.9583 128.5833
129.0000 129.7500
1950 131.2500 133.0833 134.9167 136.4167 137.4167 138.7500 140.9167 143.1667 145.7083 148.4167
151.5417 154.7083
1951 157.1250 159.5417 161.8333 164.1250 166.6667 169.0833 171.2500 173.5833 175.4583 176.8333
178.0417 180.1667
1952 183.1250 186.2083 189.0417 191.2917 193.5833 195.8333 198.0417 199.7500 202.2083 206.2500
210.4167 213.3750
1953 215.8333 218.5000 220.9167 222.9167 224.0833 224.7083 225.3333 225.3333 224.9583 224.5833
224.4583 225.5417
1954 228.0000 230.4583 232.2500 233.9167 235.6250 237.7500 240.5000 243.9583 247.1667 250.2500
253.5000 257.1250
1955 261.8333 266.6667 271.1250 275.2083 278.5000 281.9583 285.7500 289.3333 293.2500 297.1667
301.0000 305.4583
1956 309.9583 314.4167 318.6250 321.7500 324.5000 327.0833 329.5417 331.8333 334.4583 337.5417
340.5417 344.0833
1957 348.2500 353.0000 357.6250 361.3750 364.5000 367.1667 369.4583 371.2083 372.1667 372.4167
372.7500 373.6250
1958 375.2500 377.9167 379.5000 380.0000 380.7083 380.9583 381.8333 383.6667 386.5000 390.3333
394.7083 398.6250
1959 402.5417 407.1667 411.8750 416.3333 420.5000 425.5000 430.7083 435.1250 437.7083 440.9583
445.8333 450.6250
1960 456.3333 461.3750 465.2083 469.3333 472.7500 475.0417       NA       NA       NA       NA
NA       NA

$random
                Jan          Feb          Mar          Apr          May          Jun          Jul
Aug          Sep          Oct          Nov          Dec
1949          NA           NA           NA           NA           NA           NA -42.6224747 -
42.0732323  -8.4785354  11.0593434  28.5934343  16.8699495
1950   8.4987374  29.1047980   8.3244949   6.6199495  -7.9103535 -25.1527778 -34.7474747 -
35.9898990  -4.2285354   5.2260101  16.0517677  13.9116162
1951  12.6237374  26.6464646  18.4078283   6.9116162   9.8396465 -26.4861111 -36.0808081 -
37.4065657  -7.9785354   5.8093434  21.5517677  14.4532828
1952  12.6237374  29.9797980   6.1994949  -2.2550505  -6.0770202 -13.2361111 -31.8724747 -
20.5732323  -9.7285354   5.3926768  15.1767677   9.2449495
1953   4.9154040  13.6881313  17.3244949  20.1199495   9.4229798 -17.1111111 -25.1641414 -
16.1565657  -4.4785354   7.0593434   9.1351010   4.0782828
1954   0.7487374  -6.2702020   4.9911616   1.1199495   2.8813131  -9.1527778  -2.3308081 -
13.7815657  -4.6868687  -0.6073232   3.0934343   0.4949495
1955   4.9154040   2.5214646  -1.8838384   1.8282828  -3.9936869  -2.3611111  14.4191919  -
5.1565657   2.2297980  -2.5239899 -10.4065657   1.1616162
1956  -1.2095960  -1.2285354   0.6161616  -0.7133838  -1.9936869  11.5138889  19.6275253
10.3434343   4.0214646 -10.8989899 -15.9482323  -9.4633838
1957  -8.5012626 -15.8118687   0.6161616  -5.3383838  -4.9936869  19.4305556  31.7108586
32.9684343  15.3131313  -4.7739899 -14.1565657  -9.0050505
```

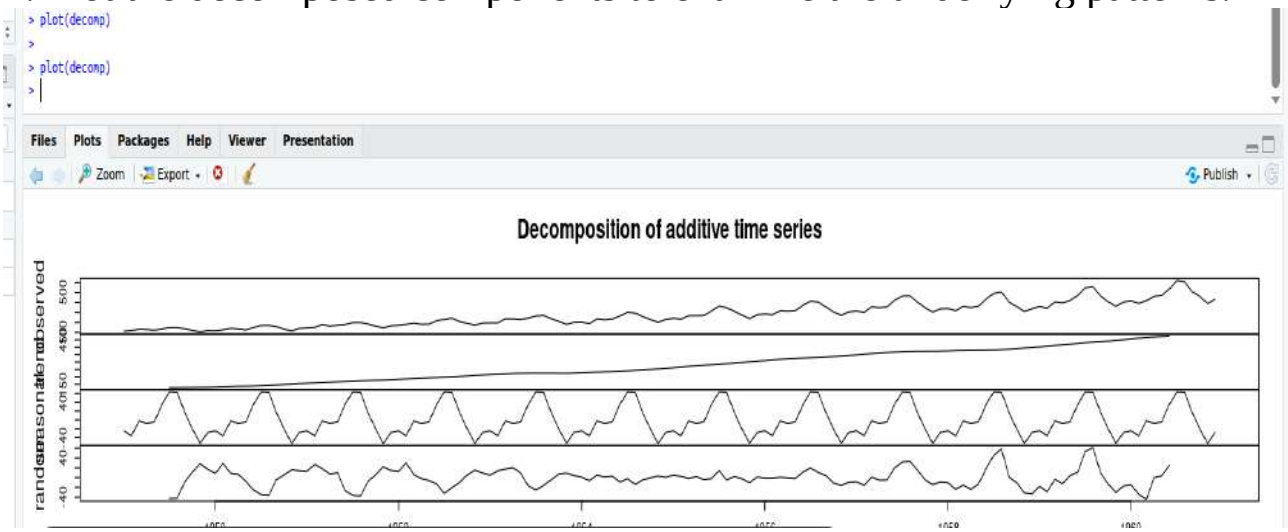
```
1958 -10.5012626 -23.7285354 -15.2588384 -23.9633838 -13.2020202  18.6388889  45.3358586
58.5101010   0.9797980 -10.6906566 -31.1148990 -33.0050505
1959 -17.7929293 -28.9785354  -3.6338384 -12.2967172   4.0063131  11.0972222  53.4608586
61.0517677   8.7714646 -13.3156566 -30.2398990 -17.0050505
1960 -14.5845960 -34.1868687 -43.9671717  -0.2967172   3.7563131  24.5555556          NA
NA           NA           NA           NA           NA

$figure
 [1] -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232
16.520202 -20.642677 -53.593434 -28.619949

$type
[1] "additive"

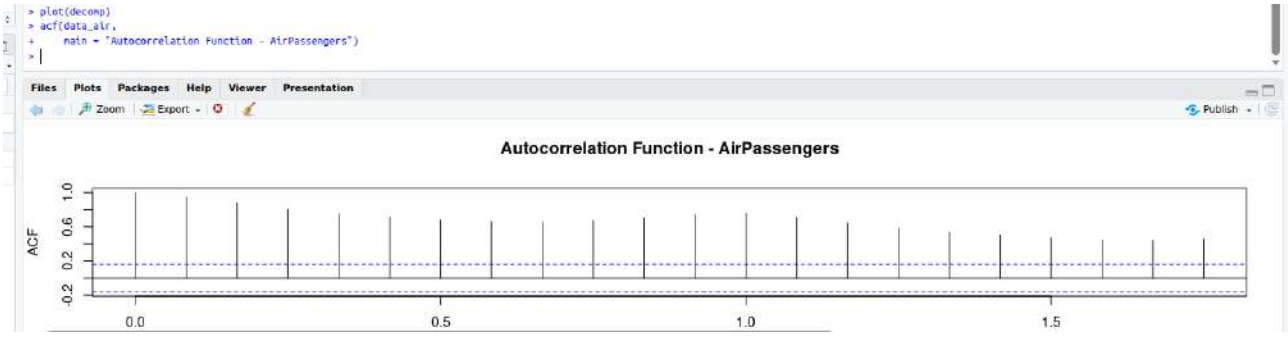
attr(,"class")
[1] "decomposed.ts"
```

7. Plot the decomposed components to examine the underlying patterns.



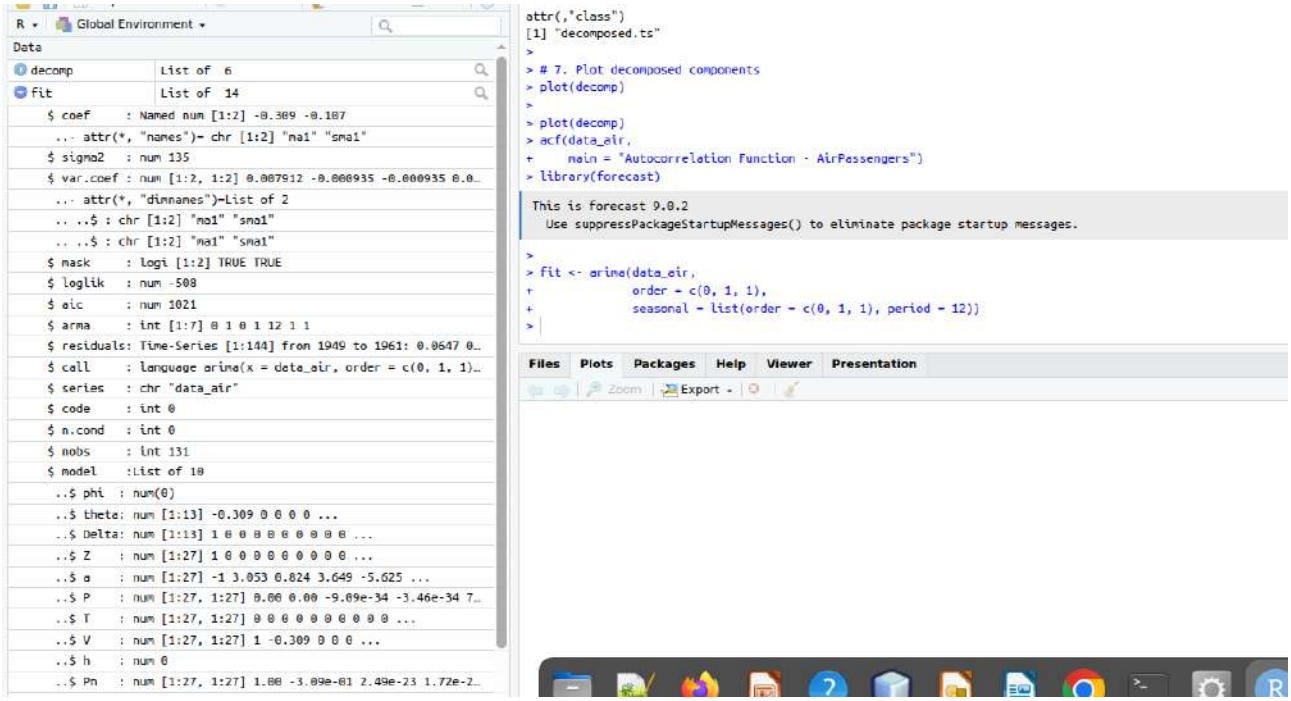
8. Analyze the autocorrelation and partial autocorrelation of the time series. (acf, pacf)

```
> plot(decomp)
> acf(data_air,
+     main = "Autocorrelation Function - AirPassengers")
```



9. Fit an ARIMA model to the AirPassengers time series. (arima)

```
> fit <- arima(data_air,
+              order = c(0, 1, 1),
+              seasonal = list(order = c(0, 1, 1), period = 12))
```



10. Display and interpret the fitted ARIMA model parameters and model summary. (summary)

```
> summary(fit)

Call:
arima(x = data_air, order = c(0, 1, 1), seasonal = list(order = c(0, 1, 1),
    period = 12))

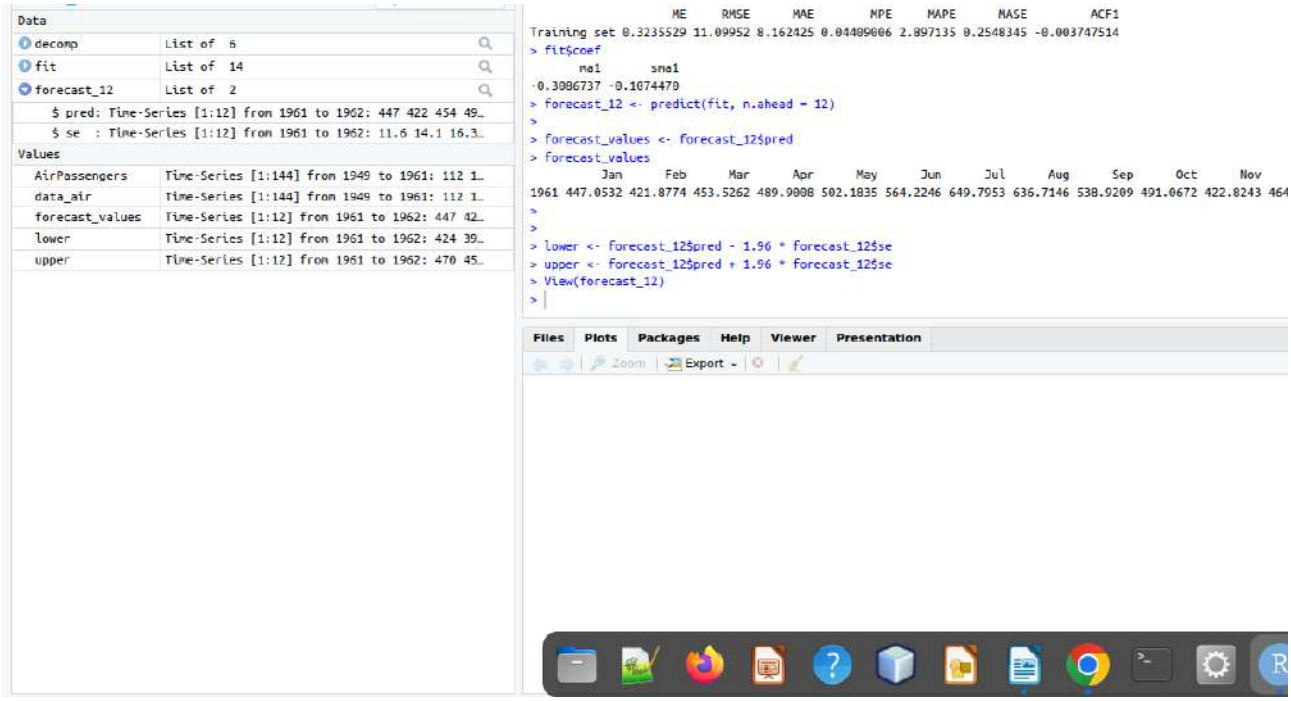
Coefficients:
          ma1     sma1
      -0.3087  -0.1074
s.e.   0.0890   0.0828

sigma^2 estimated as 135.4:  log likelihood = -507.5,  aic = 1021

Training set error measures:
                     ME     RMSE      MAE        MPE     MAPE      MASE          ACF1
Training set 0.3235529 11.09952 8.162425 0.04409006 2.897135 0.2548345 -0.003747514
> fit$coef
        ma1       sma1
-0.3086737 -0.1074470
>
```

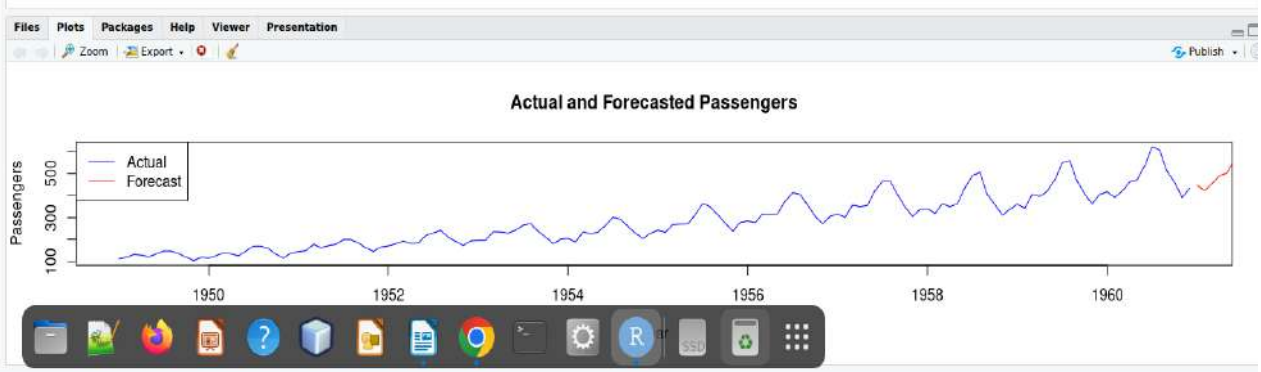
11. Forecast the number of passengers for the next 12 months using the fitted ARIMA model. (predict)

```
> forecast_12 <- predict(fit, n.ahead = 12)
>
> forecast_values <- forecast_12$pred
> forecast_values
            Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct      Nov
1961 447.0532 421.8774 453.5262 489.9008 502.1835 564.2246 649.7953 636.7146 538.9209 491.0672 422.8243 464.
>
>
> lower <- forecast_12$pred - 1.96 * forecast_12$se
> upper <- forecast_12$pred + 1.96 * forecast_12$se
>
```



12. Visualize the actual and forecasted passenger values with appropriate labels and title. (plot, lines)

```
> plot(data_air, main="Actual and Forecasted Passengers",
+      xlab="Year", ylab="Passengers", col="blue")
> lines(forecast_12$pred, col="red")
> legend("topleft", c("Actual", "Forecast"),
+        col=c("blue","red"), lty=1)
>
```



SCREENSHOTS

```
> #23BDS0081
> #VYOM KHURANA
>
> data("AirPassengers")
> data_air <- AirPassengers
>
>
> head(data_air,5)
     Jan Feb Mar Apr May
1949 112 118 132 129 121
> tail(data_air,5)
     Aug Sep Oct Nov Dec
1960 606 508 461 390 432
> start(data_air)
[1] 1949    1
> end(data_air)
[1] 1960   12
> frequency(data_air)
[1] 12
> class(data_air)
[1] "ts"
>
> plot(data_air,
+      main = "Monthly AirPassengers Trend (1949-1960)",
+      xlab = "Year",
+      ylab = "Number of Passengers",
+      col = "blue",
+      lwd = 2)
>
> decomp <- decompose(AirPassengers)
> print(decomp)
$x
     Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
1949 112 118 132 129 121 135 148 148 136 119 104 118
1950 115 126 141 135 125 149 170 170 158 133 114 140
1951 145 150 178 163 172 178 199 199 184 162 146 166
1952 171 180 193 181 183 218 230 242 209 191 172 194
1953 196 196 236 235 229 243 264 272 237 211 180 201
1954 204 188 235 227 234 264 302 293 259 229 203 229
```

Files | Plots | Packages | Help | Viewer | Presentation

```
> # 7. Plot decomposed components
> plot(decomp)
>
> plot(decomp)
> acf(data_air,
+     main = "Autocorrelation Function - AirPassengers")
> library(forecast)
```

```
This is forecast 9.0.2
  Use suppressPackageStartupMessages() to eliminate package startup messages.
```

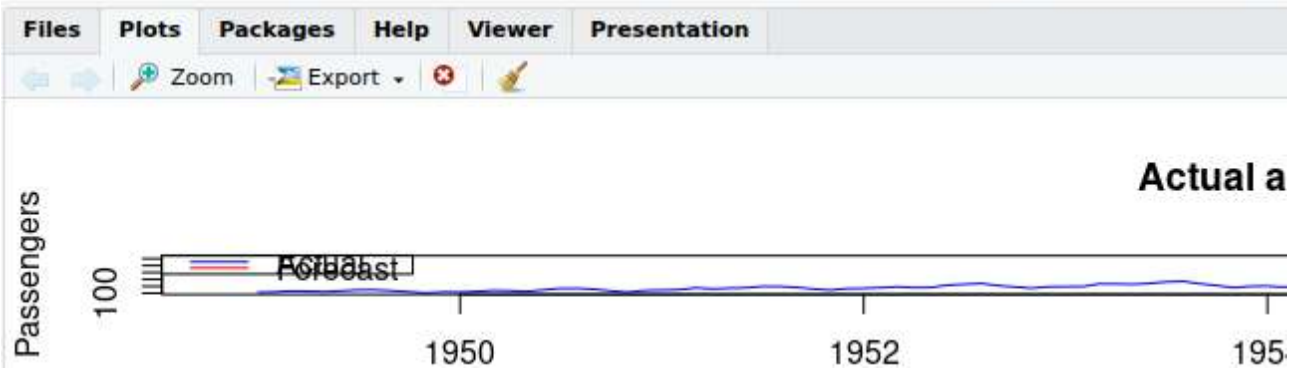
```
>
> fit <- arima(data_air,
+              order = c(0, 1, 1),
+              seasonal = list(order = c(0, 1, 1), period = 12))
> fit <- arima(data_air,order = c(0, 1, 1),seasonal = list(order = c(0, 1, 1), period = 12))
> summary(fit)

Call:
arima(x = data_air, order = c(0, 1, 1), seasonal = list(order = c(0, 1, 1),
    period = 12))

Coefficients:
          ma1     sma1
      -0.3087  -0.1074
s.e.   0.0890   0.0828

sigma^2 estimated as 135.4:  log likelihood = -507.5,  aic = 1021

Training set error measures:
                    ME     RMSE      MAE        MPE     MAPE      MASE         ACF1
Training set 0.3235529 11.09952 8.162425 0.04409006 2.897135 0.2548345 -0.003747514
> fit$coef
       ma1       sma1
-0.3086737 -0.1074470
> forecast_12 <- predict(fit, n.ahead = 12)
>
> forecast_values <- forecast_12$pred
> forecast_values
          Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct
```



```
+                seasonal = list(order = c(0, 1, 1), period = 12))
> fit <- arima(data_air,order = c(0, 1, 1),seasonal = list(order = c(0, 1, 1), period = 12))
> summary(fit)

Call:
arima(x = data_air, order = c(0, 1, 1), seasonal = list(order = c(0, 1, 1),
    period = 12))

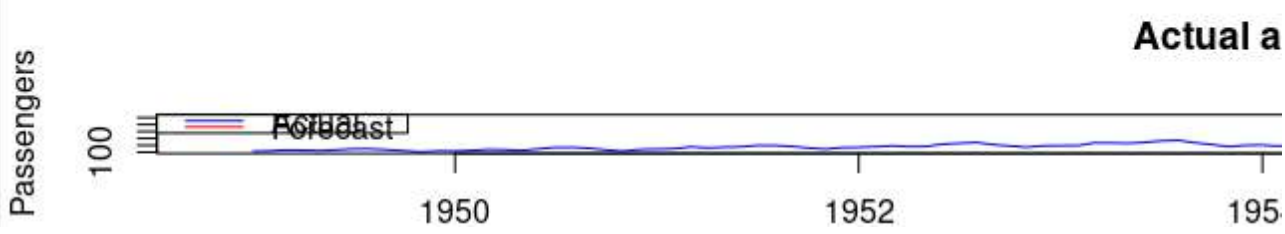
Coefficients:
          ma1     sma1
      -0.3087  -0.1074
s.e.   0.0890   0.0828

sigma^2 estimated as 135.4:  log likelihood = -507.5,  aic = 1021

Training set error measures:
                    ME     RMSE      MAE        MPE     MAPE      MASE         ACF1
Training set 0.3235529 11.09952 8.162425 0.04409006 2.897135 0.2548345 -0.003747514
> fit$coef
       ma1       sma1
-0.3086737 -0.1074470
> forecast_12 <- predict(fit, n.ahead = 12)
>
> forecast_values <- forecast_12$pred
> forecast_values
          Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct
1961 447.0532 421.8774 453.5262 489.9008 502.1835 564.2246 649.7953 636.7146 538.9209 491.0672 4
>
>
> lower <- forecast_12$pred - 1.96 * forecast_12$se
> upper <- forecast_12$pred + 1.96 * forecast_12$se
> View(forecast_12)
> plot(data_air, main="Actual and Forecasted Passengers",
+      xlab="Year", ylab="Passengers", col="blue")
> lines(forecast_12$pred, col="red")
> legend("topleft", c("Actual", "Forecast"),
+        col=c("blue","red"), lty=1)
>
>
```

Files | Plots | Packages | Help | Viewer | Presentation

Zoom | Export



EDA LAB

ASSESSMENT – 7

VYOM KHURANA

23BDS0081

1 D, 2 D and N D Analysis

1. Load the dataset shared via email

```
> # 23BDS0081
> # VYOM KHURANA
>
> library(ggplot2)
> library(e1071)
>
> suicide_data <- read.csv(
+     "https://raw.githubusercontent.com/salemprakash/EDA/main/Data/SuicideChina.csv",
+     stringsAsFactors = FALSE
+ )
>
> head(suicide_data)
  rownames Person_ID Hospitalised Died Urban Year Month    Sex Age Education Occupatio
1        1         1          yes   no    no 2010    12 female  39 Secondary  househol
2        2         2           no  yes    no 2009     3   male  83   primary    farmin
3        3         3           no  yes    no 2010     2   male  60   primary    farmin
4        4         4           no  yes    no 2011     1   male  73   primary    farmin
5        5         5          yes   no    no 2009     8   male  51 Secondary    farmin
6        6         6           no  yes    no 2009    11   male  62 iliterate    farmin
> str(suicide_data)
'data.frame':  2571 obs. of  12 variables:
 $ rownames    : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Person_ID   : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Hospitalised: chr  "yes" "no" "no" "no" ...
 $ Died        : chr  "no" "yes" "yes" "yes" ...
 $ Urban       : chr  "no" "no" "no" "no" ...
 $ Year        : int  2010 2009 2010 2011 2009 2009 2010 2010 2010 2011 ...
```

```
 $ Month       : int  12 3 2 1 8 11 1 10 7 1 ...
 $ Sex         : chr  "female" "male" "male" "male" ...
 $ Age         : int  39 83 60 73 51 62 90 54 66 30 ...
 $ Education   : chr  "Secondary" "primary" "primary" "primary" ...
 $ Occupation  : chr  "household" "farming" "farming" "farming" ...
 $ method      : chr  "Other poison" "Hanging" "Hanging" "Hanging" ...

>
```

2. Compute summary statistics: Mean, Median, Minimum, Maximum, Range, Variance, Standard Deviation, Interquartile Range (IQR), Skewness (using e1071::skewness), and Kurtosis (using e1071::kurtosis).

```
> age <- suicide_data$Age
>
> mean_age <- mean(age, na.rm = TRUE)
> median_age <- median(age, na.rm = TRUE)
> min_age <- min(age, na.rm = TRUE)
> max_age <- max(age, na.rm = TRUE)
> range_age <- max_age - min_age
> variance_age <- var(age, na.rm = TRUE)
> sd_age <- sd(age, na.rm = TRUE)
> iqr_age <- IQR(age, na.rm = TRUE)
> skewness_age <- e1071::skewness(age, na.rm = TRUE)
> kurtosis_age <- e1071::kurtosis(age, na.rm = TRUE)
>
> summary_statistics <- data.frame(
+     Statistic = c(
+         "Mean",
+         "Median",
+         "Minimum",
+         "Maximum",
+         "Range",
+         "Variance",
+         "Standard Deviation",
+         "IQR",
+         "Skewness",
+         "Kurtosis"
+     ),
+     Value = c(
+         mean_age,
+         median_age,
+         min_age,
+         max_age,
+         range_age,
+         variance_age,
+         sd_age,
+         iqr_age,
+         skewness_age,
+         kurtosis_age
+     )
+ )
>
```

```
> print(summary_statistics)
            Statistic        Value
1                Mean  52.63088292
2              Median  53.00000000
3             Minimum  12.00000000
4             Maximum 100.00000000
5               Range  88.00000000
6            Variance 391.40183187
7  Standard Deviation  19.78387808
8                 IQR  32.00000000
9            Skewness   0.01423941
10           Kurtosis  -1.00517926
```

3. Store these values in a single structured data.frame or named vector and print it.

```
> summary_statistics <- data.frame(
+     Mean = mean(suicide_data$Age, na.rm = TRUE),
+     Median = median(suicide_data$Age, na.rm = TRUE),
+     Minimum = min(suicide_data$Age, na.rm = TRUE),
+     Maximum = max(suicide_data$Age, na.rm = TRUE),
+     Range = diff(range(suicide_data$Age, na.rm = TRUE)),
+     Variance = var(suicide_data$Age, na.rm = TRUE),
+     Standard_Deviation = sd(suicide_data$Age, na.rm = TRUE),
+     IQR = IQR(suicide_data$Age, na.rm = TRUE),
+     Skewness = e1071::skewness(suicide_data$Age, na.rm = TRUE),
+     Kurtosis = e1071::kurtosis(suicide_data$Age, na.rm = TRUE)
+ )
>
> print(summary_statistics)
      Mean Median Minimum Maximum Range Variance Standard_Deviation IQR
Skewness  Kurtosis
1 52.63088     53      12     100    88 391.4018           19.78388  32 0.
01423941 -1.005179
```

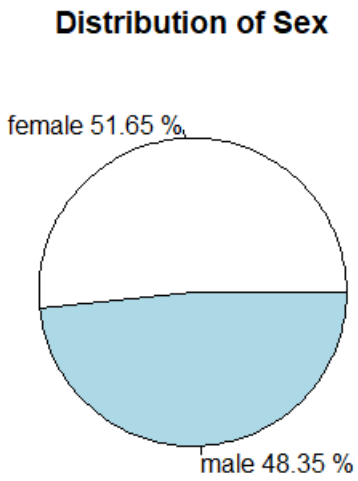
4. Create a frequency distribution table and a relative frequency distribution table for a categorical variable. Generate a Pie Chart and a Bar Plot to represent the distribution.

```
> sex_frequency <- table(suicide_data$Sex)
>
> print("Frequency Distribution:")
[1] "Frequency Distribution:"
> print(sex_frequency)

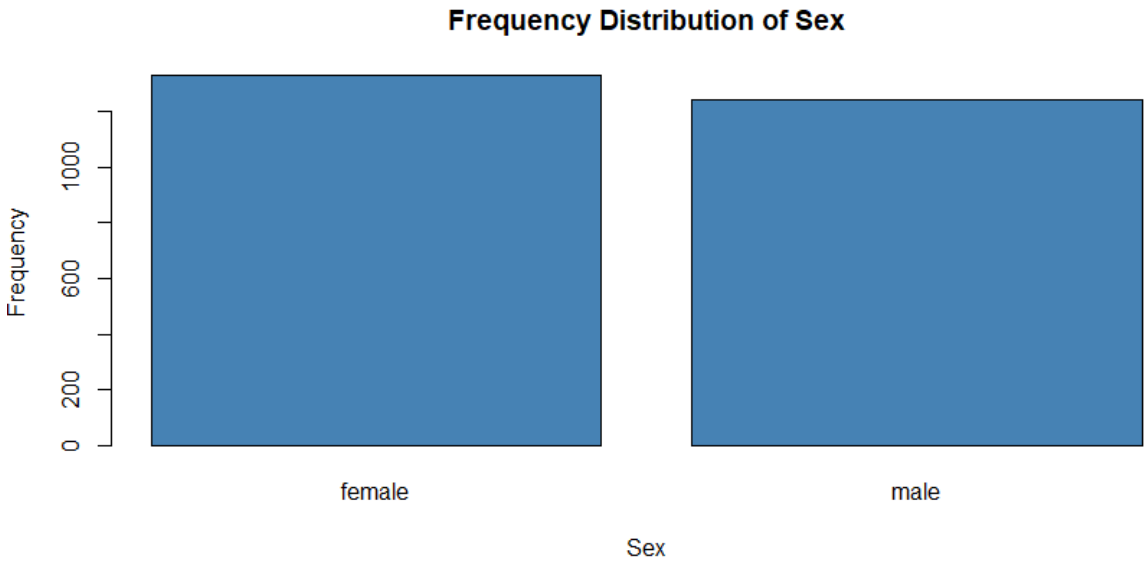
female   male
  1328   1243
> sex_relative_frequency <- prop.table(sex_frequency)
>
> print("Relative Frequency Distribution:")
[1] "Relative Frequency Distribution:"
> print(sex_relative_frequency)
```

```
   female      male
0.5165305 0.4834695
```

```
> pie(
+     sex_frequency,
+     main = "Distribution of Sex",
+     labels = paste(
+         names(sex_frequency),
+         round(100 * sex_frequency / sum(sex_frequency), 2),
+         "%"
+     )
+ )
```



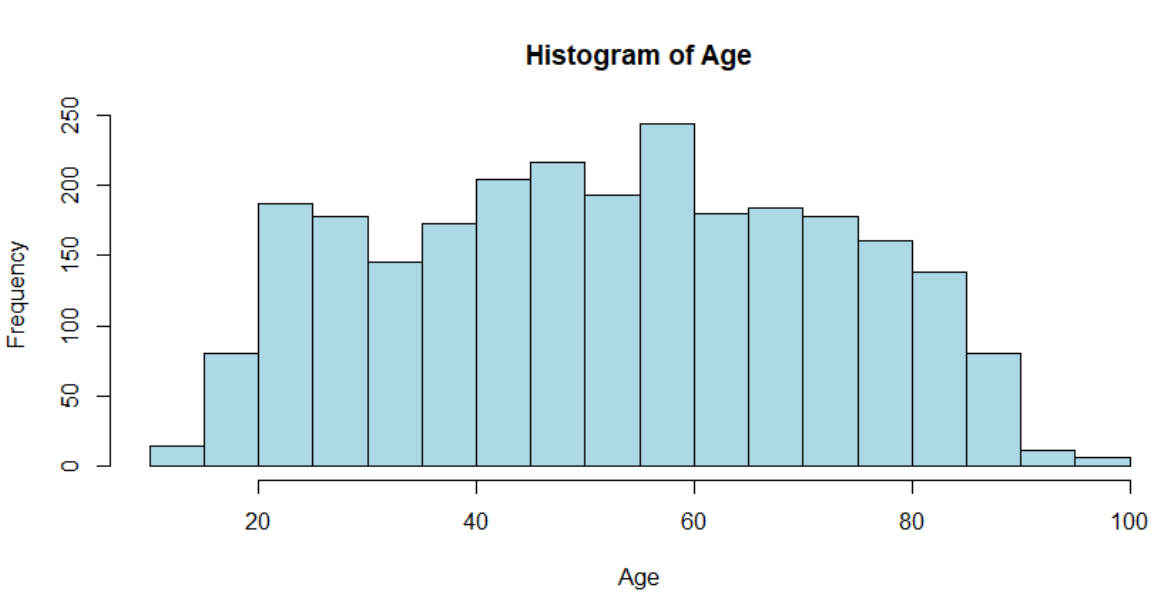

```
> barplot(
+     sex_frequency,
+     main = "Frequency Distribution of Sex",
+     xlab = "Sex",
+     ylab = "Frequency",
+     col = "steelblue"
+ )
```



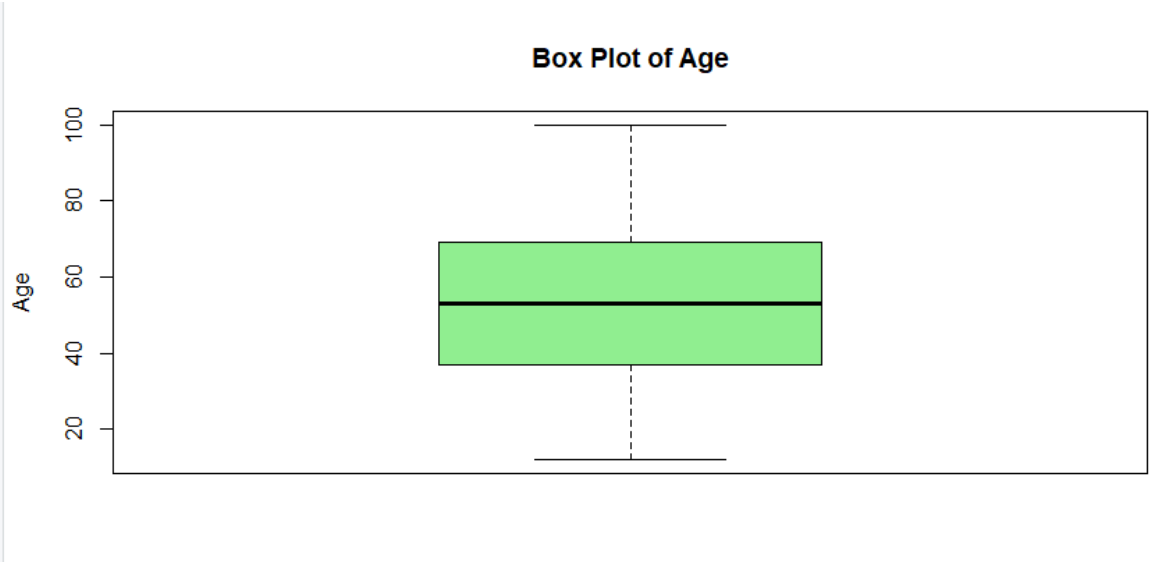
5. Create a frequency distribution table with bin intervals for a numerical variable. Plot its Histogram and a Box Plot to check for outliers.

```
> age_breaks <- seq(
+     floor(min(suicide_data$Age)),
+     ceiling(max(suicide_data$Age)),
+     by = 10
+ )
>
> age_frequency <- table(
+     cut(
+         suicide_data$Age,
+         breaks = age_breaks,
+         right = FALSE
+     )
+ )
>
> print("Age Frequency Distribution:")
[1] "Age Frequency Distribution:"
> print(age_frequency)

[12,22) [22,32) [32,42) [42,52) [52,62) [62,72) [72,82) [82,92)
    126     364     332     410     436     357     341     190
> # Histogram of Age
>
> hist(
+     suicide_data$Age,
+     main = "Histogram of Age",
+     xlab = "Age",
+     ylab = "Frequency",
+     col = "lightblue",
+     border = "black"
+ )
```




```
> boxplot(
+     suicide_data$Age,
+     main = "Box Plot of Age",
+     ylab = "Age",
+     col = "lightgreen"
+ )
```



6. Choose two categorical variables: Construct a Contingency Table (Cross-tabulation) and compute the row/column percentages using prop.table()

```
> contingency_table <- table(
+     suicide_data$Sex,
```

```
+     suicide_data$Hospitalised
+ )
>
> print("Contingency Table:")
[1] "Contingency Table:"
> print(contingency_table)

          no yes
  female 469 859
  male   549 694

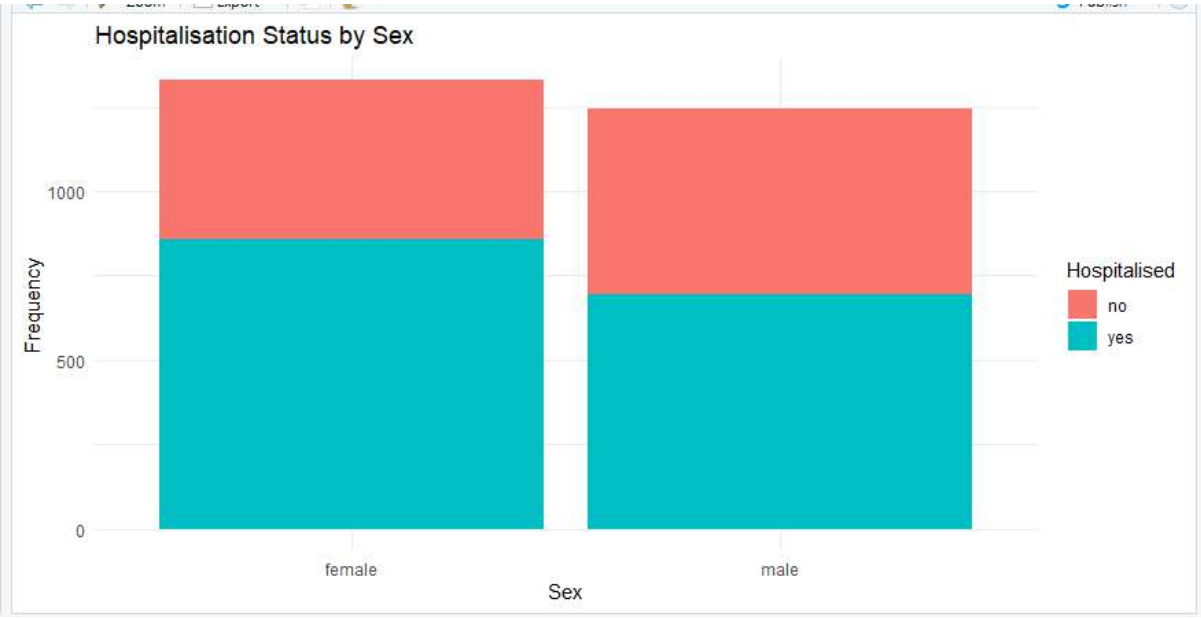
>
> row_percentages <- prop.table(
+     contingency_table,
+     margin = 1
+ )
>
> print("Row Percentages:")
[1] "Row Percentages:"
> print(round(row_percentages * 100, 2))

           no   yes
  female 35.32 64.68
  male   44.17 55.83
> column_percentages <- prop.table(
+     contingency_table,
+     margin = 2
+ )
>
> print("Column Percentages:")
[1] "Column Percentages:"
> print(round(column_percentages * 100, 2))

           no   yes
  female 46.07 55.31
  male   53.93 44.69
```

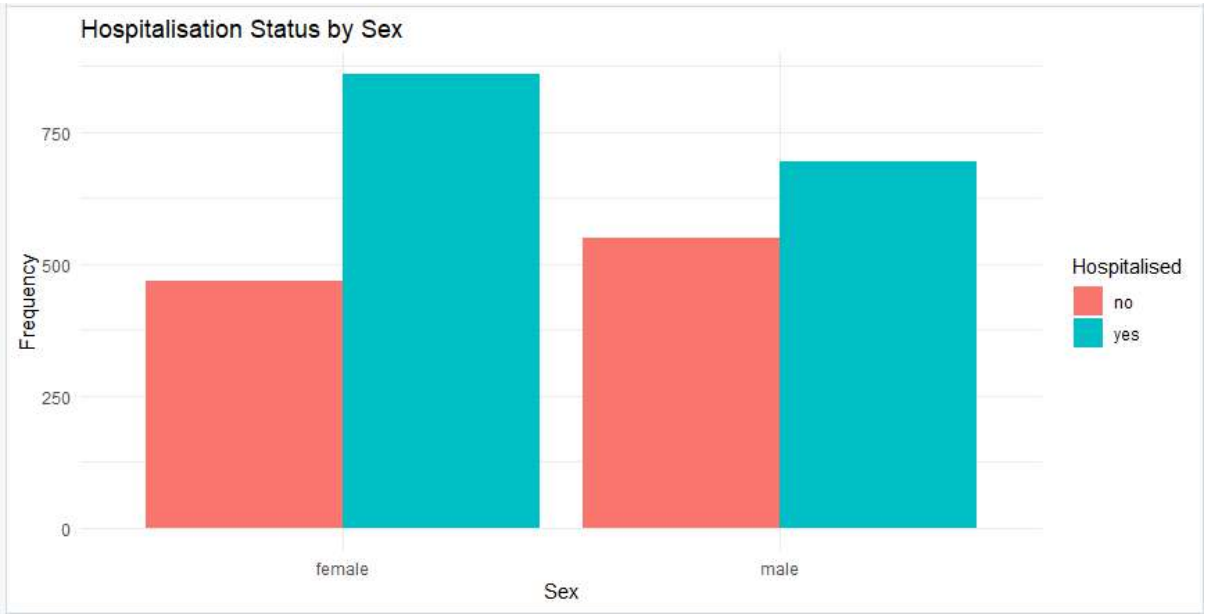
7. Create a Stacked Bar Plot or Grouped Bar Plot to visualize the relationship.

```
> ggplot(
+     suicide_data,
+     aes(x = Sex, fill = Hospitalised)
+ ) +
+     geom_bar() +
+     labs(
+         title = "Hospitalisation Status by Sex",
+         x = "Sex",
+         y = "Frequency",
+         fill = "Hospitalised"
+     ) +
+     theme_minimal()
```




```
> ggplot(
+     suicide_data,
+     aes(x = Sex, fill = Hospitalised)
+ ) +
+     geom_bar(position = "dodge") +
+     labs(
+         title = "Hospitalisation Status by Sex",
+         x = "Sex",
+         y = "Frequency",
+         fill = "Hospitalised"
+     ) +
+     theme_minimal()

>
```



8. Select one categorical variable and one numerical variable: Compute the mean, median, and standard deviation of the numerical variable grouped by each category using aggregate()

```
> age_mean <- aggregate(
+     Age ~ Sex,
+     data = suicide_data,
+     FUN = mean
+ )
>
> print("Mean Age by Sex:")
[1] "Mean Age by Sex:"
> print(age_mean)
     Sex      Age
1 female 51.53012
2   male 53.80692

>
> age_median <- aggregate(
+     Age ~ Sex,
+     data = suicide_data,
+     FUN = median
+ )
>
> print("Median Age by Sex:")
[1] "Median Age by Sex:"
> print(age_median)
     Sex Age
1 female  50
2   male  55


> age_sd <- aggregate(
+     Age ~ Sex,
+     data = suicide_data,
+     FUN = sd
+ )
>
> print("Standard Deviation of Age by Sex:")
[1] "Standard Deviation of Age by Sex:"
> print(age_sd)
     Sex      Age
1 female 20.11888
2   male 19.35859

>
> age_statistics <- aggregate(
+     Age ~ Sex,
+     data = suicide_data,
+     FUN = function(x) {
+         c(
+             Mean = mean(x),
+             Median = median(x),
+             SD = sd(x)
+         )
+     }
```

```
+ )
>
> print("Age Statistics by Sex:")
[1] "Age Statistics by Sex:"
> print(age_statistics)
     Sex Age.Mean Age.Median   Age.SD
1 female 51.53012   50.00000 20.11888
2   male 53.80692   55.00000 19.35859
```

9. Generate Side-by-Side Box Plots and a Violin Plot to visualize the numerical distribution across categories.

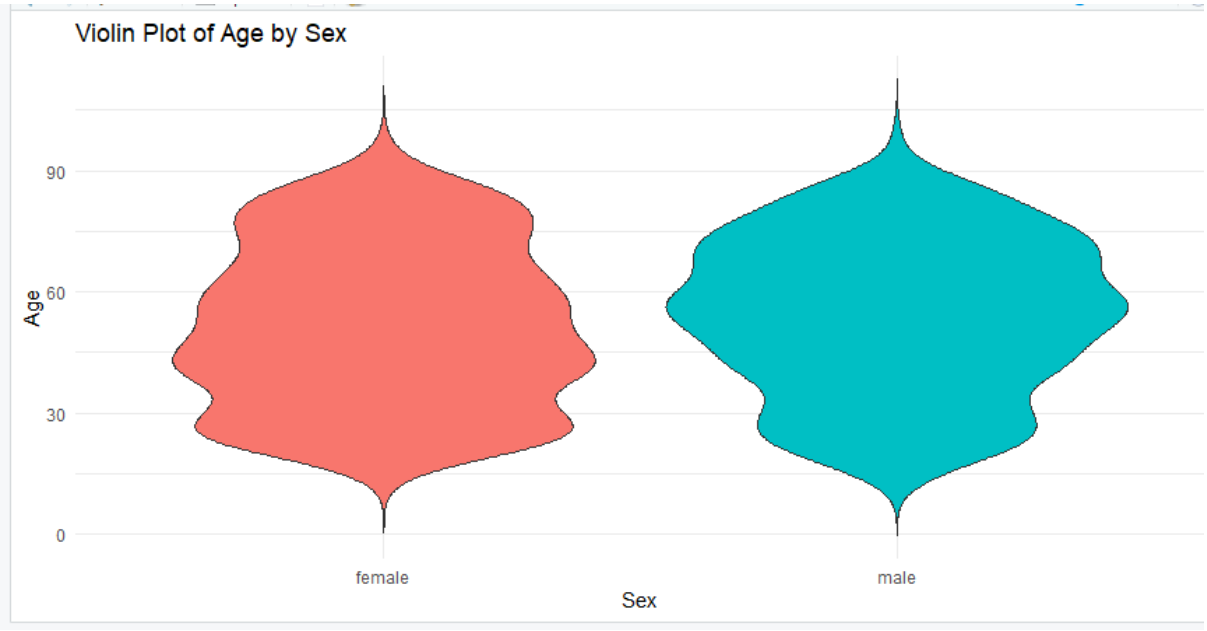
```
> ggplot(
+     suicide_data,
+     aes(x = Sex, y = Age, fill = Sex)
+ ) +
+     geom_boxplot() +
+     labs(
+         title = "Age Distribution by Sex",
+         x = "Sex",
+         y = "Age"
+     ) +
+     theme_minimal() +
+     theme(legend.position = "none")
```




```
> ggplot(
+     suicide_data,
+     aes(x = Sex, y = Age, fill = Sex)
+ ) +
+     geom_violin(trim = FALSE) +
```

```
+     labs(
+         title = "Violin Plot of Age by Sex",
+         x = "Sex",
+         y = "Age"
+     ) +
+     theme_minimal() +
+     theme(legend.position = "none")

>
```



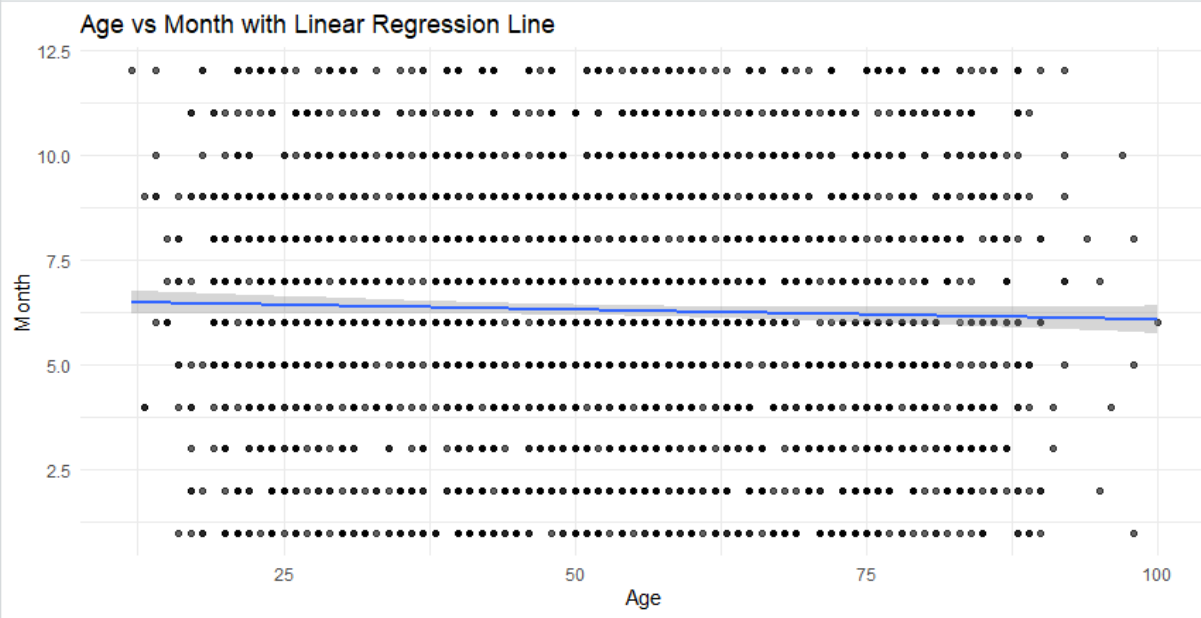
10. Select two numerical variables: Calculate the Pearson and Spearman Correlation Coefficients (cor()) and Construct a Scatter Plot with a fitted linear regression trend line.

```
> pearson_correlation <- cor(
+     suicide_data$Age,
+     suicide_data$Month,
+     method = "pearson",
+     use = "complete.obs"
+ )
>
> print("Pearson Correlation:")
[1] "Pearson Correlation:"
> print(pearson_correlation)
[1] -0.02839694


> spearman_correlation <- cor(
+     suicide_data$Age,
+     suicide_data$Month,
+     method = "spearman",
+     use = "complete.obs"
+ )
>
```

```
> print("Spearman Correlation:")
[1] "Spearman Correlation:"
> print(spearman_correlation)
[1] -0.03135478
```

```
> ggplot(
+     suicide_data,
+     aes(x = Age, y = Month)
+ ) +
+     geom_point(alpha = 0.6) +
+     geom_smooth(
+         method = "lm",
+         se = TRUE
+     ) +
+     labs(
+         title = "Age vs Month with Linear Regression Line",
+         x = "Age",
+         y = "Month"
+     ) +
+     theme_minimal()
`geom_smooth()` using formula = 'y ~ x'
```



11. Select two categorical features and one numerical feature: Calculate summary statistics (Mean and Standard Deviation) for the numerical variable across all combinations of the two categorical variables and Create a Faceted Box Plot and a Grouped Bar Chart with Error Bars.

```
> group_statistics <- aggregate(
+     Age ~ Sex + Education,
+     data = suicide_data,
+     FUN = function(x) {
+         c(
+             Mean = mean(x),
```

```
+             SD = sd(x)
+         )
+     }
+ )
>
> print("Mean and Standard Deviation by Sex and Education:")
[1] "Mean and Standard Deviation by Sex and Education:"
> print(group_statistics)
      Sex Education Age.Mean   Age.SD
1  female iliterate 71.69550 14.73318
2    male iliterate 70.72951 13.52488
3  female   primary 60.44904 14.15610
4    male   primary 62.88116 14.59354
5  female Secondary 38.96018 14.55342
6    male Secondary 42.40698 15.56690
7  female  Tertiary 30.11111 11.86849
8    male  Tertiary 42.30000 21.07684
9  female   unknown 53.81579 19.04597
10   male   unknown 47.09524 21.16233

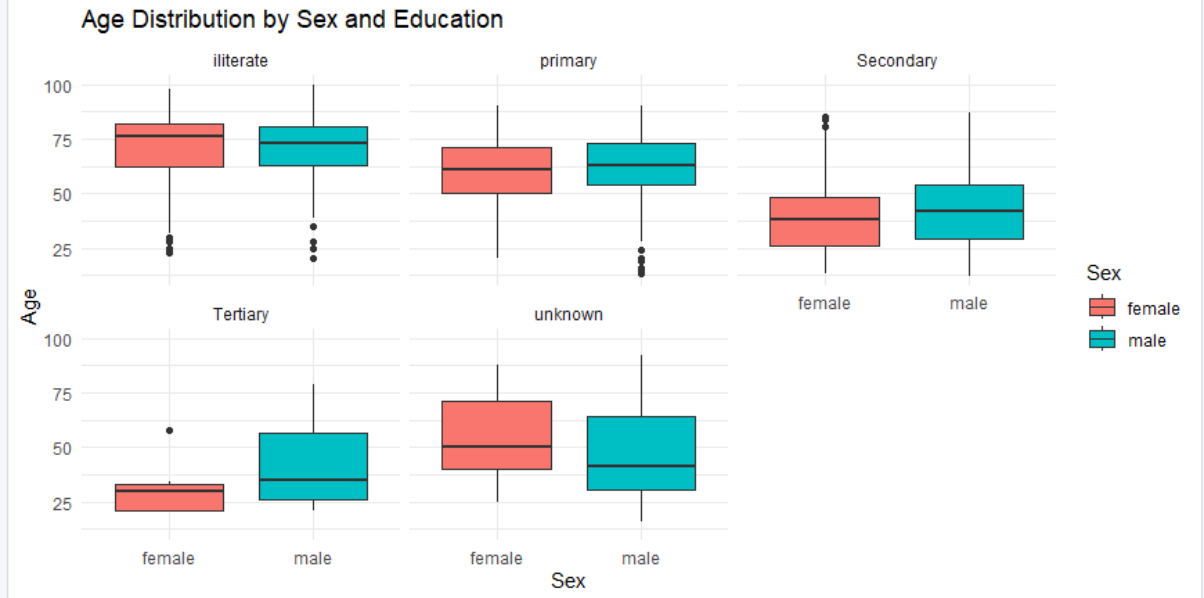
>
> mean_table <- aggregate(
+     Age ~ Sex + Education,
+     data = suicide_data,
+     FUN = mean
+ )
>
> print("Mean Age:")
[1] "Mean Age:"
> print(mean_table)
      Sex Education      Age
1  female iliterate 71.69550
2    male iliterate 70.72951
3  female   primary 60.44904
4    male   primary 62.88116
5  female Secondary 38.96018
6    male Secondary 42.40698
7  female  Tertiary 30.11111
8    male  Tertiary 42.30000
9  female   unknown 53.81579
10   male   unknown 47.09524


> sd_table <- aggregate(
+     Age ~ Sex + Education,
+     data = suicide_data,
+     FUN = sd
+ )
>
> print("Standard Deviation of Age:")
[1] "Standard Deviation of Age:"
> print(sd_table)
      Sex Education      Age
1  female iliterate 14.73318
2    male iliterate 13.52488
3  female   primary 14.15610
```

```
4    male   primary 14.59354
5  female Secondary 14.55342
6    male Secondary 15.56690
7  female  Tertiary 11.86849
8    male  Tertiary 21.07684
9  female   unknown 19.04597
10   male   unknown 21.16233

>
```

```
> ggplot(
+     suicide_data,
+     aes(x = Sex, y = Age, fill = Sex)
+ ) +
+     geom_boxplot() +
+     facet_wrap(~Education) +
+     labs(
+         title = "Age Distribution by Sex and Education",
+         x = "Sex",
+         y = "Age"
+     ) +
+     theme_minimal()
```

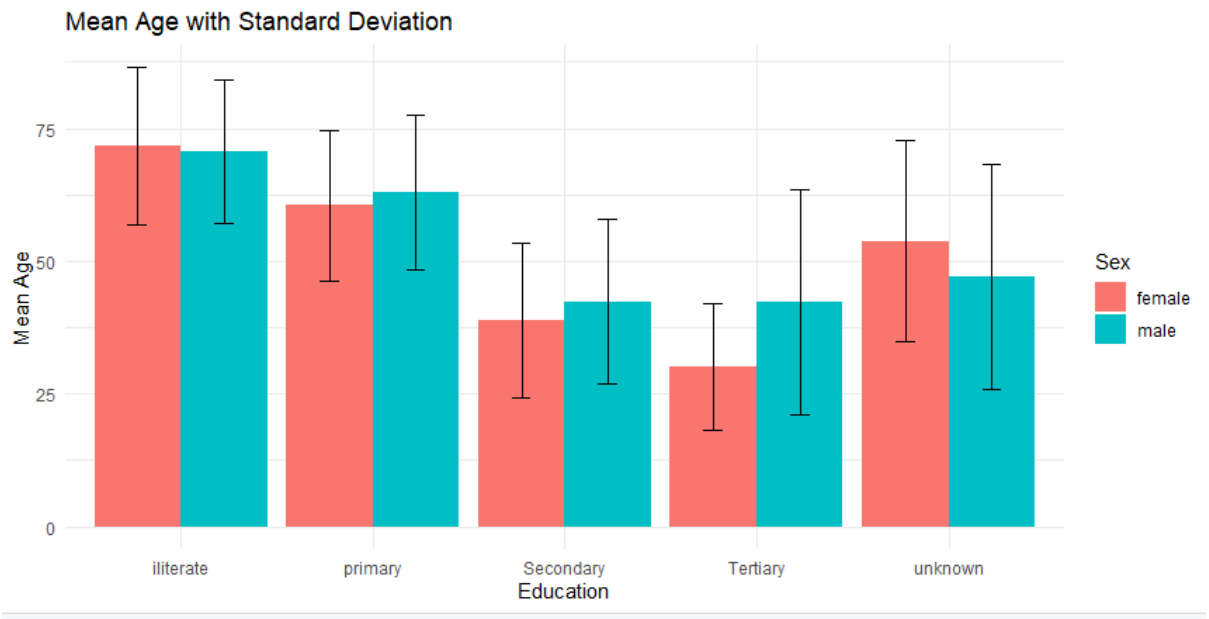



```
> group_summary <- aggregate(
+     Age ~ Sex + Education,
+     data = suicide_data,
+     FUN = function(x) {
+         c(
+             Mean = mean(x),
+             SD = sd(x)
+         )
+     }
+ )
>
> group_summary <- do.call(
+     data.frame,
+     group_summary
+ )
>
```

```
> names(group_summary) <- c(
+     "Sex",
+     "Education",
+     "Mean",
+     "SD"
+ )
>
> print(group_summary)
      Sex Education     Mean       SD
1  female iliterate 71.69550 14.73318
2    male iliterate 70.72951 13.52488
3  female   primary 60.44904 14.15610
4    male   primary 62.88116 14.59354
5  female Secondary 38.96018 14.55342
6    male Secondary 42.40698 15.56690
7  female  Tertiary 30.11111 11.86849
8    male  Tertiary 42.30000 21.07684
9  female   unknown 53.81579 19.04597
10   male   unknown 47.09524 21.16233
```

```
> ggplot(
+     group_summary,
+     aes(
+         x = Education,
+         y = Mean,
+         fill = Sex
+     )
+ ) +
+     geom_bar(
+         stat = "identity",
+         position = position_dodge()
+     ) +
+     geom_errorbar(
+         aes(
+             ymin = Mean - SD,
+             ymax = Mean + SD
+         ),
+         position = position_dodge(0.9),
+         width = 0.2
+     ) +
+     labs(
+         title = "Mean Age with Standard Deviation",
+         x = "Education",
+         y = "Mean Age",
+         fill = "Sex"
+     ) +
+     theme_minimal()

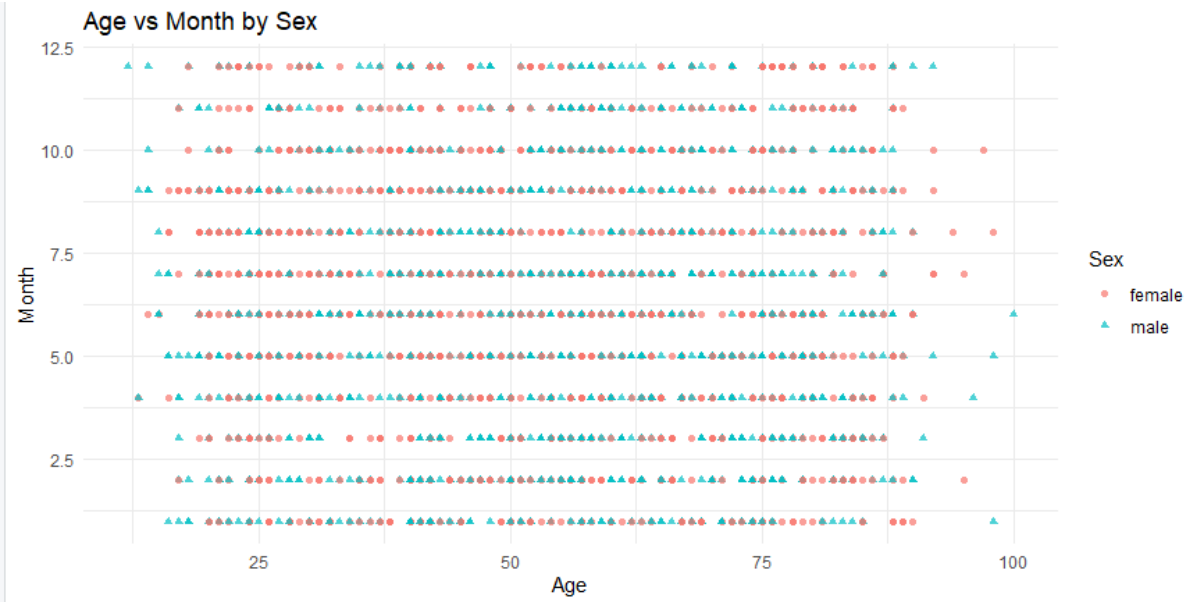
>
```



12. Select one categorical variable and two numerical variables: Construct a Scatter Plot of the two numerical features where points are colored/shaped by the categorical feature. compute a correlation matrix between the two numerical variables separately for each category level.

```
> ggplot(
+     suicide_data,
+     aes(
+         x = Age,
+         y = Month,
+         color = Sex,
+         shape = Sex
+     )
+ ) +
+     geom_point(alpha = 0.7) +
+     labs(
+         title = "Age vs Month by Sex",
+         x = "Age",
+         y = "Month",
+         color = "Sex",
+         shape = "Sex"
+     ) +
+     theme_minimal()

>
```




```
> male_data <- subset(
+     suicide_data,
+     Sex == "male"
+ )
>
> female_data <- subset(
+     suicide_data,
+     Sex == "female"
+ )
>
> male_correlation <- cor(
+     male_data[, c("Age", "Month")],
+     use = "complete.obs"
+ )
>
> female_correlation <- cor(
+     female_data[, c("Age", "Month")],
+     use = "complete.obs"
+ )
>
> print("Correlation Matrix - Male:")
[1] "Correlation Matrix - Male:"
> print(male_correlation)
              Age       Month
Age    1.00000000 -0.00970353
Month -0.00970353  1.00000000
>
> print("Correlation Matrix - Female:")
[1] "Correlation Matrix - Female:"
> print(female_correlation)
              Age       Month
Age    1.00000000 -0.03975826
Month -0.03975826  1.00000000


> male_correlation_value <- cor(
+     male_data$Age,
+     male_data$Month,
```

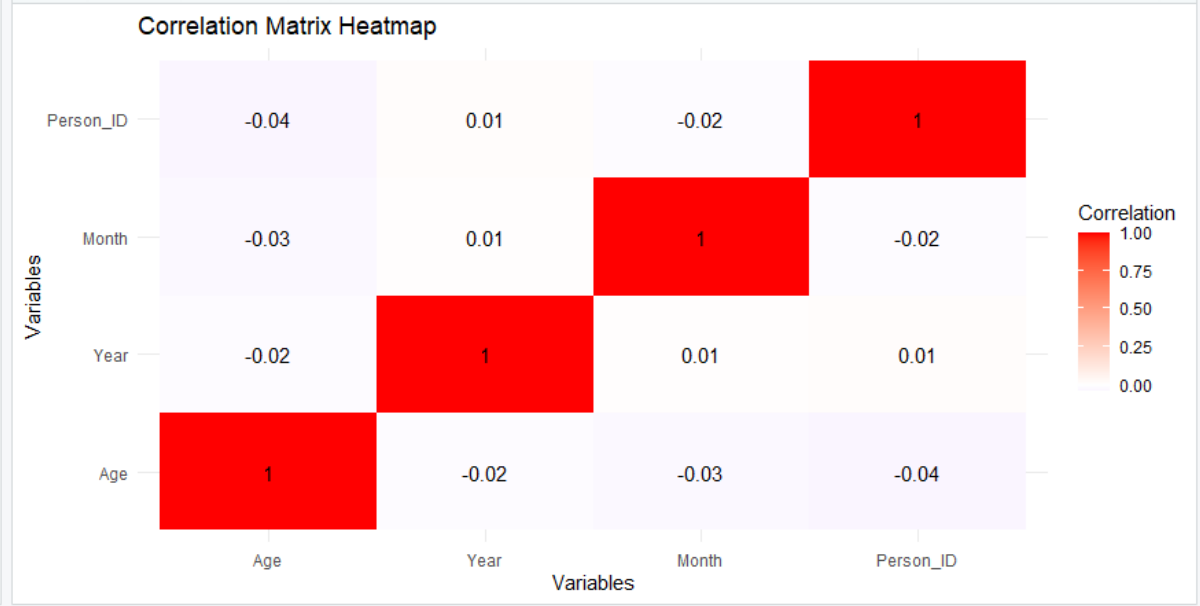
```
+     method = "pearson",
+     use = "complete.obs"
+ )
>
> female_correlation_value <- cor(
+     female_data$Age,
+     female_data$Month,
+     method = "pearson",
+     use = "complete.obs"
+ )
>
> print(paste(
+     "Male Age-Month Correlation:",
+     male_correlation_value
+ ))
[1] "Male Age-Month Correlation: -0.00970353033472517"
>
> print(paste(
+     "Female Age-Month Correlation:",
+     female_correlation_value
+ ))
[1] "Female Age-Month Correlation: -0.0397582583794294"
```

13. Select at least 3 numerical features from the dataset: Generate a full Correlation Matrix and plot it as a Heatmap

```
> numerical_data <- suicide_data[
+     ,
+     c("Age", "Year", "Month", "Person_ID")
+ ]
>
> correlation_matrix <- cor(
+     numerical_data,
+     use = "complete.obs",
+     method = "pearson"
+ )
>
> print("Full Correlation Matrix:")
[1] "Full Correlation Matrix:"
> print(round(correlation_matrix, 3))
             Age   Year  Month Person_ID
Age        1.000 -0.017 -0.028    -0.041
Year      -0.017  1.000  0.009     0.015
Month     -0.028  0.009  1.000    -0.017
Person_ID -0.041  0.015 -0.017     1.000
```

```
> correlation_df <- as.data.frame(
+     as.table(correlation_matrix)
+ )
>
> ggplot(
+     correlation_df,
```

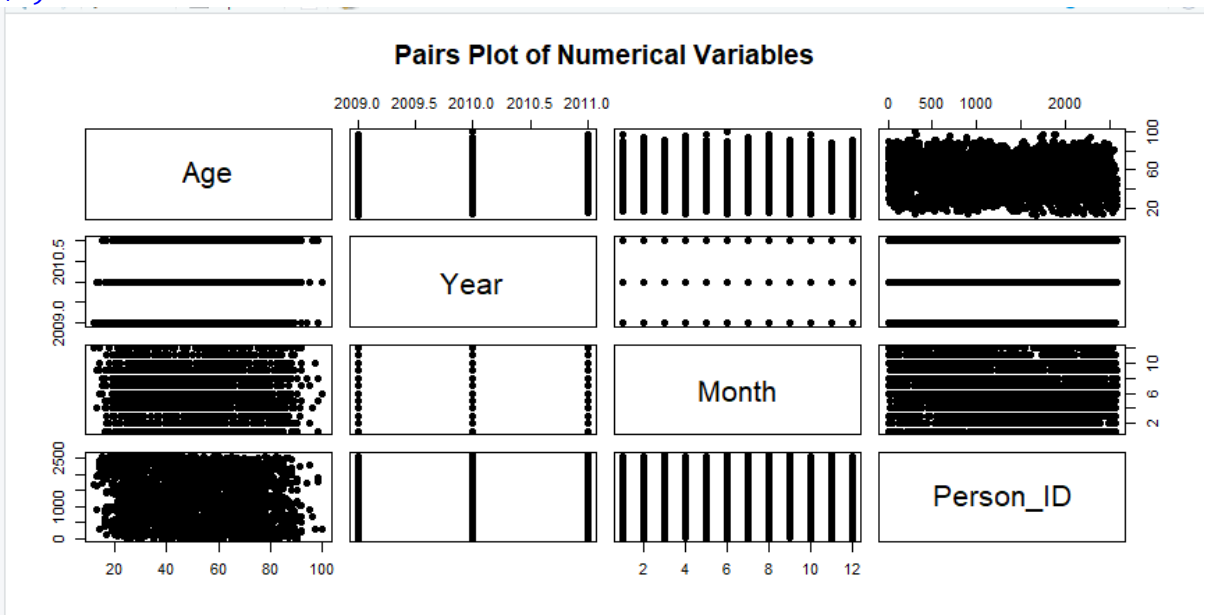
```
+     aes(
+         x = Var1,
+         y = Var2,
+         fill = Freq
+     )
+ ) +
+     geom_tile() +
+     geom_text(
+         aes(label = round(Freq, 2)),
+         color = "black"
+     ) +
+     scale_fill_gradient2(
+         low = "blue",
+         mid = "white",
+         high = "red",
+         midpoint = 0
+     ) +
+     labs(
+         title = "Correlation Matrix Heatmap",
+         x = "Variables",
+         y = "Variables",
+         fill = "Correlation"
+     ) +
+     theme_minimal()
```



14. Produce a Pairs Plot / Scatterplot Matrix to examine pairwise relationships and distributions simultaneously

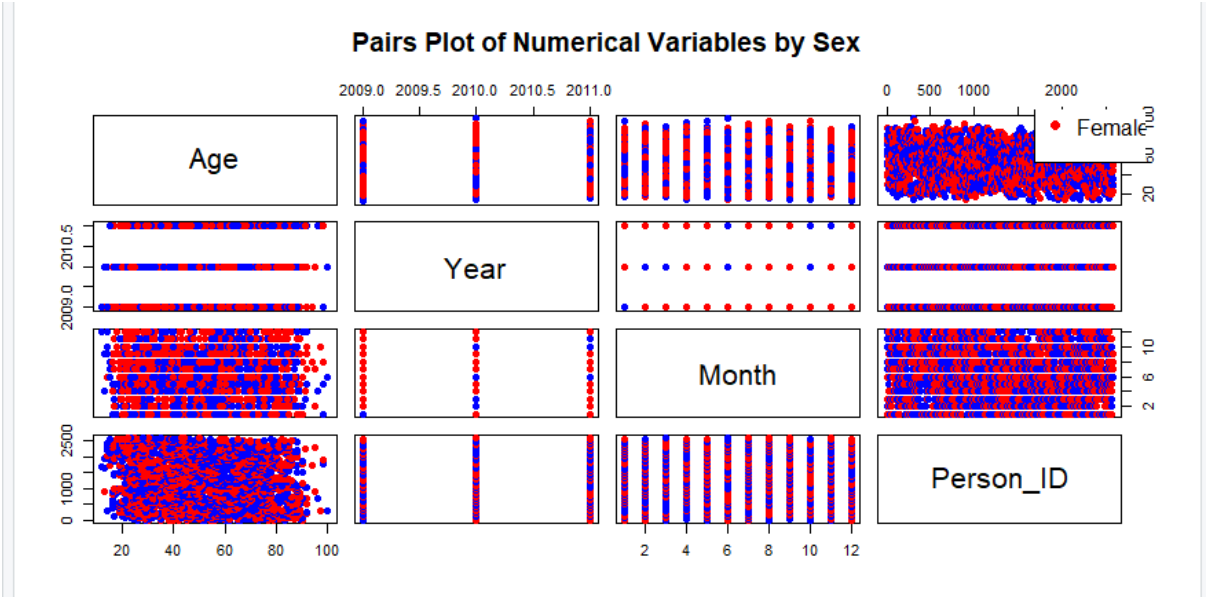
```
> pairs(
+     suicide_data[
+         ,
+         c("Age", "Year", "Month", "Person_ID")
+     ],
+     main = "Pairs Plot of Numerical Variables",
```

```
+     pch = 19
+ )
```




```
> point_colors <- ifelse(
+     suicide_data$Sex == "male",
+     "blue",
+     "red"
+ )
>
> pairs(
+     suicide_data[
+         ,
+         c("Age", "Year", "Month", "Person_ID")
+     ],
+     main = "Pairs Plot of Numerical Variables by Sex",
+     pch = 19,
+     col = point_colors
+ )
>
> legend(
+     "topright",
+     legend = c("Male", "Female"),
+     col = c("blue", "red"),
+     pch = 19
+ )

>
```

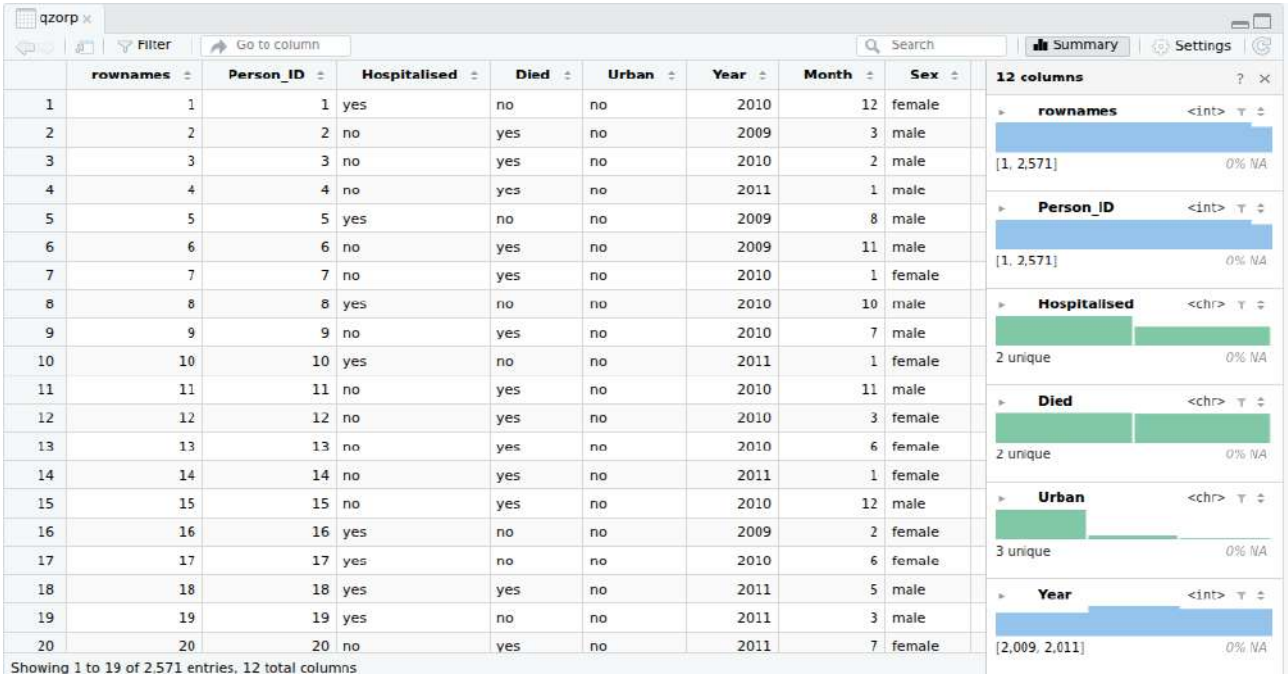

Pairs Plot of Numerical Variables by Sex
Age
Year
Month
Person_ID
Female
2009.0
2009.5
2010.0
2010.5
2011.0
0
500
1000
2000
20
40
60
80
100
2
4
6
8
10
12

EDA LAB 8
VYOM KHURANA
23BDS0081

Assessment 8: K-Means Clustering 1. Load the dataset shared through e-mail. 2. View the dataset 3. Check missing values 4. Check for the dependent variable distribution. (table) 5. Select only numerical features for K-Means clustering (Others remove it). 6. Standardize the data. (scale) 7. Apply K-Means Clustering (kmeans) for center as 2. 8. Display K-Means results 9. Display cluster centers 10. Calculate WCSS for K = 1 to 10 11. Display the elbow plot. 12. Apply K-means clustering based on the elbow recommendation. Hierarchical Clustering 1. Load the dataset shared through e-mail. 2. View the dataset 3. Check missing values 4. Select only numerical attributes 5. Standardize the data (scale) 6. Calculate Euclidean distance (dist) 7. Perform Hierarchical Clustering using Ward's method (hclust) 8. Plot the dendrogram. (plot) 9. Divide the data into 3 clusters and Display cluster assignments(cutree) 10. Display number of samples in each cluster. (table)

ANSWER :

```
> #23BDS0081
> #VYOM KHURANA
> # 1. Load the dataset shared through e-mail
> qzorp <- read.csv(
+     "https://raw.githubusercontent.com/salemprakash/EDA/main/Data/SuicideChina.csv",
+     stringsAsFactors = FALSE
+ )
> # 2. View the dataset
> head(qzorp)
  rownames Person_ID Hospitalised Died Urban Year Month    Sex Age Education Occupation
method
1        1         1          yes   no    no 2010    12 female  39 Secondary  household Other
poison
2        2         2           no  yes    no 2009     3   male  83   primary    farming
Hanging
3        3         3           no  yes    no 2010     2   male  60   primary    farming
Hanging
4        4         4           no  yes    no 2011     1   male  73   primary    farming
Hanging
5        5         5          yes   no    no 2009     8   male  51 Secondary    farming
Pesticide
6        6         6           no  yes    no 2009    11   male  62 iliterate    farming
Pesticide
>
> # 2. View the dataset
> View(qzorp)
>
```



qzorp

Filter | Go to column | Search | Summary | Settings

	rownames	Person_ID	Hospitalised	Died	Urban	Year	Month	Sex
1	1	1	yes	no	no	2010	12	female
2	2	2	no	yes	no	2009	3	male
3	3	3	no	yes	no	2010	2	male
4	4	4	no	yes	no	2011	1	male
5	5	5	yes	no	no	2009	8	male
6	6	6	no	yes	no	2009	11	male
7	7	7	no	yes	no	2010	1	female
8	8	8	yes	no	no	2010	10	male
9	9	9	no	yes	no	2010	7	male
10	10	10	yes	no	no	2011	1	female
11	11	11	no	yes	no	2010	11	male
12	12	12	no	yes	no	2010	3	female
13	13	13	no	yes	no	2010	6	female
14	14	14	no	yes	no	2011	1	female
15	15	15	no	yes	no	2010	12	male
16	16	16	yes	no	no	2009	2	female
17	17	17	yes	no	no	2010	6	female
18	18	18	yes	yes	no	2011	5	male
19	19	19	yes	no	no	2011	3	male
20	20	20	no	yes	no	2011	7	female

Showing 1 to 19 of 2,571 entries, 12 total columns

12 columns

rownames <int> [1, 2,571] 0% NA

Person_ID <int> [1, 2,571] 0% NA

Hospitalised <chr> 2 unique 0% NA

Died <chr> 2 unique 0% NA

Urban <chr> 3 unique 0% NA

Year <int> [2,009, 2,011] 0% NA

```
> # 3. Check missing values
> colSums(is.na(qzorp))
    rownames    Person_ID Hospitalised         Died        Urban         Year        Month
Sex           Age    Education   Occupation
           0            0            0            0            0            0            0
0            0            0            0
       method
            0
>
> # 4. Check for the dependent variable distribution
> table(qzorp$Died)

  no  yes
1315 1256
>
> # 5. Select only numerical features for K-Means clustering
> vexlora <- qzorp[sapply(qzorp, is.numeric)]
>
> # 5. Remove non-numerical features and unnecessary ID attributes
> vexlora <- vexlora[, !(names(vexlora) %in% c("rownames", "Person_ID"))]
>
> # 5. Display selected numerical features
> head(vexlora)
  Year Month Age
1 2010    12  39
2 2009     3  83
3 2010     2  60
4 2011     1  73
5 2009     8  51
6 2009    11  62
>
> # 6. Standardize the data
> zuntrix <- scale(vexlora)
>
> # 6. Display standardized data
> head(zuntrix)
              Year      Month         Age
[1,] -0.05750176  1.780373 -0.68898943
```

```
[2,] -1.32106616 -1.029918  1.53504368
[3,] -0.05750176 -1.342172  0.37248092
[4,]  1.20606263 -1.654427  1.02958161
[5,] -1.32106616  0.531355 -0.08243495
[6,] -1.32106616  1.468119  0.47357333
>
> # 7. Apply K-Means Clustering for center as 2
> set.seed(23)
>
> kravix <- kmeans(
+     zuntrix,
+     centers = 2,
+     nstart = 25
+ )
>
> # 8. Display K-Means results
> kravix
K-means clustering with 2 clusters of sizes 866, 1705

Cluster means:
          Year       Month         Age
1  1.2060626  0.04530518 -0.04292019
2 -0.6125808 -0.02301131  0.02179993

Clustering vector:
   [1] 2 2 2 1 2 2 2 2 2 1 2 2 2 1 2 2 2 1 1 1 2 1 2 2 2 1 1 2 2 1 2 2 2 2 1 2 2 2 2 2 1 2 2 2 1 2
2 2 2 2 1 2 2 2 1 2 1 1 2 2 1 1 2 2 1 2 1 2 2 2
  [71] 1 2 2 2 2 2 2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 1 1 2 1 1 1 2 2 1 2 2 1 2 1 2 2 1 2 1 1 2 2 1 2
2 2 1 1 2 2 2 1 1 2 2 2 2 2 1 2 1 2 1 2 2 1 2 1
 [141] 1 2 2 2 1 1 2 2 2 2 2 1 1 1 1 2 2 1 1 1 1 2 1 1 1 2 2 2 2 1 2 1 2 2 1 2 2 2 1 2 2 1 1 1 2 2
2 2 2 1 2 1 2 2 2 2 1 2 1 2 1 2 1 2 2 1 2 1 2 2
 [211] 2 2 1 1 2 2 2 2 2 2 2 2 1 2 2 2 2 1 2 2 1 2 2 1 1 1 1 2 2 2 2 2 1 2 2 2 2 2 1 2 2 1 2 2 1 1
2 2 2 2 1 2 1 1 1 1 1 2 2 1 1 1 1 1 1 1 1 2 1 1
 [281] 2 1 1 2 1 2 1 2 2 2 2 2 2 1 1 2 2 2 1 1 1 1 2 2 2 2 2 2 2 2 2 2 1 1 1 2 2 1 2 2 2 2 1 2 1 1
1 1 2 2 2 2 2 2 2 1 1 2 1 2 2 1 2 2 1 2 2 2 1 2
 [351] 1 2 1 2 2 2 2 2 2 2 2 2 2 1 1 2 1 1 2 2 1 2 2 2 1 1 1 1 2 1 2 2 2 1 1 2 2 2 2 2 2 2 2 2 2 2
2 2 1 1 1 2 2 1 1 2 2 2 1 2 1 2 2 2 1 2 2 1 1 2
 [421] 2 1 1 2 2 2 2 1 1 1 2 2 1 2 2 2 2 2 1 2 2 2 1 2 2 2 2 2 1 2 2 1 2 2 2 1 2 1 1 1 1 2 2 2 2 1
2 2 2 1 2 2 2 2 2 1 1 1 2 1 2 2 2 1 2 2 2 2 2 1
 [491] 1 2 1 2 1 2 2 1 2 1 1 2 1 1 2 2 2 2 2 2 2 2 2 2 1 1 1 2 2 1 1 1 2 2 2 2 2 1 2 2 2 2 2 2 2 2
2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
 [561] 2 2 1 1 2 1 1 2 2 2 2 2 2 2 1 2 2 1 2 2 2 2 1 1 1 2 2 2 2 1 2 1 2 2 1 1 2 2 1 1 1 2 2 2 1 2
1 1 1 2 2 2 1 1 2 1 2 1 2 2 2 1 1 2 1 1 1 1 1 2
 [631] 2 1 1 1 2 1 2 2 1 2 1 2 1 1 1 1 2 2 1 1 1 1 2 2 2 2 1 2 2 2 2 1 2 1 2 2 1 1 2 1 2 2 2 2 2 2
2 2 2 2 2 1 2 2 2 2 2 2 2 2 2 1 1 1 1 1 2 2 2 2
 [701] 1 2 2 2 2 2 1 1 2 1 2 2 2 1 2 1 2 2 2 1 1 2 2 1 2 2 1 1 1 2 2 2 2 2 2 2 1 2 2 2 1 2 2 1 1 1
1 1 2 2 2 2 1 1 2 2 2 2 2 2 2 2 2 2 1 2 2 1 2 2
 [771] 2 1 2 2 2 1 2 2 2 1 1 2 1 2 1 2 2 1 2 1 2 2 2 2 2 2 1 2 2 2 1 2 2 2 2 2 1 2 1 2 1 2 2 2 2 1
1 2 1 1 1 2 1 1 2 1 1 1 2 2 1 1 2 2 1 2 2 2 1 2
 [841] 2 1 2 1 2 1 2 1 2 1 2 1 2 2 1 2 1 2 2 2 2 1 2 2 1 2 1 2 2 2 2 2 2 1 1 1 2 1 2 1 1 2 1 2 2 1
2 1 1 2 2 1 1 1 2 2 2 2 2 2 2 2 1 2 2 2 1 2 2 2
 [911] 1 1 1 1 2 2 1 2 2 2 2 1 2 2 2 2 2 2 2 1 2 2 2 1 2 2 2 1 2 2 2 1 2 1 2 2 1 2 2 2 1 1 2 2 2 2
2 2 2 2 2 2 2 2 2 2 2 1 2 1 1 2 2 1 2 2 1 2 2 2
 [981] 2 1 2 1 2 2 2 1 2 2 1 2 2 2 2 2 2 2 1 1
 [ reached 'max' / getOption("max.print") -- omitted 1571 entries ]

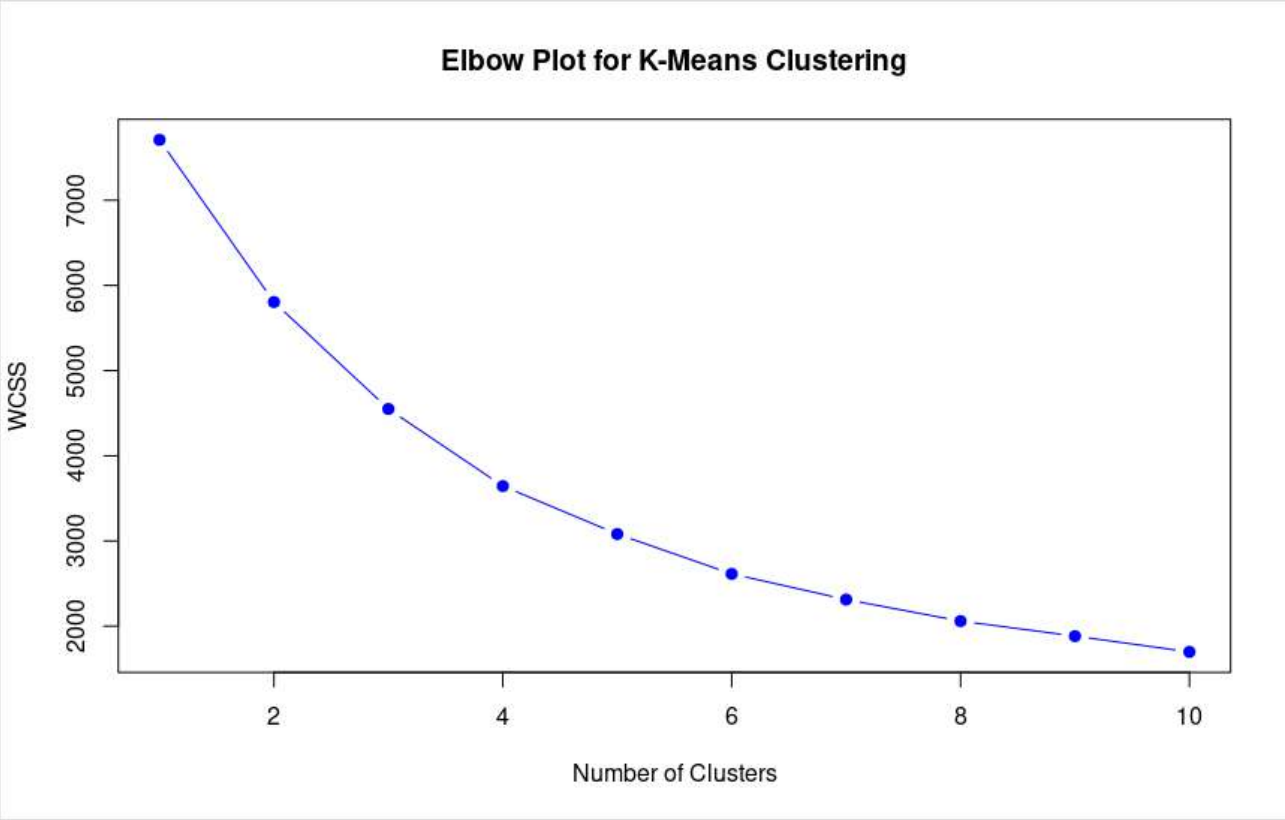
Within cluster sum of squares by cluster:
[1] 1727.050 4078.382
 (between_SS / total_SS =  24.7 %)
```

```
Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss" "betweenss"
"size"         "iter"         "ifault"
>
> # 9. Display cluster centers
> kravix$centers
         Year       Month         Age
1  1.2060626  0.04530518 -0.04292019
2 -0.6125808 -0.02301131  0.02179993
>
> # 9. Display number of samples in each cluster
> table(kravix$cluster)

   1    2
 866 1705
>
> # 10. Calculate WCSS for K = 1 to 10
> glimora <- numeric(10)
>
> set.seed(23)
>
> for (k in 1:10) {
+     glimora[k] <- kmeans(
+         zuntrix,
+         centers = k,
+         nstart = 25
+     )$tot.withinss
+ }
>
> glimora
 [1] 7710.000 5805.432 4549.630 3644.390 3080.971 2613.001 2311.816 2059.002 1882.261 1697.230
>
> # 10. Display WCSS values
> data.frame(
+     Clusters = 1:10,
+     WCSS = glimora
+ )
   Clusters     WCSS
1         1 7710.000
2         2 5805.432
3         3 4549.630
4         4 3644.390
5         5 3080.971
6         6 2613.001
7         7 2311.816
8         8 2059.002
9         9 1882.261
10       10 1697.230
>
> # 11. Display the elbow plot
> plot(
+     1:10,
+     glimora,
+     type = "b",
+     pch = 19,
+     col = "blue",
+     xlab = "Number of Clusters",
+     ylab = "WCSS",
+     main = "Elbow Plot for K-Means Clustering"
```

```
+ )
```




```
> # 12. Apply K-Means clustering based on the elbow recommendation
> set.seed(23)
>
> nuvexa <- kmeans(
+     zuntrix,
+     centers = 3,
+     nstart = 25
+ )
>
> # 12. Display final K-Means results
> nuvexa
K-means clustering with 3 clusters of sizes 923, 899, 749

Cluster means:
         Year       Month          Age
1  0.5735959  0.81316228 -0.149629092
2  0.5117347 -0.85729217  0.155902509
3 -1.3210662  0.02691172 -0.002735252

Clustering vector:
   [1] 1 3 2 2 3 3 2 1 1 2 1 2 2 2 1 3 1 2 2 1 1 2 3 1 3 1 1 3 2 2 2 2 3 2 2 3 1 3 3 1 2 1 3 2 1 2 2 2 3 3
  [71] 2 3 3 1 3 1 3 1 3 1 3 2 3 3 1 1 3 2 3 3 3 1 2 1 2 1 1 2 2 3 1 1 1 1 3 1 3 3 1 3 2 2 1 3 2 3 3 1 1 1
 [141] 1 1 2 2 2 1 3 2 3 3 3 2 2 2 1 2 3 1 2 1 2 3 1 1 1 3 1 1 3 2 1 2 3 3 1 3 3 3 2 1 2 2 1 1 1 3 1 3 3 2 1
 [211] 1 2 2 1 2 1 2 3 1 1 1 3 2 3 3 3 1 1 1 1 2 2 1 1 1 1 1 3 3 1 2 3 2 1 3 2 2 2 2 1 2 2 3 3 2 2 1 3 1 3 1
 [281] 3 1 2 3 2 2 1 2 2 3 3 1 1 2 1 3 3 1 1 2 2 1 3 3 2 2 2 2 1 2 1 3 1 1 2 3 2 1 3 2 3 2 2 2 2 2 2 2 1 2 3
 [351] 2 2 1 3 1 1 1 3 3 3 1 1 3 2 2 3 2 1 3 1 2 3 3 2 1 2 2 1 1 3 1 3 1 1 2 2 1 1 1 1 2 1 1 2 1 1 2 3 1 2 2
 [421] 1 2 1 2 3 1 3 1 1 1 1 3 1 2 3 1 2 3 1 3 3 3 2 1 3 2 3 3 1 3 1 2 2 2 1 1 2 1 2 1 1 1 3 3 2 1 3 1 1 2 1
 [491] 2 2 1 3 1 1 1 2 1 1 1 3 1 2 1 3 1 3 3 1 3 3 3 3 2 2 2 1 3 2 2 2 3 2 1 1 3 2 2 2 2 2 2 1 1 2 1 2 2 1 2
 [561] 3 3 1 1 3 1 1 3 1 2 1 1 3 1 1 3 3 1 3 2 3 2 2 1 1 3 3 3 3 2 1 2 3 1 2 2 3 2 1 1 2 1 3 3 1 3 1 1 1 3 2
 [631] 3 2 1 1 1 1 3 2 1 3 2 3 1 1 1 1 3 3 2 2 2 1 2 3 3 2 2 3 3 1 1 3 1 1 3 1 2 3 1 3 3 2 1 2 2 2 2 3 3 2 1
```

```
 [701] 1 3 3 3 3 3 1 1 2 2 3 3 3 2 3 1 1 3 1 2 2 1 3 2 1 2 1 1 1 3 2 2 2 3 1 1 2 3 3 3 1 2 3 2 1 2 1 1 2 1 3
 [771] 1 2 3 1 2 1 2 2 1 1 1 1 1 3 1 3 3 1 3 1 3 3 2 1 3 1 2 3 2 3 2 3 1 3 3 2 1 2 2 3 1 2 1 2 3 1 2 2 1 1 2
 [841] 3 1 3 1 3 2 1 2 3 2 2 1 2 3 1 3 1 1 3 3 3 2 2 3 1 3 1 3 1 3 2 3 1 2 2 1 3 1 1 1 1 1 1 2 3 2 2 2 1 2 3
 [911] 2 1 2 1 1 3 1 1 3 2 2 1 3 2 3 1 2 3 3 1 3 2 3 1 1 3 2 1 2 3 1 2 2 2 2 3 1 1 2 1 2 1 1 3 2 3 1 2 3 3 3
 [981] 2 2 2 1 1 2 2 2 2 3 1 3 3 1 3 3 3 2 1 2
 [ reached 'max' / getOption("max.print") -- omitted 1571 entries ]

Within cluster sum of squares by cluster:
[1] 1548.572 1491.128 1509.930
 (between_SS / total_SS =  41.0 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss" "betweenss"    "size"
>
>

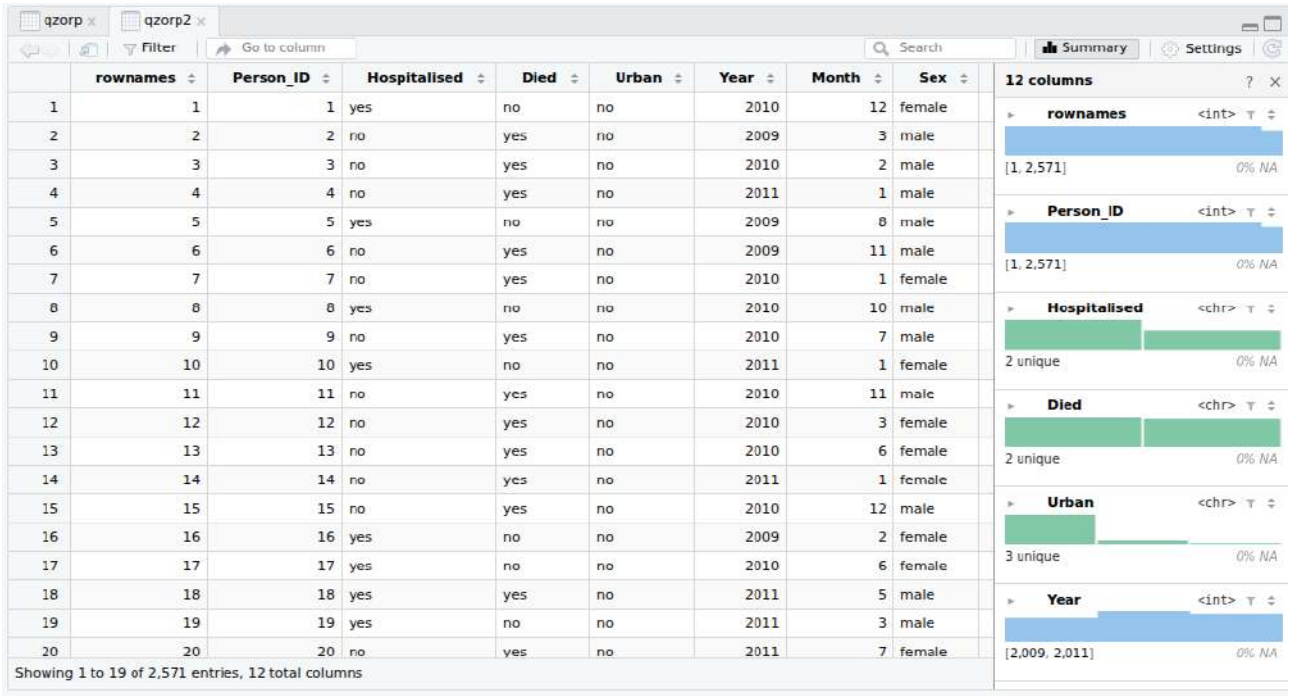
> # 12. Display final cluster centers
> nuvexa$centers
          Year       Month          Age
1  0.5735959  0.81316228 -0.149629092
2  0.5117347 -0.85729217  0.155902509
3 -1.3210662  0.02691172 -0.002735252
>
> # 12. Display final cluster assignments
> nuvexa$cluster
   [1] 1 3 2 2 3 3 2 1 1 2 1 2 2 2 1 3 1 2 2 1 1 2 3 1 3 1 1 3 2 2 2 2 3 2 2 3 1 3 3 1 2 1 3 2 1 2
2 2 3 3 1 2 1 3 2 2 1 1 3 2 1 1 3 3 2 3 2 3 2 3
  [71] 2 3 3 1 3 1 3 1 3 1 3 2 3 3 1 1 3 2 3 3 3 1 2 1 2 1 1 2 2 3 1 1 1 1 3 1 3 3 1 3 2 2 1 3 2 3
3 1 1 1 3 1 3 2 1 1 3 1 3 2 2 1 2 2 1 3 3 2 1 1
 [141] 1 1 2 2 2 1 3 2 3 3 3 2 2 2 1 2 3 1 2 1 2 3 1 1 1 3 1 1 3 2 1 2 3 3 1 3 3 3 2 1 2 2 1 1 1 3
1 3 3 2 1 1 1 2 3 2 1 3 1 3 1 3 1 1 1 2 3 2 2 2
 [211] 1 2 2 1 2 1 2 3 1 1 1 3 2 3 3 3 1 1 1 1 2 2 1 1 1 1 1 3 3 1 2 3 2 1 3 2 2 2 2 1 2 2 3 3 2 2
1 3 1 3 1 1 2 1 2 2 2 1 3 2 2 2 2 2 1 2 1 1 2 1
 [281] 3 1 2 3 2 2 1 2 2 3 3 1 1 2 1 3 3 1 1 2 2 1 3 3 2 2 2 2 1 2 1 3 1 1 2 3 2 1 3 2 3 2 2 2 2 2
2 2 1 2 3 1 2 2 2 1 1 1 2 3 3 1 3 1 1 2 3 1 1 1
 [351] 2 2 1 3 1 1 1 3 3 3 1 1 3 2 2 3 2 1 3 1 2 3 3 2 1 2 2 1 1 3 1 3 1 1 2 2 1 1 1 1 2 1 1 2 1 1
2 3 1 2 2 1 3 2 2 3 2 2 1 1 1 2 1 1 1 3 2 1 2 2
 [421] 1 2 1 2 3 1 3 1 1 1 1 3 1 2 3 1 2 3 1 3 3 3 2 1 3 2 3 3 1 3 1 2 2 2 1 1 2 1 2 1 1 1 3 3 2 1
3 1 1 2 1 2 1 2 3 1 2 1 2 1 3 2 1 2 1 3 3 1 3 1
 [491] 2 2 1 3 1 1 1 2 1 1 1 3 1 2 1 3 1 3 3 1 3 3 3 3 2 2 2 1 3 2 2 2 3 2 1 1 3 2 2 2 2 2 2 1 1 2
1 2 2 1 2 1 2 1 2 2 1 1 1 2 1 1 1 2 1 2 1 1 1 2
 [561] 3 3 1 1 3 1 1 3 1 2 1 1 3 1 1 3 3 1 3 2 3 2 2 1 1 3 3 3 3 2 1 2 3 1 2 2 3 2 1 1 2 1 3 3 1 3
1 1 1 3 2 3 1 1 1 1 3 1 2 3 2 2 1 3 2 1 1 2 1 3
 [631] 3 2 1 1 1 1 3 2 1 3 2 3 1 1 1 1 3 3 2 2 2 1 2 3 3 2 2 3 3 1 1 3 1 1 3 1 2 3 1 3 3 2 1 2 2 2
2 3 3 2 1 1 2 3 2 3 2 3 3 1 1 1 2 1 1 2 2 1 2 3
 [701] 1 3 3 3 3 3 1 1 2 2 3 3 3 2 3 1 1 3 1 2 2 1 3 2 1 2 1 1 1 3 2 2 2 3 1 1 2 3 3 3 1 2 3 2 1 2
1 1 2 1 3 2 2 2 2 3 3 3 3 2 3 3 3 3 2 3 2 2 2 3
 [771] 1 2 3 1 2 1 2 2 1 1 1 1 1 3 1 3 3 1 3 1 3 3 2 1 3 1 2 3 2 3 2 3 1 3 3 2 1 2 2 3 1 2 1 2 3 1
2 2 1 1 2 1 2 2 1 2 1 1 3 3 1 1 3 1 1 2 2 3 1 1
 [841] 3 1 3 1 3 2 1 2 3 2 2 1 2 3 1 3 1 1 3 3 3 2 2 3 1 3 1 3 1 3 2 3 1 2 2 1 3 1 1 1 1 1 1 2 3 2
2 2 1 2 3 1 2 1 3 2 3 3 3 3 2 2 2 1 3 3 1 2 3 3
 [911] 2 1 2 1 1 3 1 1 3 2 2 1 3 2 3 1 2 3 3 1 3 2 3 1 1 3 2 1 2 3 1 2 2 2 2 3 1 1 2 1 2 1 1 3 2 3
1 2 3 3 3 3 1 1 2 1 3 1 2 2 1 3 3 2 2 3 2 3 3 2
 [981] 2 2 2 1 1 2 2 2 2 3 1 3 3 1 3 3 3 2 1 2
 [ reached 'max' / getOption("max.print") -- omitted 1571 entries ]
>

> # 12. Display number of samples in each cluster
> table(nuvexa$cluster)
```

```
  1   2   3
923 899 749
>
>
```

2)

```
> #Hierarchical Clustering
> #23bds0081 vyom khurana
> # 1. Load the dataset shared through e-mail
> qzorp2 <- read.csv(
+     "https://raw.githubusercontent.com/salemprakash/EDA/main/Data/SuicideChina.csv",
+     stringsAsFactors = FALSE
+ )
>
> # 2. View the dataset
> head(qzorp2)
  rownames Person_ID Hospitalised Died Urban Year Month    Sex Age Education Occupation
method
1        1         1          yes   no    no 2010    12 female  39 Secondary  household Other
poison
2        2         2           no  yes    no 2009     3   male  83   primary    farming
Hanging
3        3         3           no  yes    no 2010     2   male  60   primary    farming
Hanging
4        4         4           no  yes    no 2011     1   male  73   primary    farming
Hanging
5        5         5          yes   no    no 2009     8   male  51 Secondary    farming
Pesticide
6        6         6           no  yes    no 2009    11   male  62 iliterate    farming
Pesticide
>
> # 2. View the dataset
> View(qzorp2)
>
```



qzorp | qzorp2

Filter | Go to column | Search | Summary | Settings

	rownames	Person_ID	Hospitalised	Died	Urban	Year	Month	Sex
1	1	1	yes	no	no	2010	12	female
2	2	2	no	yes	no	2009	3	male
3	3	3	no	yes	no	2010	2	male
4	4	4	no	yes	no	2011	1	male
5	5	5	yes	no	no	2009	8	male
6	6	6	no	yes	no	2009	11	male
7	7	7	no	yes	no	2010	1	female
8	8	8	yes	no	no	2010	10	male
9	9	9	no	yes	no	2010	7	male
10	10	10	yes	no	no	2011	1	female
11	11	11	no	yes	no	2010	11	male
12	12	12	no	yes	no	2010	3	female
13	13	13	no	yes	no	2010	6	female
14	14	14	no	yes	no	2011	1	female
15	15	15	no	yes	no	2010	12	male
16	16	16	yes	no	no	2009	2	female
17	17	17	yes	no	no	2010	6	female
18	18	18	yes	yes	no	2011	5	male
19	19	19	yes	no	no	2011	3	male
20	20	20	no	yes	no	2011	7	female

Showing 1 to 19 of 2,571 entries, 12 total columns

12 columns

rownames <int> [1, 2,571] 0% NA

Person_ID <int> [1, 2,571] 0% NA

Hospitalised <chr> 2 unique 0% NA

Died <chr> 2 unique 0% NA

Urban <chr> 3 unique 0% NA

Year <int> [2,009, 2,011] 0% NA

```
 # 3. Check missing values
> colSums(is.na(qzorp2))
    rownames    Person_ID Hospitalised         Died        Urban         Year        Month
Sex           Age    Education   Occupation
             0            0            0            0            0            0            0
0            0            0            0
       method
            0
>
> # 4. Select only numerical attributes
> bravixo <- qzorp2[sapply(qzorp2, is.numeric)]
>
> # 4. Remove unnecessary ID attributes
> bravixo <- bravixo[, !(names(bravixo) %in% c("rownames", "Person_ID"))]
>
> # 4. Display numerical attributes
> head(bravixo)
  Year Month Age
1 2010    12  39
2 2009     3  83
3 2010     2  60
4 2011     1  73
5 2009     8  51
6 2009    11  62
>
> # 5. Standardize the data
> xelvori <- scale(bravixo)
>
> # 5. Standardize the data
> xelvori <- scale(bravixo)
>
> # 5. Display standardized data
> head(xelvori)
              Year      Month         Age
[1,] -0.05750176  1.780373 -0.68898943
[2,] -1.32106616 -1.029918  1.53504368
```

```
[3,] -0.05750176 -1.342172  0.37248092
[4,]  1.20606263 -1.654427  1.02958161
[5,] -1.32106616  0.531355 -0.08243495
[6,] -1.32106616  1.468119  0.47357333
>
> # 6. Calculate Euclidean distance
> draxilo <- dist(
+     xelvori,
+     method = "euclidean"
+ )
>
> # 6. Display Euclidean distance
> draxilo
                  1          2          3          4          5          6          7          8
9         10         11         12
2    3.80008624
                 13         14         15         16         17         18         19         20
21         22         23         24
2
                 25         26         27         28         29         30         31         32
33         34         35         36
2
                 37         38         39         40         41         42         43         44
45         46         47         48
2
                 49         50         51         52         53         54         55         56
57         58         59         60
2
                 61         62         63         64         65         66         67         68
69         70         71         72
2
                 73         74         75         76         77         78         79         80
81         82         83         84
2
                 85         86         87         88         89         90         91         92
93         94         95         96
2
                 97         98         99        100        101        102        103        104
105        106        107        108
2
                109        110        111        112        113        114        115        116
117        118        119        120
2
                121        122        123        124        125        126        127        128
129        130        131        132
2
                133        134        135        136        137        138        139        140
141        142        143        144
2
                145        146        147        148        149        150        151        152
153        154        155        156
2
                157        158        159        160        161        162        163        164
165        166        167        168
2
                169        170        171        172        173        174        175        176
177        178        179        180
2
                181        182        183        184        185        186        187        188
189        190        191        192
```

2
193 194 195 196 197 198 199 200
201 202 203 204
2
205 206 207 208 209 210 211 212
213 214 215 216
2
217 218 219 220 221 222 223 224
225 226 227 228
2
229 230 231 232 233 234 235 236
237 238 239 240
2
241 242 243 244 245 246 247 248
249 250 251 252
2
253 254 255 256 257 258 259 260
261 262 263 264
2
265 266 267 268 269 270 271 272
273 274 275 276
2
277 278 279 280 281 282 283 284
285 286 287 288
2
289 290 291 292 293 294 295 296
297 298 299 300
2
301 302 303 304 305 306 307 308
309 310 311 312
2
313 314 315 316 317 318 319 320
321 322 323 324
2
325 326 327 328 329 330 331 332
333 334 335 336
2
337 338 339 340 341 342 343 344
345 346 347 348
2
349 350 351 352 353 354 355 356
357 358 359 360
2
361 362 363 364 365 366 367 368
369 370 371 372
2
373 374 375 376 377 378 379 380
381 382 383 384
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385 386 387 388 389 390 391 392
393 394 395 396
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397 398 399 400 401 402 403 404
405 406 407 408
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409 410 411 412 413 414 415 416
417 418 419 420
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429 430 431 432

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433 434 435 436 437 438 439 440
441 442 443 444
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445 446 447 448 449 450 451 452
453 454 455 456
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457 458 459 460 461 462 463 464
465 466 467 468
2
469 470 471 472 473 474 475 476
477 478 479 480
2
481 482 483 484 485 486 487 488
489 490 491 492
2
493 494 495 496 497 498 499 500
501 502 503 504
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505 506 507 508 509 510 511 512
513 514 515 516
2
517 518 519 520 521 522 523 524
525 526 527 528
2
529 530 531 532 533 534 535 536
537 538 539 540
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541 542 543 544 545 546 547 548
549 550 551 552
2
553 554 555 556 557 558 559 560
561 562 563 564
2
565 566 567 568 569 570 571 572
573 574 575 576
2
577 578 579 580 581 582 583 584
585 586 587 588
2
589 590 591 592 593 594 595 596
597 598 599 600
2
601 602 603 604 605 606 607 608
609 610 611 612
2
613 614 615 616 617 618 619 620
621 622 623 624
2
625 626 627 628 629 630 631 632
633 634 635 636
2
637 638 639 640 641 642 643 644
645 646 647 648
2
649 650 651 652 653 654 655 656
657 658 659 660
2
661 662 663 664 665 666 667 668
669 670 671 672

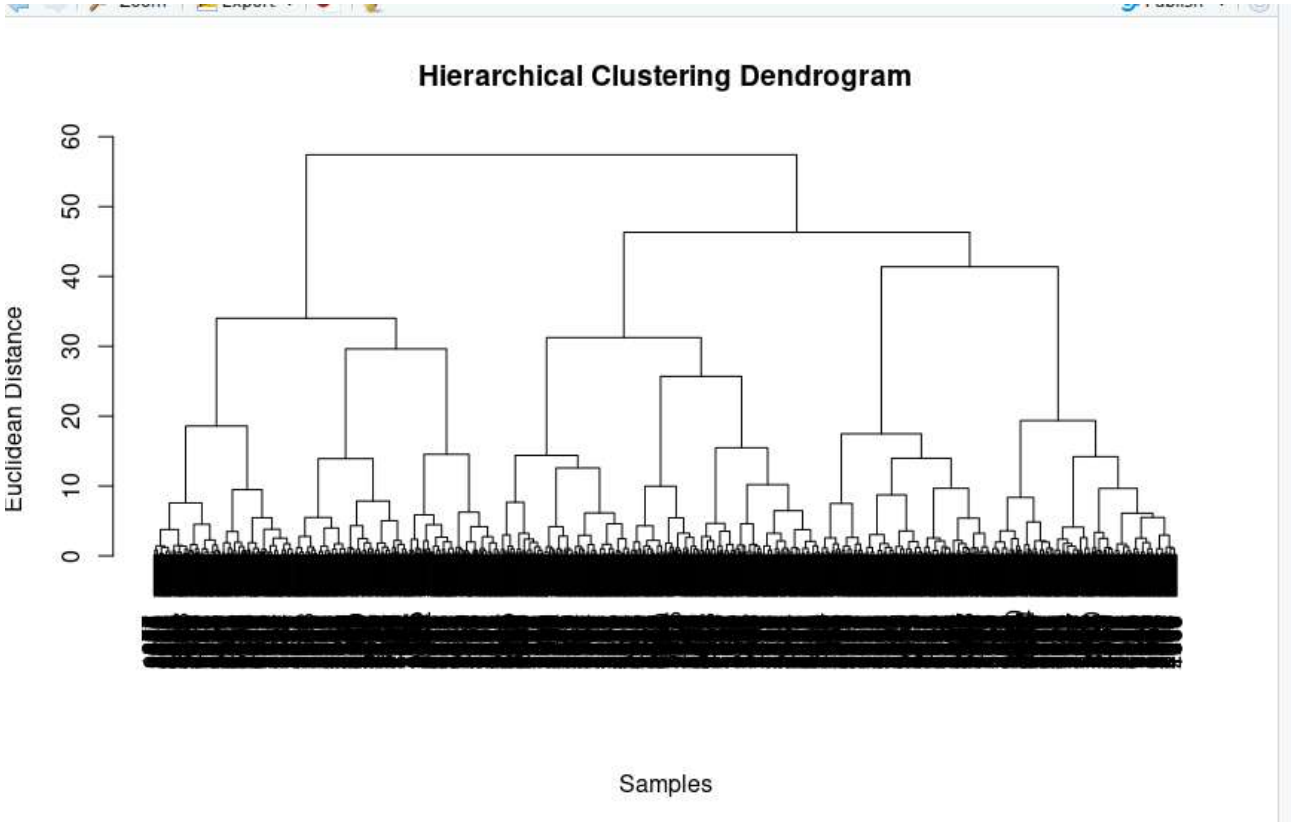
2
673 674 675 676 677 678 679 680
681 682 683 684
2
685 686 687 688 689 690 691 692
693 694 695 696
2
697 698 699 700 701 702 703 704
705 706 707 708
2
709 710 711 712 713 714 715 716
717 718 719 720
2
721 722 723 724 725 726 727 728
729 730 731 732
2
733 734 735 736 737 738 739 740
741 742 743 744
2
745 746 747 748 749 750 751 752
753 754 755 756
2
757 758 759 760 761 762 763 764
765 766 767 768
2
769 770 771 772 773 774 775 776
777 778 779 780
2
781 782 783 784 785 786 787 788
789 790 791 792
2
793 794 795 796 797 798 799 800
801 802 803 804
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805 806 807 808 809 810 811 812
813 814 815 816
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817 818 819 820 821 822 823 824
825 826 827 828
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837 838 839 840
2
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849 850 851 852
2
853 854 855 856 857 858 859 860
861 862 863 864
2
865 866 867 868 869 870 871 872
873 874 875 876
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877 878 879 880 881 882 883 884
885 886 887 888
2
889 890 891 892 893 894 895 896
897 898 899 900
2
901 902 903 904 905 906 907 908
909 910 911 912

```
2
              913         914         915         916         917         918         919         920
921         922         923         924
2
              925         926         927         928         929         930         931         932
933         934         935         936
2
              937         938         939         940         941         942         943         944
945         946         947         948
2
              949         950         951         952         953         954         955         956
957         958         959         960
2
              961         962         963         964         965         966         967         968
969         970         971         972
2
              973         974         975         976         977         978         979         980
981         982         983         984
2
              985         986         987         988         989         990         991         992
993         994         995         996
2
              997         998         999        1000
2
 [ reached 'max' / getOption("max.print") -- omitted 2569 rows and 1570 columns ]
> 
> # 7. Perform Hierarchical Clustering using Ward's method
> hierzol <- hclust(
+     draxilo,
+     method = "ward.D2"
+ )
> 
> 
> # 7. Display Hierarchical Clustering result
> hierzol

Call:
hclust(d = draxilo, method = "ward.D2")

Cluster method   : ward.D2
Distance         : euclidean
Number of objects: 2571

> 
> # 8. Plot the dendrogram
> plot(
+     hierzol,
+     main = "Hierarchical Clustering Dendrogram",
+     xlab = "Samples",
+     ylab = "Euclidean Distance",
+     sub = ""
+ )
> 
```




```
> # 10. Display number of samples in each cluster
> table(clustara)
clustara
  1   2   3
807 893 871
>
> # 10. Display cluster assignment table
> data.frame(
+     Sample = 1:length(clustara),
+     Cluster = clustara
+ )
     Sample Cluster
1         1       1
2         2       2
3         3       2
4         4       3
5         5       1
6         6       1
7         7       2
8         8       1
9         9       2
10       10       2
11       11       3
12       12       2
13       13       2
14       14       2
15       15       3
16       16       1
17       17       2
18       18       3
19       19       2
20       20       3
```

21	21	3
22	22	2
23	23	1
24	24	2
25	25	1
26	26	3
27	27	3
28	28	1
29	29	2
30	30	3
31	31	2
32	32	2
33	33	2
34	34	2
35	35	3
36	36	2
37	37	3
38	38	1
39	39	1
40	40	2
41	41	3
42	42	1
43	43	1
44	44	2
45	45	3
46	46	2
47	47	2
48	48	2
49	49	1
50	50	1
51	51	3
52	52	2
53	53	2
54	54	1
55	55	2
56	56	2
57	57	3
58	58	3
59	59	2
60	60	2
61	61	3
62	62	3
63	63	1
64	64	1
65	65	3
66	66	1
67	67	3
68	68	1
69	69	2
70	70	1
71	71	2
72	72	1
73	73	1
74	74	2
75	75	1
76	76	2
77	77	1
78	78	3
79	79	2
80	80	3

81	81	1
82	82	2
83	83	1
84	84	1
85	85	1
86	86	1
87	87	1
88	88	2
89	89	1
90	90	1
91	91	2
92	92	3
93	93	3
94	94	3
95	95	2
96	96	3
97	97	3
98	98	3
99	99	2
100	100	1
101	101	3
102	102	1
103	103	2
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106	106	3
107	107	1
108	108	1
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115	115	3
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117	117	1
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120	120	3
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127	127	1
128	128	1
129	129	2
130	130	2
131	131	3
132	132	1
133	133	3
134	134	2
135	135	3
136	136	1
137	137	1
138	138	3
139	139	1
140	140	3

141	141	3
142	142	2
143	143	2
144	144	2
145	145	3
146	146	3
147	147	1
148	148	2
149	149	1
150	150	2
151	151	1
152	152	3
153	153	3
154	154	3
155	155	3
156	156	2
157	157	1
158	158	3
159	159	2
160	160	3
161	161	3
162	162	1
163	163	3
164	164	3
165	165	3
166	166	1
167	167	1
168	168	1
169	169	1
170	170	3
171	171	1
172	172	2
173	173	1
174	174	2
175	175	3
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177	177	1
178	178	1
179	179	2
180	180	1
181	181	2
182	182	3
183	183	3
184	184	3
185	185	2
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187	187	2
188	188	2
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190	190	3
191	191	1
192	192	3
193	193	2
194	194	2
195	195	1
196	196	2
197	197	3
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199	199	3
200	200	2

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203	203	3
204	204	1
205	205	1
206	206	3
207	207	1
208	208	3
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211	211	2
212	212	2
213	213	2
214	214	3
215	215	2
216	216	2
217	217	2
218	218	1
219	219	1
220	220	1
221	221	2
222	222	1
223	223	3
224	224	1
225	225	1
226	226	1
227	227	3
228	228	3
229	229	2
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235	235	3
236	236	3
237	237	3
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247	247	2
248	248	2
249	249	3
250	250	1
251	251	2
252	252	3
253	253	1
254	254	2
255	255	3
256	256	3
257	257	3
258	258	1
259	259	1
260	260	1

261	261	3
262	262	2
263	263	3
264	264	3
265	265	3
266	266	3
267	267	3
268	268	3
269	269	2
270	270	3
271	271	3
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293	293	3
294	294	3
295	295	3
296	296	2
297	297	1
298	298	3
299	299	3
300	300	3
301	301	3
302	302	3
303	303	1
304	304	1
305	305	2
306	306	2
307	307	2
308	308	2
309	309	3
310	310	2
311	311	1
312	312	1
313	313	3
314	314	3
315	315	3
316	316	1
317	317	2
318	318	3
319	319	1
320	320	2

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327	327	3
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329	329	1
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331	331	1
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335	335	2
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370	370	1
371	371	3
372	372	1
373	373	1
374	374	3
375	375	3
376	376	2
377	377	3
378	378	3
379	379	3
380	380	1

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383	383	3
384	384	3
385	385	2
386	386	2
387	387	3
388	388	1
389	389	1
390	390	1
391	391	2
392	392	2
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394	394	2
395	395	1
396	396	1
397	397	2
398	398	1
399	399	3
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402	402	1
403	403	1
404	404	3
405	405	2
406	406	1
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410	410	1
411	411	3
412	412	2
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414	414	1
415	415	3
416	416	1
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419	419	3
420	420	2
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422	422	3
423	423	3
424	424	2
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481	481	1
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483	483	1
484	484	2
485	485	3
486	486	1
487	487	1
488	488	1
489	489	1
490	490	3
491	491	3
492	492	2
493	493	3
494	494	1
495	495	3
496	496	2
497	497	1
498	498	3
499	499	1
500	500	3

```
 [ reached 'max' / getOption("max.print") -- omitted 2071 rows ]
> 
> 
```

EDA LAB ASSESSMENT – 9
VYOM KHURANA
23BDS0081

Lab Assessment - 9 Principal Component Analysis

1. Load the dataset sent to your email into R, and verify that the packages ggplot2, ggfortify, FactoMineR, and factoextra are loaded.

```
> # Name: Vyom Khurana
> # Registration Number: 23BDS0081
>
> library(ggplot2)
>
> suicide_data <- read.csv(
+     "https://raw.githubusercontent.com/salemprakash/EDA/main/Data/SuicideChina.csv",
+     stringsAsFactors = FALSE
+ )
>
```

2. Display the total number of rows and columns in the dataset.

```
 dim(suicide_data)
[1] 2571   12
>
> nrow(suicide_data)
[1] 2571
>
> ncol(suicide_data)
[1] 12
>
```

3. Extract only the numeric columns, clean the data, and normalize it using the scale() function into an object named data_scaled.

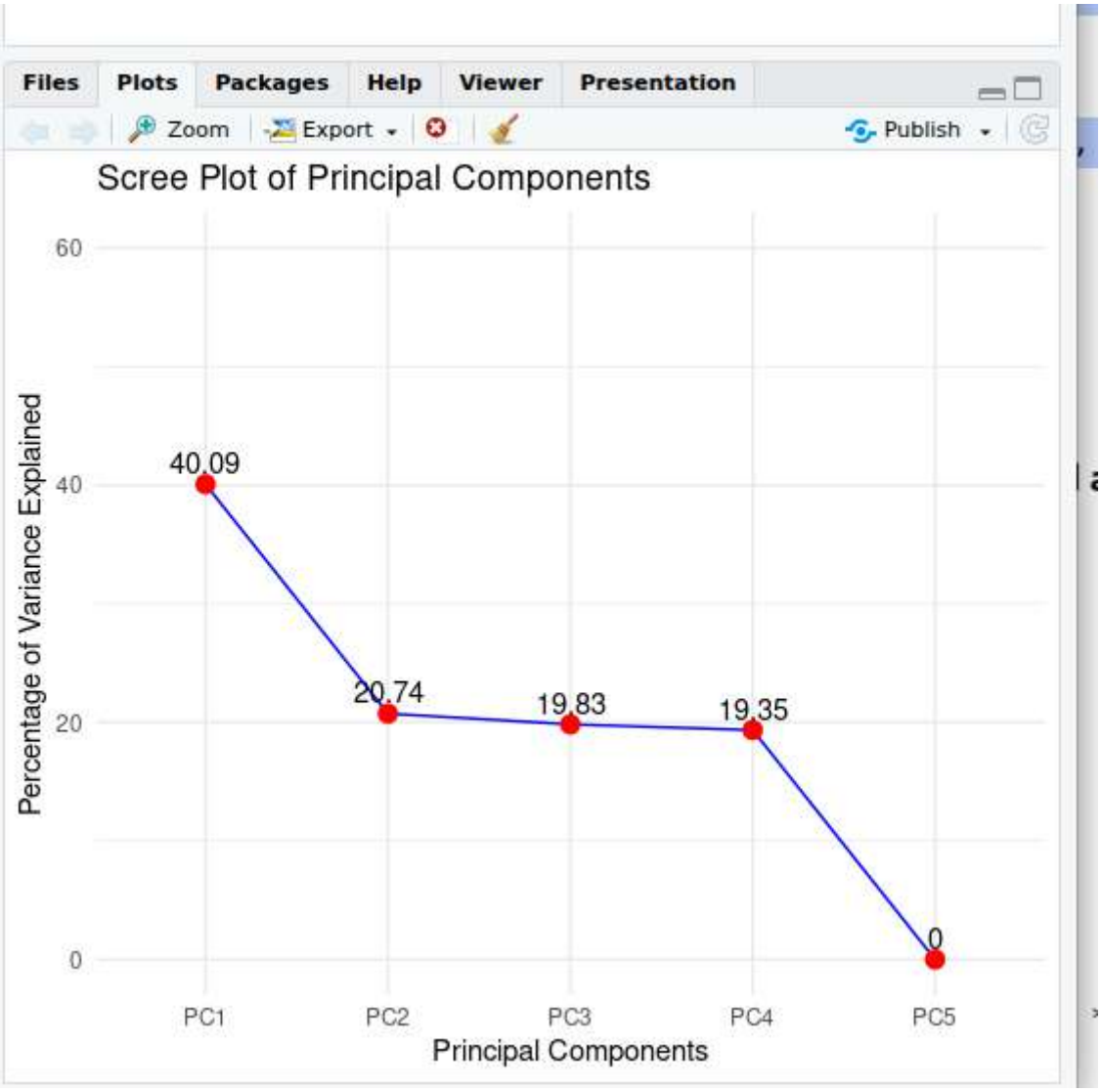
```
> data_numeric <- suicide_data[, sapply(suicide_data, is.numeric)]
>
> data_numeric <- na.omit(data_numeric)
>
> data_scaled <- scale(data_numeric)
>
> head(data_scaled)
       rownames Person_ID        Year     Month         Age
[1,] -1.731041 -1.731041 -0.05750176  1.780373 -0.68898943
[2,] -1.729693 -1.729693 -1.32106616 -1.029918  1.53504368
[3,] -1.728346 -1.728346 -0.05750176 -1.342172  0.37248092
[4,] -1.726999 -1.726999  1.20606263 -1.654427  1.02958161
[5,] -1.725652 -1.725652 -1.32106616  0.531355 -0.08243495
[6,] -1.724305 -1.724305 -1.32106616  1.468119  0.47357333
>
```

4. Perform Principal Component Analysis on data_scaled using either the PCA() or prcomp() function, and print its summary.

```
> pca_result <- prcomp(data_scaled, center = TRUE, scale. = FALSE)
>
> summary(pca_result)
Importance of components:
                            PC1    PC2    PC3    PC4       PC5
Standard deviation     1.4157 1.0183 0.9957 0.9836 1.034e-15
Proportion of Variance 0.4009 0.2074 0.1983 0.1935 0.000e+00
Cumulative Proportion  0.4009 0.6082 0.8065 1.0000 1.000e+00
>
```

5. Plot a scree plot showing the percentage of variance for each component with data labels, setting the y-axis limit from 0 to 60.

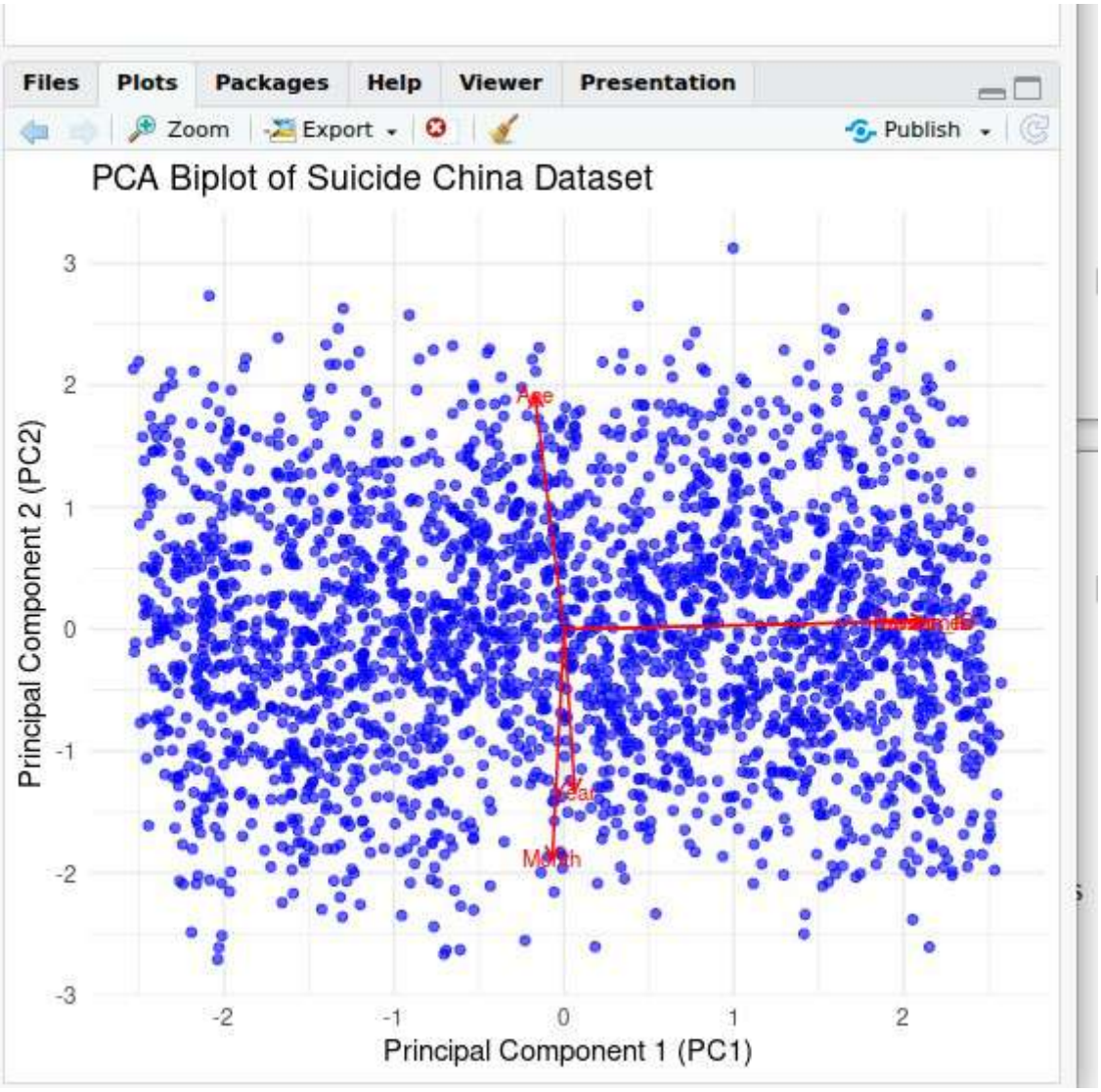
```
> variance_percent <- (pca_result$sdev^2 / sum(pca_result$sdev^2)) * 100
>
> scree_data <- data.frame(
+     Component = paste0("PC", seq_along(variance_percent)),
+     Variance = variance_percent
+ )
>
> ggplot(scree_data, aes(x = Component, y = Variance)) +
+     geom_line(aes(group = 1), color = "blue") +
+     geom_point(color = "red", size = 3) +
+     geom_text(aes(label = round(Variance, 2)), vjust = -0.5) +
+     scale_y_continuous(limits = c(0, 60)) +
+     labs(
+         title = "Scree Plot of Principal Components",
+         x = "Principal Components",
+         y = "Percentage of Variance Explained"
+     ) +
+     theme_minimal()
>
```



6. Create a PCA biplot with non-overlapping labels and a proper title.

```
> scores <- as.data.frame(pca_result$x)
>
> loadings <- as.data.frame(pca_result$rotation)
>
> ggplot(scores, aes(x = PC1, y = PC2)) +
+     geom_point(color = "blue", alpha = 0.6) +
+     geom_segment(
+         data = loadings,
+         aes(x = 0, y = 0, xend = PC1 * 3, yend = PC2 * 3),
+         inherit.aes = FALSE,
+         arrow = arrow(length = unit(0.2, "cm")),
+         color = "red"
+     ) +
+     geom_text(
+         data = loadings,
+         aes(x = PC1 * 3, y = PC2 * 3, label = rownames(loadings)),
+         inherit.aes = FALSE,
+         color = "red",
+         size = 3
+     ) +
+     labs(
+         title = "PCA Biplot of Suicide China Dataset",
```

```
+         x = "Principal Component 1 (PC1)",
+         y = "Principal Component 2 (PC2)"
+     ) +
+     theme_minimal()
>
```



7. Extract the PCA scores into a data frame named pca_scores and display the first six rows.

```
> pca_scores <- as.data.frame(pca_result$x)
>
> head(pca_scores, 6)
         PC1        PC2        PC3         PC4          PC5
1 -2.445098 -1.6109594 -0.9130999  0.48301213 1.857851e-15
2 -2.533654  2.1348671 -0.3926516  0.12143955 1.944129e-15
3 -2.431072  1.0280227  0.6556650 -0.72180329 1.938845e-15
4 -2.432785  1.0864472  2.0186668 -0.20732033 1.914621e-15
5 -2.470775  0.1196948 -1.3651156 -0.08660786 1.803920e-15
6 -2.522061 -0.1119261 -1.6845791  0.92746300 2.059178e-15
>
```

8. Draw a scatter plot of the first two principal components (PC1 vs. PC2) with blue circular points and your registration number in the title

```
> ggplot(pca_scores, aes(x = PC1, y = PC2)) +
```

```
+      geom_point(color = "blue", shape = 16, size = 3) +
+      labs(
+          title = "PCA Scatter Plot - 23BDS0081",
+          x = "Principal Component 1 (PC1)",
+          y = "Principal Component 2 (PC2)"
+      ) +
+      theme_minimal()
>
```

variance_percent num [1:5] 4.01e+01 2.07e+01 1.98e+01 1.93e+01 2.14e-…

